

stat_reversekl_two_coverage

Specialization: linear_exact_shell

TL;DR. For every fixed finite public experiment with a nonempty exact shell and every $0 < \epsilon < 1$, an explicit measurable idealized global-optimization learner achieves the all-temperature sample-complexity bound $\tilde{O}(dD \min\{\eta/\epsilon, \epsilon^{-2}\})$ without using C , D , or the law-specific context distribution as inputs. The numerical indices do not determine the fixed-design minimax frontier: one experiment can have zero risk for every $\epsilon > 0$, while another has a pointwise local $\Omega_{C,D,\eta}(d/\epsilon)$ lower bound. There is no uniform lower dependence on D or η , and no positive quadratic lower branch persists through all sufficiently small accuracies.

1 Project justification

Gap. Zhao et al. already give a single-policy fast upper bound for general realizable reverse-KL policy learning and an existential density-ratio lower bound. In a directionally rich linear parameter class their single-policy quantity becomes the trace of the target-versus-logging covariance ratio, whereas the present D_P is its largest eigenvalue; the bounded prediction polytope alone does not guarantee that trace identity. What is not supplied by the cited literature is the bounded shared-linear exact-shell combination developed here: an all-temperature expected-minimax upper bound of order dD_P up to logarithms, uniform over a fixed exact shell; a finite fixed-experiment feasibility certificate for simultaneous exact (C_P, D_P) ; and a same-index comparison between a zero-risk public experiment and an experiment with pointwise-local d/ϵ difficulty.

Niche. The note studies reverse-KL-regularized offline linear contextual-bandit policy learning on exact shells indexed jointly by pointwise coverage C and generalized-eigenvalue feature coverage D , conditional on a fixed public experiment.

Fill. It gives a finite exponential nonlinear-semidefinite certificate for fixed-design shell nonemptiness, an explicit measurable idealized global-optimization learner with the universal clean dD upper order at every temperature, and experiment-specific witnesses proving that the exact (d, C, D, η) indices do not determine the fixed-design frontier. The contribution is specialized to reverse-KL exact shells: it does not claim the general principle that coverage summaries can be insufficient, and its only informative lower comparison is pointwise and local in d/ϵ on a constructed experiment.

2 Related work

Zhao et al. are the closest general-function reverse-KL comparator, not a tabular-only result. In their offline i.i.d. model the reference policy equals the behavior policy and the reward class $\mathcal{G}$ is general and realizable. Their Theorem 2.10 gives, for sufficiently small ϵ , the high-probability sample-complexity upper bound $\tilde{O}(\eta D_{\pi^*}^2 \epsilon^{-1} \log N_{\mathcal{G}}(\epsilon))$. For a linear class $\mathcal{G}_{\Theta} = \{\phi^{\top} \vartheta : \vartheta \in \Theta\}$, their pointwise coefficient is

$$D_{\mathcal{G}}^2((x, a); \pi_{\text{ref}}) = \sup_{\delta \in \Theta - \Theta} \frac{\{\phi(x, a)^{\top} \delta\}^2}{\delta^{\top} \Lambda_{\text{ref}} \delta}.$$

Only when $\Theta - \Theta$ contains a nonzero multiple of every needed direction does this equal $\phi(x, a)^{\top} \Lambda_{\text{ref}}^{-1} \phi(x, a)$. Under that directional-richness condition, $D_{\pi^*}^2 = \text{tr}(M)$, where $M = \Lambda_{\text{ref}}^{-1/2} \Lambda_P^* \Lambda_{\text{ref}}^{-1/2}$, and therefore $D_P \leq D_{\pi^*}^2 \leq dD_P$. The bounded prediction polytope alone does not imply directional richness; without it, only $D_{\mathcal{G}}^2((x, a); \pi_{\text{ref}}) \leq \phi(x, a)^{\top} \Lambda_{\text{ref}}^{-1} \phi(x, a)$ is automatic, not the trace identity or the lower comparison with D_P . Thus the spectral factor can be up to d smaller than the trace factor in the directionally rich specialization, but the present bounded shared-linear, all-temperature, expected-minimax exact-shell theorem does not wholesale dominate Zhao et al.'s general-function, sufficiently-small-accuracy, high-probability result. Their Theorem 2.11 is existential over a

density-ratio upper bound C^* , not a lower bound on an exact (C_P, D_P) shell. Ji et al. provide the complementary matched tabular multi-armed-bandit frontier.

Li–Ma–Srebro study unregularized offline linear contextual bandits with fixed logged covariates and a known test distribution. Their ℓ_p -confidence-set rules maximize a pessimistic unregularized value over deterministic policies; for PUNC, Theorem 1 controls $V(\pi^*) - V(\hat{\pi})$ at an $n^{-1/2}$ rate through the whitened first-moment quantity $\|\Sigma_{\mathcal{D}}^{-1/2} \mathbb{E}_{\rho} \phi(s, \pi^*(s))\|_1$. The learner here instead selects a confidence-set member minimizing a worst empirical Gibbs-weighted squared radius and deploys its stochastic Gibbs policy; its loss is reverse-KL-regularized welfare, and its coverage quantity D_P is the largest generalized eigenvalue of target and logging feature second moments. Li–Ma–Srebro therefore supply the linear-pessimism precedent but do not imply the bound $\tilde{O}(\min\{\eta d D/n, \sqrt{d D/n}\})$ or its all-temperature sample-complexity inversion.

Nguyen–Tang–Arora characterize broad unregularized offline contextual-bandit classes using policy-transfer coefficients and pseudo-dimension: Theorem 4.1 gives an information-theoretic class lower bound and Theorem 5.1 analyzes the OfDM-Hedge actor–critic learner. Their Example 3.3 already demonstrates that zero policy suboptimality can coexist with a vacuous concentrability coefficient. Accordingly, the present note does not claim novelty for the general observation that coarse coverage can overstate difficulty. Its narrower theorem fixes the reverse-KL Gibbs objective and exact (C_P, D_P) equalities, gives a finite fixed-design shell-feasibility certificate, and constructs two public experiments with the same (d, C, D, η) but different minimax frontiers; those conclusions do not follow from the policy-transfer theorems.

Chen et al.’s Theorem 1 develops an interactive Fano framework that recovers classical Fano, Le Cam, and Assouad arguments, while their Theorems 10 and 15 characterize interactive model classes through fractional covering and DEC quantities. The paired sign-cube argument used here belongs to that information-theoretic lineage, but the present data are passive i.i.d. observations and Chen et al. do not establish exact reverse-KL-shell feasibility or same-index public-experiment nonuniqueness. Vannella et al.’s passive contextual-linear BPI Theorem 2 gives a fixed-confidence PAC-learnability condition and an instance-dependent information lower bound determined by the logged feature span and design matrix. That result concerns identifying an approximately optimal unregularized deterministic policy with confidence $1 - \delta$, rather than expected reverse-KL-regularized regret, exact (C, D) shells, or a zero-risk-versus-local- d/ϵ minimax comparison.

The primary-source basis for these comparisons is: Zhao et al., Section 2.1, Definitions 2.4–2.5, Remark 2.6, Assumptions 2.7–2.8, Theorems 2.10–2.11, and Appendix B, Li–Ma–Srebro, Problem Setup and Theorem 1, Nguyen–Tang–Arora, Sections 3–5, Chen et al., Theorems 1, 10, and 15, and Vannella et al..

3 Setup

Target (estimand / structural parameter): $N^*(\epsilon; d, C, D, \eta)$. Primary functional / estimator / diagnostic: $N^*(\epsilon; d, C, D, \eta) = \inf\{n \in \mathbb{N} : \mathfrak{R}_n(\mathcal{M}(d, C, D, \eta)) \leq \epsilon\}$.

- **d** — integer; $\{4, 5, \dots\}$; shell index

Definition. linear feature dimension

- **n** — integer; $\mathbb{N}$; sample-size index

Definition. number of logged observations

- **\epsilon** — scalar; $(0, 1)$; accuracy index

Definition. target regularized regret

- **\eta** — scalar; $\mathbb{R}_{>0}$; regularization index

Definition. reverse-KL inverse temperature

- **C** — scalar; $[1, \infty)$; shell index

Definition. exact pointwise policy-coverage index

- **D** — scalar; $(0, \infty)$; shell index

Definition. exact lambda-max feature-coverage index

- δ_n — scalar; $(0,1]$; failure probability
Definition. $\delta_n = n^{-2}$
- $\mathcal{X}$ — set; finite set; primitive support
Definition. context support
- $\mathcal{A}$ — set; finite set; primitive support
Definition. action support
- S — set; $\mathcal{X} \times \mathcal{A}$; prediction domain
Definition. context-action support
- X — random variable; $\mathcal{X}$; data coordinate
Definition. logged context
- A — random variable; $\mathcal{A}$; data coordinate
Definition. logged action
- Y — random variable; $[0,1]$; data coordinate
Definition. observed reward
- ρ — probability measure; $\Delta(\mathcal{X})$; sampling primitive
Definition. context distribution
- ϕ — map; $\phi : \mathcal{X} \times \mathcal{A} \rightarrow \mathbb{R}^d$; linear structure
Definition. known reward feature map
- θ — vector; $\mathbb{R}^d$; law primitive
Definition. linear reward coefficient
- r_θ — function; $r_\theta : S \rightarrow [0,1]$; conditional reward mean
Definition. $r_\theta(x, a) = \phi(x, a)^\top \theta$
- π_{ref} — Markov kernel; $\pi_{\text{ref}} : \mathcal{X} \rightarrow \Delta(\mathcal{A})$; sampling and regularization primitive
Definition. known reference and logging policy
- $\mathfrak{E}$ — statistical experiment; public contextual-bandit designs; common experiment
Definition. $\mathfrak{E} = (\mathcal{X}, \mathcal{A}, \phi, \pi_{\text{ref}}, \eta)$ is fixed and publicly supplied to every learner.
- P — contextual-bandit law; laws of X, A, Y ; model member
Definition. one bounded linear contextual-bandit specification
- $P^{\otimes n}$ — probability measure; laws on $(S \times [0,1])^n$; sample law
Definition. the product observation law induced by the one-observation law P
- Z_i — random vector; $S \times [0,1]$; observed datum
Definition. $Z_i = (X_i, A_i, Y_i)$
- $\mathcal{D}_n$ — random sample; $(S \times [0,1])^n$; learner input
Definition. $\mathcal{D}_n = (Z_i)_{i=1}^n$

- $J_{\eta,P}$ — extended-real functional; $J_{\eta,P} : \Delta(\mathcal{A})^{\mathcal{X}} \rightarrow \mathbb{R} \cup \{-\infty\}$; extended regularized welfare with a finite reference-supported restriction

Definition. The paper uses the standard extended reverse-KL convention $p \log(p/0) = +\infty$ for $p > 0$ and $0 \log(0/q) = 0$. Thus

$$J_{\eta,P}(\pi) = \mathbb{E}_{X \sim \rho, A \sim \pi(\cdot | X)} r_{\theta}(X, A) - \eta^{-1} \mathbb{E}_{X \sim \rho} \text{KL}(\pi(\cdot | X) \| \pi_{\text{ref}}(\cdot | X)).$$

If $\rho(x) > 0$ and $\pi(a | x) > 0 = \pi_{\text{ref}}(a | x)$ for some action, then $J_{\eta,P}(\pi) = -\infty$. On policies satisfying $\text{supp } \pi(\cdot | x) \subseteq \text{supp } \pi_{\text{ref}}(\cdot | x)$ for every context, this functional is finite and its restriction is real valued.

- π_P^* — policy; $\pi_P^* : \mathcal{X} \rightarrow \Delta(\mathcal{A})$; population Gibbs optimum

Definition.

$$\pi_P^*(a | x) = \pi_{\text{ref}}(a | x) \exp\{\eta r_{\theta}(x, a)\} / \sum_{b \in \mathcal{A}} \pi_{\text{ref}}(b | x) \exp\{\eta r_{\theta}(x, b)\}$$

- Λ_{ref} — matrix; $\mathbb{R}^{d \times d}$; logging feature covariance

Definition.

$$\Lambda_{\text{ref}} = \mathbb{E}_{X \sim \rho, A \sim \pi_{\text{ref}}(\cdot | X)} [\phi(X, A) \phi(X, A)^{\top}]$$

- Λ_P^* — matrix; $\mathbb{R}^{d \times d}$; target feature covariance

Definition. $\Lambda_P^* = \mathbb{E}_{X \sim \rho, A \sim \pi_P^*(\cdot | X)} [\phi(X, A) \phi(X, A)^{\top}]$

- C_P — scalar; $[1, \infty)$; pointwise coverage

Definition.

$$C_P = \text{ess sup}_{(x, a) \sim \rho \times \pi_{\text{ref}}} \pi_P^*(a | x) / \pi_{\text{ref}}(a | x)$$

- D_P — scalar; $(0, \infty)$; lambda-max feature coverage

Definition. $D_P = \lambda_{\max}(\Lambda_{\text{ref}}^{-1/2} \Lambda_P^* \Lambda_{\text{ref}}^{-1/2})$

- $D_{2,P}$ — scalar; $[1, \infty)$; trace-type feature coverage

Definition. $D_{2,P} = \text{tr}(\Lambda_{\text{ref}}^{-1} \Lambda_P^*)$

- $\mathcal{F}$ — compact set; $[0, 1]^S$; learner domain

Definition. bounded linear prediction polytope

- L_n — functional; $L_n : \mathcal{F} \rightarrow \mathbb{R}_+$; learner criterion

Definition. empirical squared loss

- $\widehat{f}$ — function; $\mathcal{F}$; prediction rule

Definition. sequentially lexicographically selected empirical-risk minimizer

- $\|\cdot\|_n$ — seminorm; functions on S ; confidence geometry

Definition. empirical logging prediction seminorm

- $\mathcal{Q}_n$ — compact set; subsets of $\mathcal{F}$; pessimistic uncertainty set

Definition. prediction confidence polytope

- $\underline{J}_n$ — functional;

$\underline{J}_n : \{\pi \in \Delta(\mathcal{A})^{\mathcal{X}} : \text{supp } \pi(\cdot | x) \subseteq \text{supp } \pi_{\text{ref}}(\cdot | x) \text{ for every } x \in \mathcal{X}\} \rightarrow \mathbb{R}$
;finite diagnostic empirical criterion

Definition. Diagnostic empirical pessimistic welfare criterion evaluated only on reference-supported policies; it does not define $\hat{\pi}$. Every Gibbs learner constructed in the note is automatically reference supported.

- $\widehat{\pi}$ — learner; $\hat{\pi} = \hat{\pi}(\mathcal{D}_n; \mathfrak{E})$; deployed lower-envelope Gibbs learner

Definition. $\hat{\pi}$ is the Gibbs policy generated by the coordinatewise lower envelope of $\mathcal{Q}_n$, as specified in def:prediction-polytope-learner.

- $\mathcal{L}_n(\mathcal{E})$ — set; measurable maps from $(S \times [0, 1])^n \times \{\mathfrak{E}\}$ to

$\{\pi \in \Delta(\mathcal{A})^{\mathcal{X}} : \text{supp } \pi(\cdot | x) \subseteq \text{supp } \pi_{\text{ref}}(\cdot | x) \text{ for every } x \in \mathcal{X}\}$
;finite-objective minimax decision class, without loss relative to the full-simplex class

Definition. The effective minimax decision class consists of sample-measurable policy learners whose every output is supported on the public reference policy. The support condition depends only on π_{ref} , hence the class is law-independent. Under the extended reverse-KL convention, its minimax infimum equals the infimum over all measurable full-simplex policy learners, because the public measurable support repair in def:minimax-risk never increases regret.

- $\mathcal{F}_{\text{hard}}^{\text{circ}(d, \eta)}$ — set; subsets of $\mathbb{R}^2$; lower-bound domain

Definition. proved interior hard-shell region

- k — integer; $\{1, 2, \dots\}$; number of hard coordinates

Definition. $k = \lfloor (d - 1)/2 \rfloor$

- $\mathbf{e}_0, \mathbf{u}_j, \mathbf{w}_j, \mathbf{z}$ — vectors; $\mathbb{R}^d$; hard-family coordinates

Definition. orthonormal basis vectors, with $\mathbf{z}$ used only when d is even

- $\mathbf{e}_{\{x, a\}}$ — vector; $\mathbb{R}^{|S|}$; tabular reduction

Definition. canonical coordinate vector indexed by $(x, a) \in S$

- $\mathbf{x}_C, \mathbf{x}_A, \mathbf{x}_j, \mathbf{x}_z$ — contexts; $\mathcal{X}$; hard-family support

Definition. pointwise-calibration, anchor, hard, and optional unused-coordinate contexts

- q — scalar; $(0, 1)$; pointwise calibration

Definition. $q = (e^n/C - 1)/(e^n - 1)$

- p — scalar; $(0, 1/4)$; hard-context reference probability

Definition. $p = (4D)^{-1}$

- T_{γ} — scalar; $\mathbb{R}_{>0}$; hard-family calibration

Definition. hard-context Gibbs calibration

- $g_0(D, \eta)$ — scalar; $[0, \infty)$; hard-family range

Definition. largest admissible hard perturbation scale

- β — scalar; $[0, 1]$; hard-family calibration

Definition. hard-context reward baseline

- $\gamma(\epsilon, d, C, D, \eta)$ — scalar; $\mathbb{R}_{>0}$; testing calibration

Definition. explicit hard-family perturbation

- $\epsilon_0(C, D, \eta)$ — scalar; $\mathbb{R}_{>0}$; hard-family range

Definition. admissible lower-bound accuracy endpoint

- Z — scalar; $\mathbb{R}_{>0}$; hard-family calibration

Definition. hard-context Gibbs normalizer

- h — function; $\text{span}(\phi)$; policy-conversion argument

Definition. linear prediction contrast

- v — sign vector; $\{-1, 1\}^k$; packing index

Definition. index of a paired-Bernoulli hard instance

- $\mathcal{H}_{\{d, C, D, \eta, \epsilon\}}$ — finite family; sets of specifications P ; lower-bound construction

Definition. explicit paired-Bernoulli exact-shell family

- T — matrix; $\text{GL}_d(\mathbb{R})$; invariance map

Definition. invertible feature reparameterization

- c_0 — scalar; $\mathbb{R}_{>0}$; learner calibration

Definition. fixed universal confidence-radius constant

- $\mathcal{M}_{\{\frac{1}{E}\}}(d, C, D, \eta)$ — class; sets of specifications P compatible with $\mathfrak{E}$; minimax model class

Definition. public-experiment-indexed exact linear reverse-KL shell

- $\mathcal{M}(d, C, D, \eta)$ — class; sets of specifications P ; fixed-experiment shell abbreviation

Definition. abbreviation for $\mathcal{M}_{\mathfrak{E}}(d, C, D, \eta)$ after $\mathfrak{E}$ is fixed and no experiments are compared

- $\mathcal{R}_{\{n, \frac{1}{E}\}}$ — functional;

$\mathfrak{R}_{n, \mathfrak{E}} : \{\mathcal{M} \neq \emptyset : \mathcal{M} \text{ consists of laws compatible with } \mathfrak{E}\} \rightarrow \mathbb{R}_+$
; risk frontier under an equivalent finite-objective decision representation

Definition. Minimax expected regularized suboptimality conditional on $\mathfrak{E}$, computed over measurable reference-supported learners. Its value is exactly the same as the full-simplex extended-reverse-KL minimax value. It is a nonnegative real number because the public reference learner is supported and has finite regret.

- $\mathcal{R}_n$ — functional; $\mathfrak{R}_n : \{\mathcal{M} \neq \emptyset\} \rightarrow \mathbb{R}_+$; fixed-experiment risk abbreviation under the equivalent finite-objective representation

Definition. Abbreviation for the nonnegative real functional $\mathfrak{R}_{n, \mathfrak{E}}$ after $\mathfrak{E}$ is fixed and no experiments are compared; it inherits the exact equivalence between the supported finite-objective and full-simplex extended-reverse-KL formulations.

- $N^{\star}_{\{\frac{1}{E}\}}$ — functional;

$N^{\star}_{\mathfrak{E}} : \{(\epsilon, d, C, D, \eta) : \mathcal{M}_{\mathfrak{E}}(d, C, D, \eta) \neq \emptyset\} \rightarrow \mathbb{N} \cup \{\infty\}$
; headline frontier

Definition. exact-shell minimax sample complexity conditional on the explicit public experiment $\mathfrak{E}$

- $N^{\star}$ — functional; $N^{\star} : \{(\epsilon, d, C, D, \eta) : \mathcal{M}(d, C, D, \eta) \neq \emptyset\} \rightarrow \mathbb{N} \cup \{\infty\}$; fixed-experiment headline-frontier abbreviation

Definition. abbreviation for $N^{\star}_{\mathfrak{E}}$ after $\mathfrak{E}$ is fixed and no experiments are compared

- $\widehat{\pi}_{\{\mathrm{ERM}\}}$ — learner; $\widehat{\pi}_{\mathrm{ERM}} = \widehat{\pi}_{\mathrm{ERM}}(\mathcal{D}_n; \mathfrak{E})$; empirical-risk-minimizer Gibbs plug-in learner

Definition. The Gibbs policy generated by the sequentially selected empirical-risk minimizer $\widehat{f}$, as specified in `def:erm-gibbs-plugin-learner`.

- κ_η — scalar; $\mathbb{R}_{>0}$; Gibbs plug-in quadratic-conversion constant

Definition. $\kappa_\eta = \eta \int_0^1 t e^{2\eta t} dt = [e^{2\eta}(2\eta - 1) + 1]/(4\eta)$

4 Assumptions

Assumption 1 (`ass:finite-contexts`). $\mathcal{X}$ is finite.

Standard (*finite contextual-bandit context support*, [1]).

Assumption 2 (`ass:finite-actions`). $\mathcal{A}$ is finite.

Standard (*finite contextual-bandit action support*, [1]).

Assumption 3 (`ass:bounded-features`). $\sup_{(x,a) \in S} \|\phi(x, a)\|_2 \leq 1$.

Standard (*bounded linear features*, [3]).

Assumption 4 (`ass:linear-realizability`). $\mathbb{E}_P[Y \mid X = x, A = a] = r_\theta(x, a)$ for every $(x, a) \in S$.

Standard (*realizable linear reward model*, [3]).

Assumption 5 (`ass:reference-logging`). $P(A = a \mid X = x) = \pi_{\mathrm{ref}}(a \mid x)$ for every $(x, a) \in S$.

Standard (*behavior equals KL reference*, [1]).

Assumption 6 (`ass:nonsingular-logging-geometry`). $\Lambda_{\mathrm{ref}} \succ 0$.

Standard (*full-rank linear design*, [3]).

Assumption 7 (`ass:pointwise-exact-shell`). $C_P = C$.

Novel. *The maximum target-to-logging likelihood ratio is an auditable finite-support property that defines a deployment coverage tier independently of an estimation procedure.*

Assumption 8 (`ass:feature-exact-shell`). $D_P = D$.

Novel. *The target-versus-logging generalized eigenvalue is a representation-aware coverage diagnostic determined by the law and feature map.*

5 Definitions

Definition 1 (`def:common-experiment`). Defined object. *common public-design experiment*

Construction. Fix the finite supports $\mathcal{X}, \mathcal{A}$, carrying their intended discrete/Borel measurable structures, the known feature map ϕ , the known reference and logging policy π_{ref} , and η as $\mathfrak{E} = (\mathcal{X}, \mathcal{A}, \phi, \pi_{\mathrm{ref}}, \eta)$. The context distribution ρ is not a component of $\mathfrak{E}$; it remains law-specific. For every compatible one-observation law P , $\mathcal{D}_n \sim P^{\otimes n}$. A learner is an element of $\mathcal{L}_n(\mathfrak{E})$, so it receives $\mathcal{D}_n$ and $\mathfrak{E}$, returns a policy, and is supported on π_{ref} at every context. Because π_{ref} is public, this support requirement is law-independent and uses only public design inputs. By `def:minimax-risk`, it is without loss relative to the class of all measurable full-simplex policy learners under the extended reverse-KL convention. Write $\mathcal{M}_{\mathfrak{E}}(d, C, D, \eta)$, $\mathfrak{R}_{n, \mathfrak{E}}$, and $N_{\mathfrak{E}}^*$ whenever public experiments are compared. The unindexed notation $\mathcal{M}(d, C, D, \eta)$, $\mathfrak{R}_n$, and N^* is an abbreviation only after one $\mathfrak{E}$ has been fixed and no cross-experiment comparison is being made.

Inputs. $\mathfrak{E}$; ρ ; P ; $P^{\otimes n}$; $\mathcal{D}_n$; $\mathcal{L}_n(\mathfrak{E})$; $\mathcal{M}_{\mathfrak{E}}(d, C, D, \eta)$; $\mathfrak{R}_{n, \mathfrak{E}}$; $N_{\mathfrak{E}}^*$.

Definition 2 (def:exact-shell). Defined object. *exact linear reverse-KL shell conditional on a public experiment*

Construction.

$\mathcal{M}_{\mathfrak{E}}(d, C, D, \eta) = \{P : P \text{ is compatible with the fixed public experiment } \mathfrak{E}, \rho \text{ is the context distribution supplied by } P, \text{ and } \text{ass:finite-contexts, ass:finite-actions, ass:bounded-features, ass:linear-realizability, ass:reference-logging, ass:nonsingular-logging-geometry, ass:pointwise-exact-shell, and ass:feature-exact-shell hold for } P\}$

. When $\mathfrak{E}$ is fixed and no experiments are compared, write $\mathcal{M}(d, C, D, \eta) = \mathcal{M}_{\mathfrak{E}}(d, C, D, \eta)$.

Member properties. *ass:finite-contexts; ass:finite-actions; ass:bounded-features; ass:linear-realizability; ass:reference-logging; ass:nonsingular-logging-geometry; ass:pointwise-exact-shell; ass:feature-exact-shell.*

Definition 3 (def:prediction-polytope-learner). Defined object. *invariant prediction-polytope pessimistic learner*

Construction. Let $S = \mathcal{X} \times \mathcal{A}$ and $\mathcal{F} = \{f \in [0, 1]^S : \exists \theta \in \mathbb{R}^d, f(x, a) = \phi(x, a)^\top \theta\}$. Fix once and for all a lexicographic order of S . Set $L_n(f) = n^{-1} \sum_{i=1}^n (Y_i - f(X_i, A_i))^2$, and let $\hat{f}$ be the sequentially lexicographically selected minimizer of L_n over the compact polytope $\mathcal{F}$. Define $\|h\|_n^2 = n^{-1} \sum_{i=1}^n h(X_i, A_i)^2$ and $\mathcal{Q}_n = \{f \in \mathcal{F} : \|f - \hat{f}\|_n^2 \leq c_0[d \log(en) + \log(2n^2)]/n\}$. The set $\mathcal{Q}_n$ is nonempty and compact because it contains $\hat{f}$. For every $(x, a) \in S$, define the unique minimum value $r_n(x, a) = \min_{f \in \mathcal{Q}_n} f(x, a)$; if an attaining witness is needed, select one sequentially lexicographically using the fixed order of S . Define the deployed policy at every context, observed or unobserved, by

$$\hat{\pi}(a \mid x) = \frac{\pi_{\text{ref}}(a \mid x) e^{\eta r_n(x, a)}}{\sum_{b \in \mathcal{A}} \pi_{\text{ref}}(b \mid x) e^{\eta r_n(x, b)}}.$$

This Gibbs policy satisfies $\text{supp } \hat{\pi}(\cdot \mid x) = \text{supp } \pi_{\text{ref}}(\cdot \mid x)$ for every context and therefore belongs to the finite-objective decision class. For reference-supported policies π , retain

$$\underline{J}_n(\pi) = \inf_{f \in \mathcal{Q}_n} [n^{-1} \sum_i \sum_a \pi(a \mid X_i) f(X_i, a) - (\eta n)^{-1} \sum_i \text{KL}(\pi(\cdot \mid X_i) \parallel \pi_{\text{ref}}(\cdot \mid X_i))]$$

as the empirical pessimistic criterion; no real-valued off-support evaluation is intended. In any feature coordinates,

$$\|f_\theta - f_{\hat{\theta}}\|_n^2 = (\theta - \hat{\theta})^\top \hat{\Sigma}_n (\theta - \hat{\theta})$$

, where $\hat{\Sigma}_n = n^{-1} \sum_i \phi(X_i, A_i) \phi(X_i, A_i)^\top$; this equality is notation only and no coefficient representative is selected.

Inputs. $\backslash \text{mathcal } D_n; \backslash \text{phi}; \backslash \text{pi}_{\text{-}\{\backslash \text{mathrm}\{\text{ref}\}\}}; \backslash \text{eta}; d; n$.

Definition 4 (def:minimax-risk). Defined object. *public-experiment-indexed exact-shell minimax risk*

Construction. For a fixed public experiment $\mathfrak{E}$ and every nonempty class $\mathcal{M}$ of laws compatible with $\mathfrak{E}$, use the standard extended reverse-KL convention in $J_{\eta, P}$. Thus a policy that assigns positive mass outside $\text{supp } \pi_{\text{ref}}(\cdot \mid x)$ at any context with positive ρ -mass has objective $-\infty$ and regret $+\infty$. For an arbitrary policy π , define the public contextwise support repair

$$(\mathcal{S}_{\text{ref}} \pi)(a \mid x) = \begin{cases} \pi(a \mid x), & \pi(b \mid x) = 0 \text{ for every } b \text{ such that } \pi_{\text{ref}}(b \mid x) = 0, \\ \pi_{\text{ref}}(a \mid x), & \text{otherwise.} \end{cases}$$

Because the supports have the standing finite discrete/Borel measurable structures, $\mathcal{S}_{\text{ref}}$ is Borel measurable and sends every measurable policy learner to an element of $\mathcal{L}_n(\mathfrak{E})$. For every law and every realized learner output, its repair leaves the objective unchanged when every offending context has zero ρ -mass, while otherwise the unrepaired regret is $+\infty$ and the repaired regret is finite. Hence repair never increases regret. Since supported learners are themselves full-simplex learners, the two infima are equal, and their common value is

$$\mathfrak{R}_{n, \mathfrak{E}}(\mathcal{M}) = \inf_{\tilde{\pi} \in \mathcal{L}_n(\mathfrak{E})} \sup_{P \in \mathcal{M}} \mathbb{E}_{P \otimes n} [J_{\eta, P}(\pi_P^*) - J_{\eta, P}(\tilde{\pi}(D_n; \mathfrak{E}))].$$

Equivalently, the right-hand side equals the same infimum taken over all measurable full-simplex policy-valued learners with the displayed loss interpreted in $[0, +\infty]$. The Gibbs optimum has exactly the support of π_{ref} , and the public reference learner has finite regret, so the common minimax value is a nonnegative real number. When $\mathfrak{E}$ is fixed and no experiments are compared, write $\mathfrak{R}_n = \mathfrak{R}_{n, \mathfrak{E}}$.

Inputs. $\mathcal{D}_n; \mathcal{M}_{\mathfrak{E}}; \mathcal{L}_n(\mathcal{M}_{\mathfrak{E}}); P^{\otimes n}; J_{\{\eta, P\}}; \pi_{P^*}; \mathcal{R}_{\{n, \mathcal{M}_{\mathfrak{E}}\}}$.

Definition 5 (def:sample-complexity). Defined object. *public-experiment-indexed exact-shell sample complexity*

Construction. For $\mathcal{M}_{\mathfrak{E}}(d, C, D, \eta) \neq \emptyset$, define

$$N_{\mathfrak{E}}^*(\epsilon; d, C, D, \eta) = \inf \left\{ n \in \mathbb{N} : \mathfrak{R}_{n, \mathfrak{E}}(\mathcal{M}_{\mathfrak{E}}(d, C, D, \eta)) \leq \epsilon \right\},$$

with the convention $\inf \emptyset = \infty$. When $\mathfrak{E}$ is fixed and no experiments are compared, write $N^* = N_{\mathfrak{E}}^*$.

Inputs. $\epsilon; d; C; D; \eta; \mathcal{M}_{\mathfrak{E}}; \mathcal{R}_{\{n, \mathcal{M}_{\mathfrak{E}}\}}; \mathcal{M}_{\mathfrak{E}}(d, C, D, \eta); N^{\star}(\mathcal{M}_{\mathfrak{E}}(d, C, D, \eta))$.

Definition 6 (def:hard-region). Defined object. *proved interior hard-shell region*

Construction. $\mathfrak{F}_{\text{hard}}^{\circ}(d, \eta) = \{(C, D) : 1 < D \leq C < e^{\eta}, 2D - 1 < e^{\eta}, d \geq 4\}$

Inputs. $d; C; D; \eta$.

Definition 7 (def:hard-accuracy-range). Defined object. *explicit hard-family accuracy range*

Construction. $\epsilon_0(C, D, \eta) = [64\eta]^{-1} \log \cosh(\eta \min\{g_0(D, \eta), 1/4\})$, indexed by $0 < \epsilon \leq \epsilon_0(C, D, \eta)$.

Inputs. $C; D; \eta; g_0(D, \eta)$.

Definition 8 (def:feasible-frontier-handle). Defined object. *feasible-frontier handle*

Construction. Handle: derive the joint Gibbs feasibility constraints for C_P and D_P , then construct covariance-preserving lower families or prove their nonexistence.

Inputs. $C_P; D_P; \pi_{P^*}; \pi_{\text{ref}}; \Lambda_{\text{ref}}; \Lambda_{P^*}$.

Definition 9 (def:explicit-hard-family). Defined object. *explicit paired-Bernoulli exact-shell family*

Construction. For $(C, D) \in \mathfrak{F}_{\text{hard}}^{\circ}(d, \eta)$, put $k = \lfloor (d-1)/2 \rfloor$, take an orthonormal basis $e_0, (u_j, w_j)_{j=1}^k$, and add z when d is even. Use actions $\{+, -, 0\}$. At x_C , set $q = (e^{\eta}/C - 1)/(e^{\eta} - 1)$, $\pi_{\text{ref}}(+ | x_C) = q$, distribute $1 - q$ positively over $-, 0$, and set $\phi(x_C, +) = e_0$, $\phi(x_C, -) = \phi(x_C, 0) = 0$, and $\theta_0 = 1$. Put $A = q(C - D)/(D - 1)$, add x_A when $C > D$ with every action feature equal to e_0 , give x_C raw mass $\tau = [4(1 + A)]^{-1}$ and x_A raw mass τA , and give the hard contexts total raw mass one. When z is present, add x_z with raw mass $1/4$, every action feature z , and coefficient zero. Normalize these raw masses to define ρ . At each x_j , of raw mass $1/k$, put $p = (4D)^{-1}$, $\pi_{\text{ref}}(+ | x_j) = \pi_{\text{ref}}(- | x_j) = p$, $\pi_{\text{ref}}(0 | x_j) = 1 - 2p$, $\phi(x_j, +) = (u_j + w_j)/\sqrt{2}$, $\phi(x_j, -) = (u_j - w_j)/\sqrt{2}$, and $\phi(x_j, 0) = 0$. For $v \in \{-1, 1\}^k$, let $T_{\gamma} = D(1 - 2p)/[e^{\eta\gamma} - 2pD \cosh(\eta\gamma)]$, $\beta = \eta^{-1} \log T_{\gamma}$, and $\theta_{u_j} = \sqrt{2}\beta$, $\theta_{w_j} = \sqrt{2}\gamma v_j$, so $r_v(x_j, \pm) = \beta \pm \gamma v_j$. Define

$$g_0(D, \eta) = \sup \left\{ g > 0 : \text{for every } s \in [0, g], e^{\eta s} - 2pD \cosh(\eta s) > 0 \text{ and } s \leq \eta^{-1} \log T_s \leq 1 - s \right\},$$

and set $\gamma(\epsilon, d, C, D, \eta) = \eta^{-1} \text{arcosh}(e^{64\eta\epsilon})$. The family $\mathcal{H}_{d, C, D, \eta, \epsilon}$ comprises these 2^k Bernoulli laws whenever $\gamma \leq \min\{g_0(D, \eta), 1/4, \beta, 1 - \beta\}$.

Inputs. $d; C; D; \eta; \epsilon; \mathcal{H}_{d, C, D, \eta, \epsilon}$.

Definition 10 (def:erm-gibbs-plugin-learner). Defined object. *empirical-risk-minimizer Gibbs plug-in learner*

Construction. Using the sequentially lexicographically selected empirical-risk minimizer $\hat{f} \in \mathcal{F}$ from def:prediction-polytope-learner, define, for every $x \in \mathcal{X}$ and $a \in \mathcal{A}$,

$$\hat{\pi}_{\text{ERM}}(a | x) = \frac{\pi_{\text{ref}}(a | x) e^{\eta \hat{f}(x, a)}}{\sum_{b \in \mathcal{A}} \pi_{\text{ref}}(b | x) e^{\eta \hat{f}(x, b)}}.$$

The learner receives only $(\mathcal{D}_n, \mathfrak{E})$; it does not receive the shell indices C or D .

Inputs. $\backslash \widehat{f}; \backslash \mathfrak{p}_{i-\{\mathrm{ref}\}}; \backslash \eta; \backslash \mathrm{mathcal{D}}_n; \backslash \mathrm{mathfrak{E}}$.

6 Main results

Theorem 1 (thm:zero-risk-boundary). *If the shell $\mathcal{M}(d, 1, 1, \eta)$ is nonempty, then every P in it has $\pi_P^* = \pi_{\mathrm{ref}}$ ρ -almost surely, $D_P = 1$, and $J_{\eta, P}(\pi_P^*) - J_{\eta, P}(\pi_{\mathrm{ref}}) = 0$. Hence $N^*(\epsilon; d, 1, 1, \eta) = 1$ for every $\epsilon > 0$.*

Proof. Fix $P \in \mathcal{M}(d, 1, 1, \eta)$, and write $w_x(a) = \pi_P^*(a \mid x) / \pi_{\mathrm{ref}}(a \mid x)$ on the support of $\pi_{\mathrm{ref}}(\cdot \mid x)$. By $C_P = 1$, $w_x(a) \leq 1$ for $\rho \times \pi_{\mathrm{ref}}$ -almost every (x, a) . On the other hand, for every x ,

$$\sum_a \pi_{\mathrm{ref}}(a \mid x) w_x(a) = \sum_a \pi_P^*(a \mid x) = 1.$$

A random variable bounded above by one and having mean one equals one almost surely. Hence $w_x(a) = 1$ for $\rho \times \pi_{\mathrm{ref}}$ -almost every (x, a) , and therefore $\pi_P^* = \pi_{\mathrm{ref}}$ ρ -almost surely. It follows directly from the covariance definitions that $\Lambda_P^* = \Lambda_{\mathrm{ref}}$, so $D_P = 1$.

For completeness, let $Z_P(x) = \sum_a \pi_{\mathrm{ref}}(a \mid x) e^{\eta r_\theta(x, a)}$. The Gibbs formula gives, for every policy π supported on π_{ref} ,

$$\mathrm{KL}(\pi(\cdot \mid x) \parallel \pi_P^*(\cdot \mid x)) = \mathrm{KL}(\pi(\cdot \mid x) \parallel \pi_{\mathrm{ref}}(\cdot \mid x)) - \eta \sum_a \pi(a \mid x) r_\theta(x, a) + \log Z_P(x).$$

Also $J_{\eta, P}(\pi_P^*) = \eta^{-1} \mathbb{E}_\rho \log Z_P(X)$. Subtraction and integration yield

$$J_{\eta, P}(\pi_P^*) - J_{\eta, P}(\pi) = \eta^{-1} \mathbb{E}_\rho \mathrm{KL}(\pi(\cdot \mid X) \parallel \pi_P^*(\cdot \mid X)).$$

Taking $\pi = \pi_{\mathrm{ref}} = \pi_P^*$ makes the right-hand side zero. The constant learner that always returns the publicly known π_{ref} belongs to $\mathcal{L}_n(\mathfrak{E})$, so $\mathfrak{R}_n(\mathcal{M}(d, 1, 1, \eta)) = 0$ for every $n \in \mathbb{N}$. Since the sample-size index begins at one, the definition of N^* gives $N^*(\epsilon; d, 1, 1, \eta) = 1$ for every $\epsilon > 0$. $\square$

Justification. A conditional likelihood ratio has reference mean one, so an essential supremum of one forces equality of the target and reference policies. *Gap.* Existing reverse-KL coverage bounds do not separately record the exact-shell $C = 1$ zero-risk boundary. *Consumer.* It prevents a coverage diagnostic from assigning a positive sample requirement when the known reference policy is already optimal.

Lemma 1 (lem:learner-invariance). *Under every invertible reparameterization $\phi \mapsto T\phi$ and $\theta \mapsto T^{-\top}\theta$, the prediction polytope $\mathcal{F}$, the selected function $\hat{f}$, the confidence set $\mathcal{Q}_n$, the criterion $\underline{J}_n$, and the selected policy $\hat{\pi}$ are unchanged.*

Proof. Let $\phi' = T\phi$ and $\theta' = T^{-\top}\theta$, where $T \in \mathrm{GL}_d(\mathbb{R})$. Then

$$\phi'(x, a)^\top \theta' = \phi(x, a)^\top T^\top T^{-\top} \theta = \phi(x, a)^\top \theta.$$

Thus the prediction functions representable in the primed coordinates form exactly the same subset $\mathcal{F} \subseteq [0, 1]^S$. The empirical loss L_n depends only on prediction values and the data, so its minimizer set is unchanged. The sequential lexicographic rule orders the values of functions on the same fixed ordered set S , rather than coefficient representatives, and therefore the selected function $\hat{f}$ is unchanged.

Likewise, $\|h\|_n^2 = n^{-1} \sum_i h(X_i, A_i)^2$ is coordinate free. Hence $\mathcal{Q}_n$ is the same compact subset of prediction space after reparameterization. For every $(x, a) \in S$, the scalar optimization problem

$$r_n(x, a) = \min_{f \in \mathcal{Q}_n} f(x, a)$$

has the same feasible prediction functions and objective values in the two coordinate systems. Its minimum value is therefore unchanged. Consequently every numerator and denominator in

$$\widehat{\pi}(a \mid x) = \frac{\pi_{\text{ref}}(a \mid x) e^{\eta \underline{r}_n(x, a)}}{\sum_b \pi_{\text{ref}}(b \mid x) e^{\eta \underline{r}_n(x, b)}}$$

is unchanged, so the deployed policy is unchanged at every context, including contexts absent from the sample. Finally, $\underline{J}_n$ is written entirely in terms of prediction values in $\mathcal{Q}_n$, the observed contexts, π_{ref} , and η . It is therefore unchanged as well. $\square$

Justification. The learner is defined entirely in prediction space, so invertible feature-coordinate changes leave every deployed probability and the diagnostic criterion unchanged. *Gap.* This is coordinate-free implementation infrastructure rather than the statistical novelty claim. *Consumer.* It permits whitening and other invertible feature changes without altering the learner at observed or unobserved contexts.

Theorem 2 (thm:localized-upper). *There are universal constants such that, after choosing c_0 sufficiently large and whenever $\mathcal{M}(d, C, D, \eta) \neq \emptyset$, the lower-envelope Gibbs learner $\widehat{\pi} \in \mathcal{L}_n(\mathfrak{E})$ uses neither C nor D as input and satisfies*

$$\sup_{P \in \mathcal{M}(d, C, D, \eta)} \mathbb{E}_{P^{\otimes n}} [J_{\eta, P}(\pi_P^*) - J_{\eta, P}(\widehat{\pi})] \leq \tilde{O} \left(\min \left\{ \frac{\eta d^2 D}{n}, \sqrt{\frac{d^2 D}{n}} \right\} \right) + \delta_n.$$

The proof derives a distribution-free d/n logging prediction radius for every member of $\mathcal{Q}_n$, a pointwise leverage majorant for the nonlinear lower envelope, the trace inequality $D_{2, P} \leq dD_P$, the Gibbs regret identity, and the endpoint-monotonicity fast and slow conversions. Consequently

$$N^*(\epsilon; d, C, D, \eta) \leq \tilde{O}(d^2 D \min\{\eta/\epsilon, \epsilon^{-2}\})$$

when the right-hand side is at least two. The additional factor d is unavoidable for the empirical-ball lower-envelope learner in general: prop:empirical-ball-dimension-witness gives exact shells on which the original dD upper rate fails.

Proof. Choose $c_0 = 64$ and put

$$a_n = d \log(en) + \log(2n^2).$$

The construction in def:prediction-polytope-learner is a finite-dimensional prediction-space procedure. Its stipulated sequential lexicographic selections make the empirical minimizer, the coordinatewise minima over the compact set $\mathcal{Q}_n$, and the resulting Gibbs normalization measurable in $(\mathcal{D}_n, \mathfrak{E})$. Thus $\widehat{\pi} \in \mathcal{L}_n(\mathfrak{E})$. Its formula uses neither C nor D .

Fix $P \in \mathcal{M}(d, C, D, \eta)$. By lem:bounded-linear-localization, there is an event $\mathcal{E}_n$, with probability at least $1 - \delta_n/2$, on which

$$r_\theta \in \mathcal{Q}_n, \quad \sup_{f \in \mathcal{Q}_n} \mathbb{E}_{\rho, \pi_{\text{ref}}} (f - r_\theta)^2 \leq 600 \frac{a_n}{n}.$$

On this event define

$$\Delta_n(x, a) = \underline{r}_n(x, a) - r_\theta(x, a).$$

Truth inclusion and $\mathcal{F} \subseteq [0, 1]^S$ imply $-1 \leq \Delta_n \leq 0$ pointwise.

Write $\Lambda = \Lambda_{\text{ref}} \succ 0$. At each (x, a) , compactness gives an attaining function $f_{x,a} \in \mathcal{Q}_n$. Put $h_{x,a} = f_{x,a} - r_\theta$, and let $b_{x,a} \in \mathbb{R}^d$ represent this linear contrast. Cauchy–Schwarz in the Λ -inner product gives

$$\begin{aligned} \Delta_n(x, a)^2 &= \{\phi(x, a)^\top b_{x,a}\}^2 \\ &\leq \{b_{x,a}^\top \Lambda b_{x,a}\} \{\phi(x, a)^\top \Lambda^{-1} \phi(x, a)\} \\ &= \mathbb{E}_{\rho, \pi_{\text{ref}}} h_{x,a}^2 \{\phi(x, a)^\top \Lambda^{-1} \phi(x, a)\} \\ &\leq 600 \frac{a_n}{n} \phi(x, a)^\top \Lambda^{-1} \phi(x, a). \end{aligned}$$

Consequently $\Delta_n^2 \leq \phi^\top M_n \phi$ pointwise for

$$M_n = 600 \frac{a_n}{n} \Lambda^{-1} \succeq 0.$$

By lem:quadratic-majorant-transfer,

$$\mathbb{E}_{\rho, \pi_P^*} \Delta_n^2 \leq 600 \frac{a_n}{n} \text{tr}(\Lambda^{-1} \Lambda_P^*) = 600 \frac{a_n}{n} D_{2,P}.$$

Every eigenvalue of $\Lambda^{-1/2} \Lambda_P^* \Lambda^{-1/2}$ is nonnegative and at most $D_P = D$. Hence

$$D_{2,P} = \text{tr}(\Lambda^{-1} \Lambda_P^*) \leq dD,$$

and therefore

$$\mathbb{E}_{\rho, \pi_P^*} \Delta_n^2 \leq 600 \frac{a_n d D}{n}.$$

This trace step is the extra dimension factor: the logging radius is of order d/n , and averaging the nonlinear envelope’s leverage majorant costs at most dD .

The deployed learner is the Gibbs policy generated by $r_\theta + \Delta_n$. Applying lem:gibbs-pessimism-conversion and then Cauchy–Schwarz gives, on $\mathcal{E}_n$,

$$\begin{aligned} J_{\eta,P}(\pi_P^*) - J_{\eta,P}(\hat{\pi}) &\leq \min \left\{ \mathbb{E}_{\rho, \pi_P^*} [-\Delta_n], \frac{\eta}{2} \mathbb{E}_{\rho, \pi_P^*} \Delta_n^2 \right\} \\ &\leq \min \left\{ \sqrt{600 a_n d D / n}, 300 \eta a_n d D / n \right\}. \end{aligned}$$

For the complement, let $r, g : S \rightarrow [0, 1]$, and denote their Gibbs policies by π_r, π_g . Their normalizers lie in $[1, e^\eta]$, so on the support of π_{ref} ,

$$\log \frac{\pi_g(a \mid x)}{\pi_r(a \mid x)} = \eta \{g(x, a) - r(x, a)\} + \log \frac{Z_r(x)}{Z_g(x)} \leq 2\eta.$$

The Gibbs regret identity yields

$$0 \leq J_{\eta,r}(\pi_r) - J_{\eta,r}(\pi_g) \leq 2.$$

Thus $\mathcal{E}_n^c$ contributes at most $2(\delta_n/2) = \delta_n$. Uniformly in the shell,

$$\sup_{P \in \mathcal{M}(d, C, D, \eta)} \mathbb{E}_{P \otimes n} [J_{\eta,P}(\pi_P^*) - J_{\eta,P}(\hat{\pi})] \leq \min \left\{ 300 \eta a_n d D / n, \sqrt{600 a_n d D / n} \right\} + \delta_n.$$

Because $a_n = \tilde{O}(d)$, this is the stated

$$\tilde{O} \left(\min \left\{ \frac{\eta d^2 D}{n}, \sqrt{\frac{d^2 D}{n}} \right\} \right) + \delta_n$$

bound.

For inversion, first note that every nonempty shell has $D \geq 1$. If $\theta = 0$, both policies coincide and $D = 1$. Otherwise, at each context let $R = r_\theta(x, A)$ under the reference policy and let R' be an independent copy. Since both $s \mapsto s^2$ and $s \mapsto e^{\eta s}$ are nondecreasing on $[0, 1]$,

$$2 \operatorname{Cov}(R^2, e^{\eta R}) = \mathbb{E}[(R^2 - R'^2)(e^{\eta R} - e^{\eta R'})] \geq 0.$$

After averaging over contexts, the generalized Rayleigh quotient in direction θ is at least one; nonsingularity makes its denominator positive. Hence $D_P = D \geq 1$.

A second deterministic bound handles very small η . For $\Delta = g - r$, the Gibbs path identity gives

$$J_{\eta, r}(\pi_r) - J_{\eta, r}(\pi_g) = \eta \int_0^1 t \operatorname{Var}_{\pi_t}(\Delta) dt \leq \frac{\eta}{2},$$

because $\Delta \in [-1, 1]$ implies $\operatorname{Var}(\Delta) \leq 1$. If $\eta \leq 2\epsilon$, the risk is already at most ϵ . Otherwise, choosing n , up to universal logarithmic factors and integer rounding, of order

$$d^2 D \min\{\eta/\epsilon, \epsilon^{-2}\}$$

makes the applicable good-event term a fixed fraction of ϵ . The complement term is also absorbed: on the fast branch one may use the sharper complement cap $(\eta/2)(\delta_n/2)$, and on the slow branch $\delta_n \leq n^{-2}$ is smaller than a fixed fraction of ϵ . Increasing the universal constant handles both branches. The minimax risk is no larger than the risk of this measurable learner, so `def:sample-complexity` gives

$$N^*(\epsilon; d, C, D, \eta) \leq \tilde{O}(d^2 D \min\{\eta/\epsilon, \epsilon^{-2}\})$$

whenever the displayed right-hand side is at least two.

Finally, `prop:empirical-ball-dimension-witness` gives exact-shell deterministic-reward experiments on which this same empirical-ball coordinatewise-envelope learner has regret bounded away from zero at $n = d^2$, although dD/n and $\sqrt{dD/n}$ vanish up to logarithms. Hence its additional dimension factor cannot be removed uniformly for this learner. The witness does not exclude a different learner, as claimed. $\square$

Justification. The lower-envelope Gibbs learner remains a valid invariant theorem-local construction with clean temperature dependence. *Gap.* Its extra factor d is learner-specific and is superseded for general achievability by `thm:clean-temperature-feature-upper`. *Consumer.* It diagnoses why coordinatewise pessimistic envelopes can lose a dimension factor.

Theorem 3 (`thm:same-shell-lower`). *For every $(C, D) \in \mathfrak{F}_{\text{hard}}^\circ(d, \eta)$, the construction in `def:explicit-hard-family` gives a nonempty shell $\mathcal{M}(d, C, D, \eta)$, has $g_0(D, \eta) > 0$, and has a positive admissible accuracy interval on which $\mathcal{H}_{d, C, D, \eta, \epsilon} \subset \mathcal{M}(d, C, D, \eta)$. Its members satisfy exactly $C_P = C$, $D_P = D$, and $\Lambda_{\text{ref}} \succ 0$. There are constants $\epsilon_1(C, D, \eta) \in (0, \epsilon_0(C, D, \eta)]$ and $c(C, D, \eta) > 0$ such that, for every $0 < \epsilon \leq \epsilon_1(C, D, \eta)$, $N^*(\epsilon; d, C, D, \eta) \geq c(C, D, \eta)dD/\epsilon$. Thus the displayed paired-Bernoulli family proves the local fast branch; it does not prove the ϵ^{-2} branch.*

Proof. Put $p = (4D)^{-1}$, $k = \lfloor (d-1)/2 \rfloor$, and write H for the sum of the raw context masses in `def:explicit-hard-family`. Since $\tau(1+A) = 1/4$,

$$H = \frac{5}{4} \quad \text{if } d \text{ is odd,} \quad H = \frac{3}{2} \quad \text{if } d \text{ is even.}$$

Thus every hard context has probability

$$\rho(x_j) = \frac{1}{Hk}.$$

At the anchor and optional unused-coordinate contexts, any reference probabilities not affecting the displayed construction may be chosen with full support; because all action features and rewards agree there, this choice does not affect any calculation below.

We first prove that the admissible perturbation interval is nonempty. For $s \geq 0$, set

$$F(s) = e^{\eta s} - 2pD \cosh(\eta s), \quad T_s = \frac{D(1-2p)}{F(s)}, \quad \beta_s = \eta^{-1} \log T_s$$

whenever $F(s) > 0$. At zero,

$$F(0) = 1 - 2pD = \frac{1}{2}, \quad T_0 = \frac{D(1-2p)}{1-2pD} = 2D - 1.$$

The hard-region conditions give $1 < 2D - 1 < e^\eta$, so

$$0 < \beta_0 = \eta^{-1} \log(2D - 1) < 1.$$

The functions F , T_s , and β_s are continuous near zero. Consequently there is a number $g_* > 0$, depending only on (D, η) , such that for every $0 \leq s \leq g_*$,

$$F(s) > 0, \quad s \leq \beta_s \leq 1 - s, \quad s \leq \frac{1}{4}, \quad \eta s \leq 1.$$

This interval is one of the intervals admitted in the defining supremum for $g_0(D, \eta)$; hence $g_0(D, \eta) \geq g_* > 0$. It follows at once from def:hard-accuracy-range that $\epsilon_0(C, D, \eta) > 0$. Define

$$\epsilon_1(C, D, \eta) = \min \left\{ \epsilon_0(C, D, \eta), \frac{1}{64\eta} \log \cosh(\eta g_*) \right\} > 0.$$

For $0 < \epsilon \leq \epsilon_1(C, D, \eta)$, monotonicity of $\cosh$ on $[0, \infty)$ and the calibration

$$\cosh(\eta\gamma) = e^{64\eta\epsilon}$$

imply $0 < \gamma \leq g_*$. In particular, all hard-context reward means $\beta_\gamma \pm \gamma$ lie in $[0, 1]$, and the family in def:explicit-hard-family is well defined.

We next verify the two exact shell equalities. At x_C ,

$$qe^\eta + 1 - q = \frac{e^\eta}{C}.$$

The Gibbs likelihood ratio of action $+$ relative to the reference policy is therefore exactly C , whereas the ratios of the other actions are $C/e^\eta < 1$. At a hard context, put $t = \eta\gamma$. Its Gibbs normalizer is

$$Z = 1 - 2p + 2pT_\gamma \cosh t.$$

The defining formula for T_γ is algebraically equivalent to

$$T_\gamma e^t = DZ.$$

Accordingly, the favored and unfavored informative actions have likelihood ratios

$$D \quad \text{and} \quad De^{-2t},$$

respectively. Since $\beta_\gamma \geq \gamma$, one has $T_\gamma e^t = e^{\eta(\beta_\gamma + \gamma)} \geq 1$, and hence the zero-action ratio satisfies $Z^{-1} \leq D$. At x_A , and at x_z when it is present, all actions have the same reward, so every likelihood ratio equals one. Because $D \leq C$, the global essential supremum is attained at $(x_C, +)$ and equals

$$C_P = C.$$

The logging and target covariances are block diagonal in the orthonormal basis used by the construction. In the e_0 direction, their generalized variance ratio is

$$\frac{qC + A}{q + A} = D, \quad A = \frac{q(C - D)}{D - 1};$$

the same identity remains valid when $C = D$, in which case $A = 0$. At hard context x_j , the logging covariance on $\text{span}(u_j, w_j)$ is a positive scalar multiple of pI_2 . The target probabilities of the favored and unfavored informative actions are

$$a = pD = \frac{1}{4}, \quad b = pDe^{-2t},$$

so, after whitening by the logging block, the two eigenvalues of the target block are D and De^{-2t} . The optional z block has generalized variance ratio one. Every basis direction receives positive logging mass: the e_0 direction receives the positive mass $\rho(x_C)q$, each pair (u_j, w_j) receives positive mass through the two hard actions, and the z direction, when present, receives the positive mass of x_z . Therefore

$$\Lambda_{\text{ref}} \succ 0, \quad D_P = D.$$

The features all have norm at most one. Taking the conditional reward at every cell to be Bernoulli with the displayed linear mean gives bounded rewards and verifies linear realizability and reference logging. Thus, for every $0 < \epsilon \leq \epsilon_1(C, D, \eta)$,

$$\mathcal{H}_{d, C, D, \eta, \epsilon} \subseteq \mathcal{M}(d, C, D, \eta).$$

In particular, the exact shell is nonempty throughout this positive interval.

It remains to establish the local minimax lower bound. Fix such an ϵ , retain $t = \eta\gamma \leq 1$, and index the family by $v \in \{-1, 1\}^k$. At any hard context, the two target Gibbs action laws associated with the two possible signs are

$$s_+ = (a, b, c), \quad s_- = (b, a, c), \quad a = \frac{1}{4}, \quad b = \frac{1}{4}e^{-2t}, \quad c = 1 - a - b.$$

For completeness, if a reward vector r has Gibbs law $s(a) = \pi_{\text{ref}}(a)e^{\eta r(a)}/Z$, direct substitution gives, for every reference-supported action law u ,

$$\eta^{-1} \text{KL}(u \| s) = \eta^{-1} \text{KL}(u \| \pi_{\text{ref}}) - \sum_a u(a)r(a) + \eta^{-1} \log Z.$$

The last term is the optimal regularized value at that context, so this is exactly the conditional regularized regret.

For the two sign laws, let

$$\alpha = \sum_a \sqrt{s_+(a)s_-(a)} = 2\sqrt{ab} + c = 1 - a(1 - e^{-t})^2.$$

If $m(a) = \sqrt{s_+(a)s_-(a)}/\alpha$, then an exact rearrangement yields

$$\text{KL}(u \| s_+) + \text{KL}(u \| s_-) = 2 \text{KL}(u \| m) - 2 \log \alpha \geq -2 \log \alpha.$$

Since $0 \leq t \leq 1$, the elementary integral bound

$$1 - e^{-t} = \int_0^t e^{-s} ds \geq \frac{t}{2}$$

holds. Using also $-\log(1 - z) \geq z$ for $0 \leq z < 1$, we obtain the pointwise separation

$$\inf_u \eta^{-1} \{ \text{KL}(u \| s_+) + \text{KL}(u \| s_-) \} \geq \frac{t^2}{8\eta}.$$

We now control the statistical affinity of neighboring vertices. By decreasing g_* if necessary, continuity and the strict inequalities at zero provide a number $m = m(D, \eta) \in (0, 1/2)$ such that

$$m \leq \beta_\gamma - \gamma \leq \beta_\gamma + \gamma \leq 1 - m$$

throughout the chosen interval. For Bernoulli laws, $\log x \leq x - 1$ gives

$$\text{kl}(u\|w) \leq \frac{(u-w)^2}{w(1-w)}.$$

Two neighboring vertices differ only at one hard context x_j , and there only for the two informative actions. Hence their one-observation Kullback–Leibler divergence is at most

$$K(D, \eta) \rho(x_j) p \gamma^2, \quad K(D, \eta) = \frac{8}{m(1-m)}.$$

Product structure multiplies this bound by n . To keep the testing reduction self-contained, for two probability laws Q_1, Q_0 , the log-sum inequality applied to the set on which the density of Q_1 exceeds that of Q_0 , followed by the scalar inequality $\text{kl}(x\|y) \geq 2(x-y)^2$, gives

$$\text{TV}(Q_1, Q_0) \leq \sqrt{\frac{1}{2} \text{KL}(Q_1\|Q_0)}.$$

Therefore there is a constant $c_1(D, \eta) > 0$ such that every pair of neighboring n -sample laws has total variation at most $1/2$ whenever

$$n \leq c_1(D, \eta) \frac{k}{p \gamma^2}.$$

Consider an arbitrary measurable reference-supported learner and decompose its regularized regret into its nonnegative contextwise contributions. For a fixed coordinate j , pair every vertex v with the vertex $v^{(j)}$ obtained by flipping its j -th sign. If q_v and $q_{v^{(j)}}$ are densities of the two sample laws with respect to their sum, integration against $\min\{q_v, q_{v^{(j)}}\}$ and the pointwise separation above give

$$\mathbb{E}_v L_{v,j} + \mathbb{E}_{v^{(j)}} L_{v^{(j)},j} \geq \rho(x_j) \frac{t^2}{8\eta} \{1 - \text{TV}(P_v^{\otimes n}, P_{v^{(j)}}^{\otimes n})\} \geq \rho(x_j) \frac{t^2}{16\eta}.$$

There are 2^{k-1} disjoint pairs for each j . Averaging over the uniform prior on the sign cube and then summing the contextwise losses over j shows that the Bayes risk of every learner is at least

$$\sum_{j=1}^k \frac{\rho(x_j) t^2}{32\eta} = \frac{t^2}{32H\eta} \geq \frac{t^2}{48\eta}.$$

For $0 \leq t \leq 1$,

$$\log \cosh t \leq \frac{t^2}{2}$$

because the derivative of the left side is $\tanh t \leq t$. The calibration $\log \cosh t = 64\eta\epsilon$ therefore implies $t^2 \geq 128\eta\epsilon$. Thus the preceding Bayes risk is at least $8\epsilon/3$, and in particular is strictly larger than ϵ .

Finally, $\cosh t \geq 1 + t^2/2$ and $\log(1+x) \geq x/(1+x)$ give, for $0 \leq t \leq 1$,

$$\log \cosh t \geq \frac{t^2}{3}.$$

Consequently

$$\gamma^2 = \frac{t^2}{\eta^2} \leq \frac{192\epsilon}{\eta}.$$

Since $p = (4D)^{-1}$ and $k \geq d/4$,

$$\frac{k}{p \gamma^2} \geq \frac{dD\eta}{192\epsilon}.$$

It follows that there is $c(C, D, \eta) > 0$ such that the total-variation condition, and hence minimax risk strictly exceeding ϵ , holds for every positive integer

$$n \leq c(C, D, \eta) \frac{dD\eta}{\epsilon}.$$

The minimax risk over the exact shell dominates the Bayes risk over its finite subfamily. By def:sample-complexity, no sample size in the last range can attain risk at most ϵ , and therefore

$$N^*(\epsilon; d, C, D, \eta) \geq c(C, D, \eta) \frac{dD\eta}{\epsilon} \quad (0 < \epsilon \leq \epsilon_1(C, D, \eta)).$$

The argument uses the locally quadratic separation $\log \cosh(\eta\gamma) \asymp \eta^2\gamma^2$, and therefore proves precisely the stated local fast branch; it supplies no ϵ^{-2} lower branch. $\square$

Justification. The paired-Bernoulli family preserves both exact shell equalities and yields the local fast lower branch on its proved interior hard region. *Gap.* The construction proves only a pointwise small-accuracy $\Omega_{C,D,\eta}(d/\epsilon)$ rate; it does not prove a uniform D - or η -dependence and does not produce a quadratic branch. *Consumer.* It supplies the informative lower headline that complements, but does not uniformly match, the all-temperature stabilization upper bound.

Proposition 1 (prop:canonical-tabular-reduction). *Under the canonical tabular specialization $d = |S|$ and $\phi(x, a) = e_{(x,a)}$, $\Lambda_{\text{ref}}^{-1/2} \Lambda_P^* \Lambda_{\text{ref}}^{-1/2}$ is diagonal with entries $\pi_P^*(a | x) / \pi_{\text{ref}}(a | x)$. Hence $D_P = C_P$, and the exact shell $\mathcal{M}(d, C, C, \eta)$ reduces to the canonical tabular reverse-KL shell.*

Proof. Fix $P \in \mathcal{M}(d, C, C, \eta)$, and index the coordinates of $\mathbb{R}^d$ by cells $s = (x, a) \in S$. By the canonical specialization, $\phi(x, a) = e_{(x,a)}$. Hence, directly from the covariance definitions,

$$\Lambda_{\text{ref}} = \sum_{x \in \mathcal{X}} \sum_{a \in \mathcal{A}} \rho(x) \pi_{\text{ref}}(a | x) e_{(x,a)} e_{(x,a)}^\top = \text{diag}_{(x,a) \in S} \{ \rho(x) \pi_{\text{ref}}(a | x) \},$$

and

$$\Lambda_P^* = \sum_{x \in \mathcal{X}} \sum_{a \in \mathcal{A}} \rho(x) \pi_P^*(a | x) e_{(x,a)} e_{(x,a)}^\top = \text{diag}_{(x,a) \in S} \{ \rho(x) \pi_P^*(a | x) \}.$$

The condition $\Lambda_{\text{ref}} \succ 0$ implies $\rho(x) \pi_{\text{ref}}(a | x) > 0$ for every $(x, a) \in S$. In particular, every denominator below is positive, and the inverse square root is

$$\Lambda_{\text{ref}}^{-1/2} = \text{diag}_{(x,a) \in S} \left\{ \left(\rho(x) \pi_{\text{ref}}(a | x) \right)^{-1/2} \right\}.$$

Multiplying the three diagonal matrices gives

$$\Lambda_{\text{ref}}^{-1/2} \Lambda_P^* \Lambda_{\text{ref}}^{-1/2} = \text{diag}_{(x,a) \in S} \left\{ \frac{\rho(x) \pi_P^*(a | x)}{\rho(x) \pi_{\text{ref}}(a | x)} \right\} = \text{diag}_{(x,a) \in S} \left\{ \frac{\pi_P^*(a | x)}{\pi_{\text{ref}}(a | x)} \right\}.$$

Therefore its largest eigenvalue is the maximum of these diagonal entries:

$$D_P = \max_{(x,a) \in S} \frac{\pi_P^*(a | x)}{\pi_{\text{ref}}(a | x)}.$$

Because S is finite and $\rho(x) \pi_{\text{ref}}(a | x) > 0$ at every cell, this maximum is exactly the essential supremum in the definition of C_P . Thus $D_P = C_P$.

Finally, under canonical features the bounded-feature condition holds with equality, linear realizability is precisely a tabular conditional-mean specification, and the two exact shell restrictions $C_P = C$ and $D_P = C$ are the same restriction because $D_P = C_P$. All remaining membership conditions in def:exact-shell are the common finite-support, reference-logging, bounded-reward, and full-support conditions of the canonical tabular model. Consequently $\mathcal{M}(d, C, C, \eta)$ is exactly the canonical tabular reverse-KL shell expressed in the shared-linear notation. $\square$

Justification. Canonical features diagonalize both the logging and target covariances. *Gap.* Ji et al., arXiv:2605.02141, study the tabular KL-regularized setting; this proposition gives the required reduction of the shared-linear notation to that setting. *Consumer.* It makes the tabular fast-slow frontier an auditable special case of the proposed exact-shell vocabulary.

Proposition 2 (prop:shared-feature-witness). *For $d = 5$, every $(C, D) \in \mathfrak{F}_{\text{hard}}^\circ(5, \eta)$, and every admissible ϵ , each law in $\mathcal{H}_{5,C,D,\eta,\epsilon}$ is a non-tabular shared-feature member: at every hard context x_j , the two action features $\phi(x_j, +) = (u_j + w_j)/\sqrt{2}$ and $\phi(x_j, -) = (u_j - w_j)/\sqrt{2}$ share both latent directions and are not canonical coordinate indicators.*

Proof. Fix $(C, D) \in \mathfrak{F}_{\text{hard}}^\circ(5, \eta)$, fix an admissible ϵ , and fix $P \in \mathcal{H}_{5,C,D,\eta,\epsilon}$. By def:explicit-hard-family,

$$k = \left\lfloor \frac{5-1}{2} \right\rfloor = 2.$$

Because $d = 5$ is odd, the construction uses the orthonormal basis

$$e_0, u_1, w_1, u_2, w_2$$

and no auxiliary vector z . For each hard context x_j , where $j \in \{1, 2\}$, the same definition gives

$$\phi(x_j, +) = \frac{u_j + w_j}{\sqrt{2}}, \quad \phi(x_j, -) = \frac{u_j - w_j}{\sqrt{2}}.$$

Orthonormality implies

$$\begin{aligned} \langle u_j, \phi(x_j, +) \rangle &= 2^{-1/2}, & \langle w_j, \phi(x_j, +) \rangle &= 2^{-1/2}, \\ \langle u_j, \phi(x_j, -) \rangle &= 2^{-1/2}, & \langle w_j, \phi(x_j, -) \rangle &= -2^{-1/2}. \end{aligned}$$

All four coordinates are nonzero. Thus both informative action features use both latent directions u_j and w_j , so the two actions share the same pair of latent directions. Each displayed feature has two nonzero coordinates in the construction's orthonormal basis and therefore is not one of its coordinate-basis vectors.

The construction is also not the canonical tabular specialization. Indeed, it contains the distinct contexts x_C, x_1, x_2 and the action set $\{+, -, 0\}$, so $|S| \geq 9 > 5 = d$. Canonical tabular features would require $d = |S|$ and a distinct canonical vector $e_{(x,a)} \in \mathbb{R}^{|S|}$ for every cell $(x, a) \in S$. Hence the two displayed vectors cannot be canonical tabular coordinate indicators. Finally, the feature map is fixed by def:explicit-hard-family and does not depend on the sign index v or on the Bernoulli reward draws. The shared-feature and non-tabular conclusions therefore hold for every $P \in \mathcal{H}_{5,C,D,\eta,\epsilon}$, as claimed. $\square$

Justification. The lower construction preserves its shell equalities while coupling the two informative action rewards through common feature directions. *Gap.* The tabular construction in Ji et al. has independent arm coordinates; this exhibits the feature sharing absent from that reduction. *Consumer.* It makes the claimed linear extension testable on a concrete finite instance rather than on a verbal distinction from tabular bandits.

Lemma 2 (lem:zhao-uniform-square-comparison). *Let $\mathcal{G} \subseteq [0, 1]^S$, let $W_1, \dots, W_n$ be independent with common law μ , and let $N_{\mathcal{G}}(u)$ be the u -covering number of $\mathcal{G}$ in supremum norm. For $0 < u < 1$, with probability at least $1 - \delta$, simultaneously for every $g_1, g_2 \in \mathcal{G}$,*

$$\mathbb{E}_\mu (g_1 - g_2)^2 \leq \frac{2}{n} \sum_{i=1}^n (g_1 - g_2)^2(W_i) + \frac{32}{3n} \log \frac{2N_{\mathcal{G}}(u)}{\delta} + 10u.$$

Lemma 3 (lem:bounded-projection-concentration). *Let $\varepsilon_1, \dots, \varepsilon_n$ be conditionally independent given a sigma-field $\mathcal{G}$, with $\mathbb{E}[\varepsilon_i | \mathcal{G}] = 0$ and $|\varepsilon_i| \leq 1$. If Π is a $\mathcal{G}$ -measurable orthogonal projector of rank at most d , then, conditionally with probability at least $1 - \zeta$,*

$$\|\Pi \varepsilon\|_2^2 \leq 8 \left\{ d \log 5 + \log \frac{2}{\zeta} \right\}.$$

Proof. For a fixed unit vector v in the range of Π , convexity of the exponential on $[-1, 1]$, conditional centering, and $\cosh u \leq e^{u^2/2}$ give

$$\mathbb{E} [e^{tv_i \varepsilon_i} \mid \mathcal{G}] \leq \cosh(tv_i) \leq e^{t^2 v_i^2/2}.$$

Conditional independence and $\sum_i v_i^2 = 1$ imply

$$\mathbb{E} [e^{tv^\top \varepsilon} \mid \mathcal{G}] \leq e^{t^2/2}.$$

For $u > 0$, multiplication by e^{-tu} , the inequality $\mathbf{1}\{v^\top \varepsilon \geq u\} \leq e^{t(v^\top \varepsilon - u)}$, and minimization at $t = u$ yield

$$\Pr\{|v^\top \varepsilon| \geq u \mid \mathcal{G}\} \leq 2e^{-u^2/2}.$$

The unit sphere of a space of dimension at most d has a one-half net $\mathcal{N}$ with $|\mathcal{N}| \leq 5^d$, by the disjoint-ball volume argument. Moreover,

$$\|\Pi \varepsilon\|_2 \leq 2 \max_{v \in \mathcal{N}} |v^\top \varepsilon|.$$

Take $u^2 = 2\{d \log 5 + \log(2/\zeta)\}$ and apply the union bound over $\mathcal{N}$. With conditional probability at least $1 - \zeta$, the last maximum is at most u , and hence $\|\Pi \varepsilon\|_2^2 \leq 4u^2$, which is the claimed inequality. $\square$

Lemma 4 (lem:multinomial-max-count). *Let $N_1, \dots, N_d$ be the cell counts from d^2 independent draws from the uniform distribution on $\{1, \dots, d\}$. Then*

$$\Pr \left\{ \max_{1 \leq j \leq d} N_j > 3d \right\} \leq d \exp\{-d(3 \log 3 - 2)\} = o(1).$$

Proof. For a fixed j and $t > 0$, independence and $1 + u \leq e^u$ give

$$\mathbb{E} e^{tN_j} = \left(1 + \frac{e^t - 1}{d}\right)^{d^2} \leq \exp\{d(e^t - 1)\}.$$

Since $\mathbf{1}\{N_j > 3d\} \leq e^{t(N_j - 3d)}$,

$$\Pr\{N_j > 3d\} \leq \exp\{-3dt + d(e^t - 1)\}.$$

Taking $t = \log 3$ yields $\Pr\{N_j > 3d\} \leq e^{-d(3 \log 3 - 2)}$. Summing these probabilities over $j = 1, \dots, d$ proves the displayed bound, which tends to zero because $3 \log 3 - 2 > 0$. $\square$

Lemma 5 (lem:feature-coverage-domination). *For every $P \in \mathcal{M}(d, C, D, \eta)$ and every $h \in \text{span}(\phi)$, $\mathbb{E}_{\rho, \pi_P^*} h(X, A)^2 \leq D_P \mathbb{E}_{\rho, \pi_{\text{ref}}} h(X, A)^2$.*

Proof. Fix $P \in \mathcal{M}(d, C, D, \eta)$ and $h \in \text{span}(\phi)$. By the definition of $\text{span}(\phi)$, there is a vector $b \in \mathbb{R}^d$ such that $h(x, a) = \phi(x, a)^\top b$ for every $(x, a) \in S$. Therefore, by the definitions of the two feature covariance matrices,

$$\mathbb{E}_{\rho, \pi_P^*} h(X, A)^2 = b^\top \Lambda_P^* b \quad \text{and} \quad \mathbb{E}_{\rho, \pi_{\text{ref}}} h(X, A)^2 = b^\top \Lambda_{\text{ref}} b.$$

Membership of P in the exact shell entails $\Lambda_{\text{ref}} \succ 0$, so its symmetric positive-definite square root and inverse square root exist. Set $u = \Lambda_{\text{ref}}^{1/2} b$. Then

$$b^\top \Lambda_P^* b = u^\top \left(\Lambda_{\text{ref}}^{-1/2} \Lambda_P^* \Lambda_{\text{ref}}^{-1/2} \right) u.$$

The matrix in parentheses is symmetric positive semidefinite. Hence the Rayleigh-quotient inequality and the definition of D_P give

$$u^\top \left(\Lambda_{\text{ref}}^{-1/2} \Lambda_P^* \Lambda_{\text{ref}}^{-1/2} \right) u \leq \lambda_{\max} \left(\Lambda_{\text{ref}}^{-1/2} \Lambda_P^* \Lambda_{\text{ref}}^{-1/2} \right) u^\top u = D_P u^\top u.$$

Finally, $u^\top u = b^\top \Lambda_{\text{ref}} b$. Combining the preceding displays yields

$$\mathbb{E}_{\rho, \pi_P^*} h(X, A)^2 \leq D_P \mathbb{E}_{\rho, \pi_{\text{ref}}} h(X, A)^2,$$

as claimed. $\square$

Lemma 6 (lem:quadratic-majorant-transfer). *Fix $P \in \mathcal{M}(d, C, D, \eta)$. If $M \succeq 0$ and a function $q : S \rightarrow \mathbb{R}$ satisfies $q(x, a)^2 \leq \phi(x, a)^\top M \phi(x, a)$ on S , then*

$$\mathbb{E}_{\rho, \pi_P^*} q^2 \leq \text{tr}(M \Lambda_P^*) \leq D_P \text{tr}(M \Lambda_{\text{ref}}) = D_P \mathbb{E}_{\rho, \pi_{\text{ref}}} \phi^\top M \phi.$$

Proof. The first inequality follows by integrating the pointwise majorant and using the covariance definition:

$$\mathbb{E}_{\rho, \pi_P^*} q^2 \leq \mathbb{E}_{\rho, \pi_P^*} \phi^\top M \phi = \text{tr}(M \Lambda_P^*).$$

By definition of D_P and $\Lambda_{\text{ref}} \succ 0$,

$$\Lambda_{\text{ref}}^{-1/2} \Lambda_P^* \Lambda_{\text{ref}}^{-1/2} \preceq D_P I_d.$$

Congruence by $\Lambda_{\text{ref}}^{1/2}$ gives

$$\Lambda_P^* \preceq D_P \Lambda_{\text{ref}}.$$

Taking the trace after multiplying by $M^{1/2}$ preserves the inequality because $M \succeq 0$, and hence

$$\text{tr}(M \Lambda_P^*) \leq D_P \text{tr}(M \Lambda_{\text{ref}}).$$

The last trace is $\mathbb{E}_{\rho, \pi_{\text{ref}}} \phi^\top M \phi$, proving the claim. Equivalently, a spectral decomposition of M reduces the middle inequality to repeated applications of lem:feature-coverage-domination to linear functions. $\square$

Lemma 7 (lem:measurable-maximum-theorem). *Let (Ω, Σ) be a measurable space, let K be a separable metrizable space, let $\Gamma : \Omega \rightrightarrows K$ be weakly measurable with nonempty compact values, and let $u : \Omega \times K \rightarrow \mathbb{R}$ be measurable in its first argument and continuous in its second. Then $\omega \mapsto \max_{k \in \Gamma(\omega)} u(\omega, k)$ is measurable, and the maximizing correspondence is measurable, nonempty, compact valued, and admits a measurable selector.*

Proposition 3 (prop:empirical-ball-dimension-witness). *For the lower-envelope Gibbs completion of the empirical ball in def:prediction-polytope-learner, the claimed $\tilde{O}(dD/n)$ fast rate is false. Specifically, fix $\eta_0 = 4 \log 3$, $C_0 = 3/2$, and $D_0 = 13/10$. There is a sequence of public experiments and deterministic-reward laws $P_d \in \mathcal{M}(d, C_0, D_0, \eta_0)$ such that, for $n_d = \lceil d^2 \rceil$,*

$$\mathbb{E}_{P_d^{\otimes n_d}} [J_{\eta_0, P_d}(\pi_{P_d}^*) - J_{\eta_0, P_d}(\hat{\pi})] \geq \frac{\log(4/3)}{8 \log 3} \{1 - o(1)\}.$$

In contrast, both dD_0/n_d and $\sqrt{dD_0/n_d}$ tend to zero up to polylogarithmic factors.

Proof. For each integer $d \geq 4$, let $\mathcal{X}_d = \{x_1, \dots, x_d\}$, let $\mathcal{A} = \{1, 0\}$, and put $\rho(x_j) = 1/d$. With $(e_j)_{j=1}^d$ the canonical basis of $\mathbb{R}^d$, define

$$\phi(x_j, 1) = e_j, \quad \phi(x_j, 0) = \frac{1}{2} e_j, \quad \pi_{\text{ref}}(1 \mid x_j) = \pi_{\text{ref}}(0 \mid x_j) = \frac{1}{2}.$$

Take $\theta = \frac{1}{2} \sum_{j=1}^d e_j$, set $\eta_0 = 4 \log 3$, and let P_d have deterministic reward $Y = r_\theta(X, A)$. Thus

$$r_\theta(x_j, 1) = \frac{1}{2}, \quad r_\theta(x_j, 0) = \frac{1}{4}.$$

The supports are finite, the feature norms are at most one, the reward belongs to $[0, 1]$, linear realizability holds, and the action law is the reference policy. Moreover,

$$\Lambda_{\text{ref}} = \frac{1}{d} \left(\frac{1}{2} + \frac{1}{8} \right) I_d = \frac{5}{8d} I_d \succ 0.$$

Because $\eta_0 \{r_\theta(x_j, 1) - r_\theta(x_j, 0)\} = \log 3$, the Gibbs odds of action 1 against action 0 equal three, whence

$$\pi_{P_d}^*(1 \mid x_j) = \frac{3}{4}, \quad \pi_{P_d}^*(0 \mid x_j) = \frac{1}{4}.$$

It follows that

$$C_{P_d} = \max \left\{ \frac{3/4}{1/2}, \frac{1/4}{1/2} \right\} = \frac{3}{2} =: C_0.$$

The target feature covariance is

$$\Lambda_{P_d}^* = \frac{1}{d} \left(\frac{3}{4} + \frac{1}{16} \right) I_d = \frac{13}{16d} I_d.$$

Consequently every eigenvalue of $\Lambda_{\text{ref}}^{-1/2} \Lambda_{P_d}^* \Lambda_{\text{ref}}^{-1/2}$ is

$$\frac{13/16}{5/8} = \frac{13}{10} =: D_0.$$

All defining conditions of $\mathcal{M}(d, C_0, D_0, \eta_0)$ hold, so $P_d \in \mathcal{M}(d, C_0, D_0, \eta_0)$.

Now take $n_d = \lceil d^2 \rceil = d^2$, and let $N_j = \sum_{i=1}^{n_d} \mathbf{1}\{X_i = x_j\}$. By `lem:multinomial-max-count`, the event

$$\mathcal{B}_d = \left\{ \max_{1 \leq j \leq d} N_j \leq 3d \right\}$$

satisfies

$$P_d^{\otimes n_d}(\mathcal{B}_d) \geq 1 - d \exp\{-d(3 \log 3 - 2)\} = 1 - o(1).$$

The true reward function belongs to $\mathcal{F}$ and has empirical loss zero. Hence every empirical-risk minimizer, including $\hat{f}$, has empirical loss zero. Write $\hat{f}(x_j, 1) = \hat{\theta}_j$ and $\hat{f}(x_j, 0) = \hat{\theta}_j/2$. If $N_j > 0$, zero empirical loss for either possible logged action implies $\hat{\theta}_j = 1/2$. If $N_j = 0$, the empirical loss and seminorm contain no term from context x_j .

For each j , let $f^{(j)}$ be obtained from $\hat{f}$ by setting coordinate j of its coefficient vector to zero and leaving all other coordinates unchanged. The resulting predictions at x_j are zero and all other predictions equal those of $\hat{f}$, so $f^{(j)} \in \mathcal{F}$. When $N_j > 0$, the preceding identification gives

$$\|f^{(j)} - \hat{f}\|_{n_d}^2 \leq \frac{N_j}{4n_d};$$

when $N_j = 0$, the left-hand side is zero, so the same inequality holds. On $\mathcal{B}_d$,

$$\|f^{(j)} - \hat{f}\|_{n_d}^2 \leq \frac{3d}{4n_d} \leq \frac{3}{4d}.$$

The squared radius defining $\mathcal{Q}_{n_d}$ is

$$R_d = c_0 \frac{d \log(en_d) + \log(2n_d^2)}{n_d}.$$

Since $dR_d \rightarrow \infty$, one has $R_d \geq 3/(4d)$ for all sufficiently large d . Thus $f^{(j)} \in \mathcal{Q}_{n_d}$ simultaneously for every j on $\mathcal{B}_d$. Every member of $\mathcal{F}$ is pointwise nonnegative, while $f^{(j)}(x_j, 1) = f^{(j)}(x_j, 0) = 0$. Coordinatewise minimization therefore yields

$$r_{n_d}(x_j, 1) = r_{n_d}(x_j, 0) = 0 \quad (j = 1, \dots, d)$$

on $\mathcal{B}_d$. The lower-envelope Gibbs formula then gives $\hat{\pi} = \pi_{\text{ref}}$ at every context on this event.

Applying `lem:gibbs-regret-identity` on $\mathcal{B}_d$, and using that all contexts have the same conditional policies, gives

$$\begin{aligned} J_{\eta_0, P_d}(\pi_{P_d}^*) - J_{\eta_0, P_d}(\hat{\pi}) &= \frac{1}{\eta_0} \text{KL}((1/2, 1/2) \parallel (3/4, 1/4)) \\ &= \frac{1}{4 \log 3} \left\{ \frac{1}{2} \log \frac{2}{3} + \frac{1}{2} \log 2 \right\} \\ &= \frac{\log(4/3)}{8 \log 3}. \end{aligned}$$

Regularized regret is nonnegative outside $\mathcal{B}_d$ because $\pi_{P_d}^*$ maximizes J_{η_0, P_d} . Hence

$$\mathbb{E}_{P_d^{\otimes n_d}} [J_{\eta_0, P_d}(\pi_{P_d}^*) - J_{\eta_0, P_d}(\hat{\pi})] \geq \frac{\log(4/3)}{8 \log 3} P_d^{\otimes n_d}(\mathcal{B}_d) = \frac{\log(4/3)}{8 \log 3} \{1 - o(1)\}.$$

Finally,

$$\frac{dD_0}{n_d} = \frac{D_0}{d} \longrightarrow 0, \quad \sqrt{\frac{dD_0}{n_d}} = \sqrt{\frac{D_0}{d}} \longrightarrow 0,$$

and either convergence remains valid after multiplication by any fixed power of $\log d$. Thus neither proposed $\tilde{O}(dD/n)$ conversion can upper-bound the regret of this empirical-ball coordinatewise-envelope learner on these exact shells. $\square$

Justification. An explicit deterministic-reward exact-shell sequence makes the coordinatewise lower envelope collapse toward the reference policy at $n = d^2$, producing constant regret. *Gap.* The witness is specific to the empirical-ball coordinatewise-envelope learner and is not an all-learner minimax lower bound. *Consumer.* It explains the extra dimension factor of thm:localized-upper; it does not obstruct the different stabilization learner in thm:clean-temperature-feature-upper.

Lemma 8 (lem:gibbs-regret-identity). *For every $P \in \mathcal{M}(d, C, D, \eta)$ and every policy π supported on π_{ref} ,*

$$J_{\eta, P}(\pi_P^*) - J_{\eta, P}(\pi) = \eta^{-1} \mathbb{E}_\rho \text{KL}(\pi(\cdot | X) \| \pi_P^*(\cdot | X))$$

Proof. Let $Z_P(x) = \sum_a \pi_{\text{ref}}(a | x) e^{\eta r_\theta(x, a)}$. Taking logarithms in the Gibbs formula gives

$$\log \frac{\pi(a | x)}{\pi_P^*(a | x)} = \log \frac{\pi(a | x)}{\pi_{\text{ref}}(a | x)} - \eta r_\theta(x, a) + \log Z_P(x).$$

Integrating with respect to $\pi(\cdot | x)$, then with respect to ρ , shows that the right-hand side in the statement equals $\eta^{-1} \mathbb{E}_\rho \log Z_P(X) - J_{\eta, P}(\pi)$. Substituting the Gibbs policy into the definition of $J_{\eta, P}$ shows that its value is $\eta^{-1} \mathbb{E}_\rho \log Z_P(X)$, completing the identity. $\square$

Theorem 4 (thm:feasible-index-region). *For every nonempty exact shell, $1 \leq D \leq C \leq e^\eta$; because the supports are finite, either $(C, D) = (1, 1)$ or $1 < D \leq C < e^\eta$. Conversely, for every $d \geq 4$ and every pair $1 < D \leq C < e^\eta$, there is a finite public experiment with bounded features and nonsingular logging covariance for which $\mathcal{M}(d, C, D, \eta) \neq \emptyset$; the pair $(1, 1)$ is also realizable. Thus the feasible index region when the public experiment may vary is exactly $\{(1, 1)\} \cup \{(C, D) : 1 < D \leq C < e^\eta\}$. Feasibility for a fixed public experiment can be strictly smaller: canonical tabular features force $D = C$, whereas shared features realize off-diagonal pairs $D < C$. This theorem answers the feasibility part of the question; it does not assert a minimax lower frontier outside the proved local hard regime.*

Proof. Fix a nonempty shell and a law $P \in \mathcal{M}(d, C, D, \eta)$. For every context of positive ρ -mass and every action in the support of $\pi_{\text{ref}}(\cdot | x)$, define

$$Z_x = \sum_b \pi_{\text{ref}}(b | x) e^{\eta r_\theta(x, b)}, \quad w_x(a) = \frac{\pi_P^*(a | x)}{\pi_{\text{ref}}(a | x)} = \frac{e^{\eta r_\theta(x, a)}}{Z_x}.$$

The Gibbs formula and normalization give

$$\sum_a \pi_{\text{ref}}(a | x) w_x(a) = 1.$$

Consequently the essential supremum of $w_x(a)$ is at least one, so $C \geq 1$. Since $0 \leq r_\theta \leq 1$, one has $e^{\eta r_\theta(x,a)} \leq e^\eta$ and $Z_x \geq 1$, whence $C \leq e^\eta$. In fact equality is impossible. Because the supports are finite, equality would be attained at some cell having positive $\rho(x)\pi_{\text{ref}}(a \mid x)$. The equalities $w_x(a) = e^\eta$, $e^{\eta r_\theta(x,a)} \leq e^\eta$, and $Z_x \geq 1$ would force $r_\theta(x,a) = 1$ and $Z_x = 1$. If $s = \pi_{\text{ref}}(a \mid x) > 0$, however, bounded rewards imply

$$Z_x \geq se^\eta + (1-s) > 1,$$

a contradiction. Thus $C < e^\eta$.

The change-of-policy identity yields

$$\Lambda_P^* = \mathbb{E}_{\rho, \pi_{\text{ref}}} [w_X(A) \phi(X, A) \phi(X, A)^\top].$$

Since $0 \leq w_X(A) \leq C$ almost surely, for every $t \in \mathbb{R}^d$,

$$t^\top \Lambda_P^* t \leq Ct^\top \Lambda_{\text{ref}} t.$$

The nonsingularity condition therefore gives $D_P \leq C$, and exact-shell membership gives $D \leq C$.

It remains to prove the lower endpoint $D \geq 1$. If $\theta = 0$, all rewards are zero, the Gibbs tilt is trivial, and $\Lambda_P^* = \Lambda_{\text{ref}}$, so $D = 1$. Suppose $\theta \neq 0$. At a fixed context, let $R = r_\theta(x, A)$ when $A \sim \pi_{\text{ref}}(\cdot \mid x)$, and let R' be an independent copy. Exponential tilting gives

$$\mathbb{E}_{\pi_P^*}(\cdot \mid x) R^2 - \mathbb{E}_{\pi_{\text{ref}}}(\cdot \mid x) R^2 = \frac{\text{Cov}(R^2, e^{\eta R})}{\mathbb{E} e^{\eta R}}.$$

Moreover,

$$2 \text{Cov}(R^2, e^{\eta R}) = \mathbb{E}[(R^2 - R'^2)(e^{\eta R} - e^{\eta R'})] \geq 0,$$

because both functions are increasing on $[0, 1]$. Averaging over x and using $r_\theta(x, a) = \phi(x, a)^\top \theta$ gives

$$\theta^\top \Lambda_P^* \theta \geq \theta^\top \Lambda_{\text{ref}} \theta > 0.$$

The final strict inequality follows from $\Lambda_{\text{ref}} \succ 0$ and $\theta \neq 0$. By the generalized Rayleigh-quotient formula,

$$D_P = \sup_{t \neq 0} \frac{t^\top \Lambda_P^* t}{t^\top \Lambda_{\text{ref}} t} \geq 1.$$

We next identify the equality case. If $D = 1$, then $\Lambda_{\text{ref}}^{-1/2} \Lambda_P^* \Lambda_{\text{ref}}^{-1/2} \preceq I$, hence $\Lambda_P^* \preceq \Lambda_{\text{ref}}$. The preceding nonnegative comparison in direction θ must consequently be an equality. Its contextwise integrand is nonnegative, so for ρ -almost every x , $\text{Cov}(R^2, e^{\eta R}) = 0$. In the independent-copy display the integrand is nonnegative and is strictly positive whenever $R \neq R'$, because $r \mapsto r^2$ and $r \mapsto e^{\eta r}$ are strictly increasing on $[0, 1]$. Thus R is constant on the reference-policy support at almost every context. Exponential tilting by a constant leaves the action law unchanged, so $\pi_P^* = \pi_{\text{ref}}$ ρ -almost surely and $C = 1$. Conversely, $C = 1$ implies $D = 1$ by thm:zero-risk-boundary. Hence every feasible pair is either $(1, 1)$ or satisfies

$$1 < D \leq C < e^\eta.$$

We now prove sufficiency when the public experiment may vary. Fix $d \geq 4$, $1 < D \leq C < e^\eta$, an orthonormal basis $e_0, \dots, e_{d-1}$ of $\mathbb{R}^d$, and the two-action set $\{1, 0\}$. For $1 < K < e^\eta$, set

$$q_K = \frac{e^\eta/K - 1}{e^\eta - 1} \in (0, 1).$$

Let $\theta = \sum_{j=0}^{d-1} e_j$. Introduce a context x_C with

$$\phi(x_C, 1) = e_0, \quad \phi(x_C, 0) = 0, \quad \pi_{\text{ref}}(1 \mid x_C) = q_C,$$

and, for $j = 1, \dots, d-1$, a context x_j with

$$\phi(x_j, 1) = e_j, \quad \phi(x_j, 0) = 0, \quad \pi_{\text{ref}}(1 \mid x_j) = q_D.$$

At any such context calibrated by K , the two rewards are one and zero, its Gibbs normalizer is

$$q_K e^\eta + 1 - q_K = e^\eta / K,$$

and the target-to-reference policy ratios of actions 1 and 0 are respectively K and K/e^η .

If $C > D$, introduce one further context x_A at which both action features equal e_0 , give both actions positive reference probability, and put

$$A_0 = \frac{q_C(C - D)}{D - 1} > 0.$$

Give x_C raw context mass one, x_A raw mass A_0 , and every x_j raw mass one, and normalize these masses to a probability distribution ρ . Both rewards at x_A equal one, so the target and reference policies coincide there. If $C = D$, omit x_A and use the other positive raw masses. Finally, let $Y = r_\theta(X, A)$ deterministically. All rewards lie in $[0, 1]$, every feature has Euclidean norm at most one, and linear realizability and reference logging hold by construction.

All covariance matrices are diagonal in the chosen basis. For $j \geq 1$, the target-to-logging covariance ratio in direction e_j equals

$$\frac{\rho(x_j)q_D D}{\rho(x_j)q_D} = D.$$

When $C = D$, the ratio in direction e_0 is likewise $C = D$. When $C > D$, the normalization of the raw context masses cancels, and that ratio is

$$\frac{q_C C + A_0}{q_C + A_0} = D,$$

where the equality follows from $A_0(D - 1) = q_C(C - D)$. Every logging diagonal entry is strictly positive, so $\Lambda_{\text{ref}} \succ 0$, and every generalized eigenvalue equals D . Thus $D_P = D$. The largest policy likelihood ratio is C , attained by action 1 at x_C ; all ratios at the D -contexts are at most $D \leq C$, and those at x_A equal one. Hence $C_P = C$, and the constructed law belongs to a nonempty exact shell $\mathcal{M}(d, C, D, \eta)$.

For $(C, D) = (1, 1)$, take d positive-mass contexts, one action at each context, features $e_0, \dots, e_{d-1}$, coefficient $\theta = 0$, and deterministic reward zero. Then the logging covariance is positive definite, the two policies coincide, and both indices equal one. This proves that the varying-public-experiment feasible region is exactly the asserted set.

Finally, consider a canonical tabular public experiment with $d = |S|$ and $\phi(x, a) = e_{(x, a)}$. Nonsingularity makes every diagonal logging weight $\rho(x)\pi_{\text{ref}}(a | x)$ positive, and direct calculation gives

$$\Lambda_{\text{ref}}^{-1/2} \Lambda_P^* \Lambda_{\text{ref}}^{-1/2} = \text{diag} \left(\frac{\pi_P^*(a | x)}{\pi_{\text{ref}}(a | x)} \right)_{(x, a) \in S}.$$

Therefore $D_P = C_P$, so such a fixed experiment cannot realize an off-diagonal pair. In contrast, when $D < C$, the construction above uses the same feature direction e_0 at x_C and x_A and realizes that off-diagonal pair. Thus fixed-experiment feasibility can indeed be strictly smaller. The argument concerns feasibility alone and supplies no minimax lower bound outside the separately stated local hard regime. $\square$

Justification. Mean-one Gibbs likelihood ratios, covariance ordering, and an explicit shared-feature construction characterize feasibility when the public experiment may vary. *Gap.* The theorem itself concerns varying-design feasibility; fixed-design feasibility is supplied separately by `lem:fixed-experiment-shell-certificate`, and statistical difficulty still depends on the public experiment. *Consumer.* It supplies $1 \leq D \leq C < e^\eta$, used by the clean upper proof, and separates impossible index pairs from realizable designs.

Proposition 4 (`prop:informative-fast-shell-all-indices`). *For every $d \geq 4$, $\eta > 0$, and $1 < D \leq C < e^\eta$, there is a fixed finite public experiment $\mathfrak{E}_1$ with bounded features and a nonempty exact shell $\mathcal{M}_{\mathfrak{E}_1}(d, C, D, \eta)$, and there are constants $c(C, D, \eta) > 0$ and $\epsilon_0(C, D, \eta) > 0$ such that*

$$N_{\mathfrak{E}_1}^*(\epsilon; d, C, D, \eta) \geq c(C, D, \eta) \frac{dD\eta}{\epsilon} \quad (0 < \epsilon \leq \epsilon_0(C, D, \eta)).$$

No condition $2D - 1 < e^\eta$ is required. The constants may depend on (C, D, η) ; hence this is a pointwise local d/ϵ lower bound with displayed factor $D\eta$, not a uniform assertion about coverage or temperature dependence.

Proof. Proof. Put $E = e^\eta$, $k = \lfloor (d-1)/2 \rfloor$, and

$$\alpha = \frac{E-D}{8(E-1)}, \quad p = \frac{\alpha}{D}.$$

Since $1 < D < E$, one has $0 < \alpha < 1/8$ and $0 < p < 1/8$. Choose an orthonormal system $e_0, (u_j, w_j)_{j=1}^k$, completed by a vector z when d is even, and take the action set to be $\{+, -, 0\}$. At each hard context x_j , assign raw context mass $1/k$ and set

$$\begin{aligned} \pi_{\text{ref}}(+ | x_j) &= \pi_{\text{ref}}(- | x_j) = p, & \pi_{\text{ref}}(0 | x_j) &= 1 - 2p, \\ \phi(x_j, +) &= \frac{u_j + w_j}{\sqrt{2}}, & \phi(x_j, -) &= \frac{u_j - w_j}{\sqrt{2}}, & \phi(x_j, 0) &= 0. \end{aligned}$$

Add a calibration context x_C of raw mass one. With

$$q = \frac{E/C - 1}{E - 1} \in (0, 1),$$

put $\pi_{\text{ref}}(+ | x_C) = q$, distribute the remaining probability strictly positively over $-$ and 0 , and set

$$\phi(x_C, +) = e_0, \quad \phi(x_C, -) = \phi(x_C, 0) = 0.$$

If $C > D$, add an anchor context x_A with raw mass

$$a_{\text{anc}} = q \frac{C-D}{D-1} > 0,$$

give all three actions positive reference probability, and set every action feature equal to e_0 . If $C = D$, omit this context and put $a_{\text{anc}} = 0$. If d is even, add a context x_z of raw mass one, give all actions positive reference probability, and set every action feature equal to z . The hard contexts have total raw mass one. Hence the normalizing constant is

$$H = 2 + a_{\text{anc}} + \mathbf{1}\{d \text{ is even}\}, \quad 2 \leq H \leq \bar{H} := 3 + a_{\text{anc}},$$

and normalization of these raw masses defines ρ . Thus H is bounded above and below by positive constants depending only on (C, D, η) , uniformly in d . The resulting finite supports, feature map, reference policy, and η define a fixed public experiment $\mathfrak{E}_1$. The distribution ρ is a law-specific component and is not supplied as part of $\mathfrak{E}_1$.

For $\gamma \geq 0$, write $t = \eta\gamma$ and define

$$T_\gamma = \frac{D(1-2p)}{e^t - 2\alpha \cosh t}, \quad \beta_\gamma = \eta^{-1} \log T_\gamma.$$

At $\gamma = 0$,

$$T_0 = \frac{D-2\alpha}{1-2\alpha} > 1.$$

Moreover, $T_0 < E$, because

$$2\alpha(E-1) = \frac{E-D}{4} < E-D.$$

The denominator in the definition of T_γ is positive at zero. Consequently, continuity gives constants $\gamma_0 > 0$ and $m \in (0, 1/2)$, depending only on (D, η) , such that, for $0 \leq \gamma \leq \gamma_0$,

$$0 \leq t \leq 1, \quad m \leq \beta_\gamma - \gamma \leq \beta_\gamma + \gamma \leq 1 - m,$$

and the denominator defining T_γ remains positive.

For each $v = (v_1, \dots, v_k) \in \{-1, 1\}^k$, define a coefficient vector by

$$\theta_{e_0} = 1, \quad \theta_{u_j} = \sqrt{2}\beta_\gamma, \quad \theta_{w_j} = \sqrt{2}\gamma v_j,$$

and, when present, $\theta_z = 0$. Thus

$$r_v(x_j, +) = \beta_\gamma + \gamma v_j, \quad r_v(x_j, -) = \beta_\gamma - \gamma v_j, \quad r_v(x_j, 0) = 0.$$

At x_C , the reward is one for action $+$ and zero for the other actions. At the anchor all actions have reward one, and at x_z all actions have reward zero. Let P_v use the common context law ρ , the logging action law π_{ref} , and

conditionally Bernoulli rewards with these means. The preceding bounds put every reward mean in $[0, 1]$; linear realizability holds by construction, and every feature has Euclidean norm at most one.

We next verify exact shell membership. At x_C , the Gibbs normalizer is

$$qE + 1 - q = E/C.$$

The target-to-reference ratios of action $+$ and of either zero-reward action are therefore C and C/E , respectively. At a hard context the normalizer is

$$Z = 1 - 2p + 2pT_\gamma \cosh t.$$

Since $pD = \alpha$, the defining equation for T_γ is equivalent to

$$T_\gamma e^t = DZ.$$

Thus the favored and unfavored informative actions have target-to-reference ratios D and De^{-2t} . Because $\beta_\gamma \geq \gamma$, one has $T_\gamma e^t = e^{\eta(\beta_\gamma + \gamma)} \geq 1$, so the zero-action ratio satisfies $1/Z \leq D$. Every action ratio at an action-invariant context is one. Since $D \leq C$, the essential maximum is exactly C , and hence $C_{P_v} = C$.

On $\text{span}\{u_j, w_j\}$, the logging covariance block is a positive scalar multiple of pI_2 . The target probabilities of the favored and unfavored informative actions are

$$pD = \alpha, \quad pDe^{-2t} = \alpha e^{-2t},$$

so the two generalized eigenvalues on this block are D and De^{-2t} . On the e_0 block, the generalized ratio is D : it equals $C = D$ when the anchor is absent, and, when the anchor is present, it equals

$$\frac{qC + a_{\text{anc}}}{q + a_{\text{anc}}} = D.$$

The optional z block has ratio one. Every basis direction receives positive logging mass, so $\Lambda_{\text{ref}} \succ 0$, and the largest generalized eigenvalue is exactly D . Therefore $D_{P_v} = D$, and all P_v belong to the same nonempty exact shell $\mathcal{M}_{\mathbf{e}_1}(d, C, D, \eta)$.

We now prove the testing lower bound, keeping the coordinatewise regret decomposition explicit. Conditional on a hard context, the target action laws for the two signs are

$$s_+ = (\alpha, \alpha e^{-2t}, 1 - \alpha - \alpha e^{-2t}), \quad s_- = (\alpha e^{-2t}, \alpha, 1 - \alpha - \alpha e^{-2t}).$$

For every action law u , direct normalization of the coordinatewise geometric mean gives

$$\text{KL}(u \| s_+) + \text{KL}(u \| s_-) = 2 \text{KL} \left(u \left\| \frac{(\sqrt{s_+(a)s_-(a)})_a}{\sum_b \sqrt{s_+(b)s_-(b)}} \right. \right) - 2 \log \sum_a \sqrt{s_+(a)s_-(a)}.$$

Hence

$$\inf_u \{\text{KL}(u \| s_+) + \text{KL}(u \| s_-)\} = -2 \log \{1 - \alpha(1 - e^{-t})^2\}.$$

For $0 \leq t \leq 1$, the inequality $e^{-t} \leq 1 - t + t^2/2$ follows by differentiating the difference of the two sides twice, and therefore $1 - e^{-t} \geq t/2$. Together with $-\log(1 - y) \geq y$, this yields

$$\inf_u \eta^{-1} \{\text{KL}(u \| s_+) + \text{KL}(u \| s_-)\} \geq \Delta, \quad \Delta := \frac{\alpha t^2}{2\eta}.$$

Let $v^{(j)}$ denote v with its j -th sign flipped. For an arbitrary learner $\tilde{\pi}$ and realized data set z , define the coordinate contribution

$$R_j(v, z) = \frac{\rho(x_j)}{\eta} \text{KL}(\tilde{\pi}_z(\cdot \mid x_j) \| s_{v_j}), \quad \rho(x_j) = \frac{1}{Hk}.$$

By lem:gibbs-regret-identity, the total regularized regret $R(v, z)$ is the sum of nonnegative context contributions, and thus

$$R(v, z) \geq \sum_{j=1}^k R_j(v, z).$$

Applying the preceding two-law separation to the common action law $\tilde{\pi}_z(\cdot \mid x_j)$ gives the pointwise neighbor inequality

$$R_j(v, z) + R_j(v^{(j)}, z) \geq \rho(x_j)\Delta.$$

It remains to control neighboring data laws. For $r, s \in [m, 1 - m]$, the elementary inequality $\log x \leq x - 1$ gives

$$\text{kl}(r \parallel s) \leq \frac{(r - s)^2}{s(1 - s)}.$$

Neighboring laws differ only in the two Bernoulli reward means at x_j . Hence their one-observation divergence satisfies

$$\text{KL}(P_v \parallel P_{v^{(j)}}) \leq \frac{8p\gamma^2}{Hkm(1 - m)}.$$

The logarithm of a product likelihood ratio is the sum of its coordinate logarithms, so the n -observation divergence is n times this quantity. For completeness, if P, Q are any two probability laws, $A = \{dP/dQ \geq 1\}$, and $\tau = P(A) - Q(A) = \text{TV}(P, Q)$, the log-sum inequality gives

$$\text{KL}(P \parallel Q) \geq \text{kl}(Q(A) + \tau \parallel Q(A)).$$

As a function of τ , the Bernoulli divergence on the right has value and first derivative zero at zero and second derivative at least four. Thus it is at least $2\tau^2$, proving

$$\text{TV}(P, Q) \leq \sqrt{\text{KL}(P \parallel Q)/2}.$$

It follows that there is a constant $c_1(C, D, \eta) > 0$ such that, simultaneously for every neighboring pair,

$$\text{TV}(P_v^{\otimes n}, P_{v^{(j)}}^{\otimes n}) \leq \frac{1}{2} \quad \text{whenever} \quad n \leq c_1(C, D, \eta) \frac{k}{p\gamma^2}.$$

Let

$$B = 2^{-k} \sum_{v \in \{-1, 1\}^k} \mathbb{E}_{P_v^{\otimes n}} R(v, \mathcal{D}_n)$$

be the Bayes regret under the uniform sign prior. Integrating the pointwise neighbor inequality against the smaller of the two neighboring data densities gives

$$\mathbb{E}_v R_j(v, \mathcal{D}_n) + \mathbb{E}_{v^{(j)}} R_j(v^{(j)}, \mathcal{D}_n) \geq \rho(x_j)\Delta \{1 - \text{TV}(P_v^{\otimes n}, P_{v^{(j)}}^{\otimes n})\} \geq \frac{\rho(x_j)\Delta}{2}.$$

For each fixed j , partitioning the 2^k signs into 2^{k-1} neighboring pairs and then summing over coordinates yields

$$\begin{aligned} B &\geq \sum_{j=1}^k 2^{-k} \sum_v \mathbb{E}_v R_j(v, \mathcal{D}_n) \\ &\geq \sum_{j=1}^k 2^{-k} 2^{k-1} \frac{\rho(x_j)\Delta}{2} = \frac{\Delta}{4H} = \frac{\alpha t^2}{8H\eta}. \end{aligned}$$

This coordinatewise decomposition is essential: pairing the full regret separately for every j and then summing would count the same full loss k times and would not justify the displayed bound.

Choose

$$t^2 = \frac{16H\eta\epsilon}{\alpha}, \quad \gamma = \frac{t}{\eta}.$$

Because $H \leq \bar{H}$, there is an $\epsilon_0(C, D, \eta) > 0$, independent of d , such that $0 < \epsilon \leq \epsilon_0(C, D, \eta)$ implies $t \leq 1$ and $\gamma \leq \gamma_0$. In this range the Bayes regret satisfies

$$B \geq 2\epsilon.$$

Furthermore,

$$p\gamma^2 = \frac{\alpha}{D} \frac{t^2}{\eta^2} = \frac{16H\epsilon}{D\eta}.$$

Since $k = \lfloor (d-1)/2 \rfloor \geq d/4$ for $d \geq 4$, the neighboring-total-variation condition holds whenever

$$n \leq c_2(C, D, \eta) \frac{dD\eta}{\epsilon}$$

for some $c_2(C, D, \eta) > 0$; one may take any $c_2 \leq c_1/(64\bar{H})$. The supremum risk over the full shell dominates the uniform-prior Bayes risk over the finite family $\{P_v\}$, for every learner. Taking the infimum over $\mathcal{L}_n(\mathfrak{E}_1)$ in def:minimax-risk therefore shows that the minimax risk is strictly larger than ϵ throughout this sample-size range. By def:sample-complexity, after decreasing the positive constant to absorb integer rounding,

$$N_{\mathfrak{E}_1}^*(\epsilon; d, C, D, \eta) \geq c(C, D, \eta) \frac{dD\eta}{\epsilon} \quad (0 < \epsilon \leq \epsilon_0(C, D, \eta)).$$

The construction used only $D < e^\eta$, not $2D - 1 < e^\eta$. Since the constants may depend on (C, D, η) , the invariant rate conclusion is the pointwise local lower bound $\Omega_{C,D,\eta}(d/\epsilon)$, rather than a uniform lower dependence on D or η . $\square$

Theorem 5 (thm:erm-plugin-upper). *Fix $c_0 = 64$, let $a_n = d \log(en) + \log(2n^2)$, and suppose $\mathcal{M}_{\mathfrak{E}}(d, C, D, \eta) \neq \emptyset$. Let $\hat{\pi}_{\text{ERM}}$ be the Gibbs policy generated by $\hat{f}$ as in def:erm-gibbs-plugin-learner. This learner belongs to $\mathcal{L}_n(\mathfrak{E})$, uses neither C nor D as input, and satisfies*

$$\sup_{P \in \mathcal{M}_{\mathfrak{E}}(d, C, D, \eta)} \mathbb{E}_{P^{\otimes n}}[J_{\eta, P}(\pi_P^*) - J_{\eta, P}(\hat{\pi}_{\text{ERM}})] \leq \min \left\{ 600\kappa_\eta \frac{Da_n}{n}, \sqrt{600}(1 + e^{2\eta}) \sqrt{\frac{Da_n}{n}} \right\} + \frac{\delta_n}{2} \min\{2, \eta/2\}.$$

Consequently, whenever the displayed sample-size order is at least two,

$$N_{\mathfrak{E}}^*(\epsilon; d, C, D, \eta) \leq \tilde{O} \left(dD \min \left\{ \frac{\kappa_\eta}{\epsilon}, \frac{(1 + e^{2\eta})^2}{\epsilon^2} \right\} \right).$$

Together with thm:localized-upper, this yields the complementary general envelope

$$N_{\mathfrak{E}}^*(\epsilon; d, C, D, \eta) \leq \tilde{O} \left(dD \min \left\{ \frac{d\eta}{\epsilon}, \frac{d}{\epsilon^2}, \frac{\kappa_\eta}{\epsilon}, \frac{(1 + e^{2\eta})^2}{\epsilon^2} \right\} \right).$$

No clean-temperature $\tilde{O}(dD \min\{\eta/\epsilon, \epsilon^{-2}\})$ conclusion is asserted.

Proof. The sequential lexicographic rule in def:prediction-polytope-learner makes $\hat{f}$ a measurable function of $(\mathcal{D}_n, \mathfrak{E})$. The formula in def:erm-gibbs-plugin-learner is a finite collection of continuous exponentials and normalizations applied to $\hat{f}$, with a strictly positive denominator. Hence $\hat{\pi}_{\text{ERM}} \in \mathcal{L}_n(\mathfrak{E})$. The formula contains neither C nor D .

Fix $P \in \mathcal{M}_{\mathfrak{E}}(d, C, D, \eta)$. On the event $\mathcal{E}_n$ from lem:bounded-linear-localization, the inclusion $\hat{f} \in \mathcal{Q}_n$ holds by the definition of $\mathcal{Q}_n$, and the uniform population bound gives

$$\mathbb{E}_{\rho, \pi_{\text{ref}}}(\hat{f} - r_\theta)^2 \leq 600 \frac{a_n}{n}.$$

The contrast $h = \hat{f} - r_\theta$ belongs to $\text{span}(\phi)$. Applying `lem:feature-coverage-domination` directly, and using $D_P = D$, yields

$$\mathbb{E}_{\rho, \pi_P^*} (\hat{f} - r_\theta)^2 \leq D \mathbb{E}_{\rho, \pi_{\text{ref}}} (\hat{f} - r_\theta)^2 \leq 600 \frac{Da_n}{n}.$$

Unlike the coordinatewise lower envelope, this single linear contrast incurs no leverage-trace factor.

Set $r = r_\theta$, $g = \hat{f}$, and $\Delta = \hat{f} - r_\theta$. Their Gibbs policies are π_P^* and $\hat{\pi}_{\text{ERM}}$. By `lem:gibbs-plugin-conversion`, on $\mathcal{E}_n$,

$$J_{\eta, P}(\pi_P^*) - J_{\eta, P}(\hat{\pi}_{\text{ERM}}) \leq \min \left\{ 600\kappa_\eta \frac{Da_n}{n}, \sqrt{600}(1 + e^{2\eta}) \sqrt{\frac{Da_n}{n}} \right\}.$$

For completeness, the regret between Gibbs policies generated by any two potentials in $[0, 1]^S$ has the deterministic bound

$$J_{\eta, r}(\pi_r) - J_{\eta, r}(\pi_g) \leq \min\{2, \eta/2\}.$$

Indeed, the likelihood-ratio identity gives $\log(\pi_g/\pi_r) \leq 2\eta$, and `lem:gibbs-regret-identity` gives the bound two. The exact Gibbs path identity used in `lem:gibbs-plugin-conversion` gives

$$J_{\eta, r}(\pi_r) - J_{\eta, r}(\pi_g) = \eta \int_0^1 t \text{Var}_{\pi_t}(g - r) dt \leq \eta/2,$$

since $g - r \in [-1, 1]$ has variance at most one. As $P^{\otimes n}(\mathcal{E}_n^c) \leq \delta_n/2$, averaging the good- and bad-event bounds proves

$$\sup_{P \in \mathcal{M}_\epsilon(d, C, D, \eta)} \mathbb{E}_{P^{\otimes n}} [J_{\eta, P}(\pi_P^*) - J_{\eta, P}(\hat{\pi}_{\text{ERM}})] \leq \min \left\{ 600\kappa_\eta \frac{Da_n}{n}, \sqrt{600}(1 + e^{2\eta}) \sqrt{\frac{Da_n}{n}} \right\} + \frac{\delta_n}{2} \min\{2, \eta/2\}.$$

The bound is uniform because all constants are independent of the member P .

We now invert it. First,

$$\kappa_\eta = \eta \int_0^1 t e^{2\eta t} dt \geq \eta/2.$$

It follows that $\min\{2, \eta/2\} \leq \kappa_\eta$: for $\eta \leq 4$ use the preceding inequality, and for $\eta > 4$ use $2 < \eta/2 \leq \kappa_\eta$. Also $D \geq 1$ on every nonempty shell by `thm:feasible-index-region`.

If $\eta \leq 2\epsilon$, the deterministic $\eta/2$ bound already gives risk at most ϵ at sample size one. Suppose $\eta > 2\epsilon$. On the quadratic branch, choose n , up to universal logarithmic factors and integer rounding, of order

$$dD \frac{\kappa_\eta}{\epsilon}.$$

Since $a_n = \tilde{O}(d)$, this makes the quadratic good-event term at most a fixed fraction of ϵ . Moreover, $dD \geq 4$ and $\kappa_\eta > \epsilon$, so increasing the universal constant makes

$$\frac{\delta_n}{2} \min\{2, \eta/2\} \leq \frac{\kappa_\eta}{2n^2} \leq \frac{\epsilon}{4}.$$

On the square-root branch, choosing n of order

$$dD \frac{(1 + e^{2\eta})^2}{\epsilon^2}$$

makes the square-root term a fixed fraction of ϵ , and, because $1 + e^{2\eta} \geq 2$, also makes the complement contribution at most a fixed fraction of ϵ . Thus the same explicit learner witnesses

$$N_\epsilon^*(\epsilon; d, C, D, \eta) \leq \tilde{O} \left(dD \min \left\{ \frac{\kappa_\eta}{\epsilon}, \frac{(1 + e^{2\eta})^2}{\epsilon^2} \right\} \right)$$

whenever this order is at least two.

Finally, thm:localized-upper supplies the separate sufficient order

$$\tilde{O}(d^2 D \min\{\eta/\epsilon, \epsilon^{-2}\}).$$

The minimax risk is bounded by the risk of either measurable learner, so one may take the smaller of the two sufficient sample sizes. Factoring out dD gives

$$\tilde{O}\left(dD \min\left\{\frac{d\eta}{\epsilon}, \frac{d}{\epsilon^2}, \frac{\kappa_\eta}{\epsilon}, \frac{(1 + e^{2\eta})^2}{\epsilon^2}\right\}\right),$$

which proves the complementary envelope. The explicit exponential temperature factors remain in the dD branch, so the theorem does not imply the clean-temperature conjecture. $\square$

Justification. The empirical-risk-minimizer Gibbs plug-in gives dD dimension dependence and the fixed-temperature fast inversion used in the quadratic-lower impossibility. *Gap.* Its explicit temperature penalties are properties of that plug-in analysis and are superseded for universal all-shell achievability by the stabilization theorem. *Consumer.* It remains the direct input to the specified-experiment $O_{d,D,\eta}(\epsilon^{-1} \log(1/\epsilon))$ inversion.

Lemma 9 (lem:bounded-linear-localization). *Fix $c_0 = 64$. For every $P \in \mathcal{M}(d, C, D, \eta)$ and every $n \in \mathbb{N}$, with $\delta_n = n^{-2}$, there is an event $\mathcal{E}_n$ satisfying $P^{\otimes n}(\mathcal{E}_n) \geq 1 - \delta_n/2$ such that*

$$r_\theta \in \mathcal{Q}_n \quad \text{and} \quad \sup_{f \in \mathcal{Q}_n} \mathbb{E}_{X \sim \rho, A \sim \pi_{\text{ref}}(\cdot|X)}[(f(X, A) - r_\theta(X, A))^2] \leq 600 \frac{d \log(en) + \log(2n^2)}{n}.$$

More generally, for any fixed universal $c_0 \geq 64$, the constant 600 may be replaced by $4c_0 + 291$.

Proof. Put

$$a_n = d \log(en) + \log(2n^2),$$

and write $W_i = (X_i, A_i)$ and $\varepsilon_i = Y_i - r_\theta(W_i)$. By ass:linear-realizability,

$$\mathbb{E}_P[\varepsilon_i | W_i] = 0,$$

while $Y_i, r_\theta(W_i) \in [0, 1]$ gives $|\varepsilon_i| \leq 1$. Conditional on $W_1, \dots, W_n$, the residuals are independent. Let $V_n \subseteq \mathbb{R}^n$ be the space of evaluation vectors $(f(W_1), \dots, f(W_n))$ with $f \in \text{span}(\phi)$, and let Π_n be its orthogonal projector. Then $\dim(V_n) \leq d$, and Π_n is measurable with respect to the logged context-action cells.

Apply lem:bounded-projection-concentration conditionally with $\zeta = \delta_n/4$. After integrating the conditional probability, there is an event $\mathcal{E}_{1,n}$ of probability at least $1 - \delta_n/4$ on which

$$\|\Pi_n \varepsilon\|_2^2 \leq 8\{d \log 5 + \log(8n^2)\}.$$

Because $r_\theta \in \mathcal{F}$, empirical-risk minimality of $\hat{f}$ gives

$$\sum_{i=1}^n \{Y_i - \hat{f}(W_i)\}^2 \leq \sum_{i=1}^n \{Y_i - r_\theta(W_i)\}^2.$$

For $h_i = \hat{f}(W_i) - r_\theta(W_i)$, expansion gives $\|h\|_2^2 \leq 2\varepsilon^\top h$. Since $h \in V_n$,

$$\varepsilon^\top h = (\Pi_n \varepsilon)^\top h \leq \|\Pi_n \varepsilon\|_2 \|h\|_2.$$

Thus $\|h\|_2 \leq 2\|\Pi_n \varepsilon\|_2$, and on $\mathcal{E}_{1,n}$,

$$\|\hat{f} - r_\theta\|_n^2 \leq \frac{32}{n} \{d \log 5 + \log(8n^2)\}.$$

For $d \geq 4$ and $n \geq 1$,

$$d \log 5 + \log(8n^2) \leq 2a_n.$$

Indeed, the difference between the right and left sides is $d\{2 + 2 \log n - \log 5\} + 2 \log n - \log 2$, which is nonnegative at $(d, n) = (4, 1)$ and nondecreasing in both variables. Hence

$$\|\hat{f} - r_\theta\|_n^2 \leq 64 \frac{a_n}{n}.$$

For every $c_0 \geq 64$, this proves $r_\theta \in \mathcal{Q}_n$ on $\mathcal{E}_{1,n}$.

It remains to transfer empirical error to the logging population. The linear space containing $\mathcal{F}$, equipped with the supremum norm on finite S , has dimension at most d , and $\mathcal{F} \subseteq [0, 1]^S$. A maximal u -separated subset and the disjoint-ball volume argument therefore give

$$N_{\mathcal{F}}(u) \leq (1 + 2/u)^d, \quad 0 < u < 1.$$

Apply `lem:zhao-uniform-square-comparison` to $\mathcal{G} = \mathcal{F}$, the logging law $\mu = \rho \times \pi_{\text{ref}}$, failure probability $\delta_n/4$, and $u = (n + 1)^{-1}$. There is an event $\mathcal{E}_{2,n}$, of probability at least $1 - \delta_n/4$, on which, simultaneously for all $f_1, f_2 \in \mathcal{F}$,

$$\mathbb{E}_\mu(f_1 - f_2)^2 \leq 2\|f_1 - f_2\|_n^2 + \frac{32}{3n} \log\{8n^2 N_{\mathcal{F}}(u)\} + \frac{10}{n+1}.$$

The covering estimate yields $N_{\mathcal{F}}(u) \leq (2n + 3)^d \leq (5n)^d$. Moreover,

$$d \log(5n) \leq 2d \log(en), \quad \log(8n^2) \leq 3 \log(2n^2),$$

so $\log\{8n^2 N_{\mathcal{F}}(u)\} \leq 3a_n$. Since $a_n \geq d \geq 4$, the final two terms in the preceding population comparison are at most $35a_n/n$.

Set $\mathcal{E}_n = \mathcal{E}_{1,n} \cap \mathcal{E}_{2,n}$. The union bound gives $P^{\otimes n}(\mathcal{E}_n) \geq 1 - \delta_n/2$. On this event, every $f \in \mathcal{Q}_n$ satisfies

$$\begin{aligned} \|f - r_\theta\|_n^2 &\leq 2\|f - \hat{f}\|_n^2 + 2\|\hat{f} - r_\theta\|_n^2 \\ &\leq (2c_0 + 128) \frac{a_n}{n}. \end{aligned}$$

Taking $f_1 = f$ and $f_2 = r_\theta$ in the uniform comparison therefore gives

$$\mathbb{E}_{\rho, \pi_{\text{ref}}}(f - r_\theta)^2 \leq (4c_0 + 291) \frac{a_n}{n}.$$

Taking the supremum over $f \in \mathcal{Q}_n$ proves the general assertion. At $c_0 = 64$, the coefficient is $547 \leq 600$, which proves the displayed bound. The same calculation covers $n = 1$, for which $u = 1/2$ and the asserted event probability is at least $1/2$. $\square$

Lemma 10 (`lem:gibbs-plugin-conversion`). *Let $r, g : S \rightarrow [0, 1]$, put $\Delta = g - r$, and let π_r and π_g be their Gibbs policies relative to π_{ref} . Write*

$$J_{\eta, r}(\pi) = \mathbb{E}_{\rho, \pi} r(X, A) - \eta^{-1} \mathbb{E}_\rho \text{KL}(\pi(\cdot \mid X) \parallel \pi_{\text{ref}}(\cdot \mid X)),$$

and define

$$\kappa_\eta = \eta \int_0^1 t e^{2\eta t} dt = \frac{e^{2\eta}(2\eta - 1) + 1}{4\eta}.$$

Then

$$J_{\eta, r}(\pi_r) - J_{\eta, r}(\pi_g) \leq \min \left\{ \kappa_\eta \mathbb{E}_{\rho, \pi_r} [\Delta^2], (1 + e^{2\eta}) \sqrt{\mathbb{E}_{\rho, \pi_r} [\Delta^2]} \right\}.$$

Proof. For a fixed context x , define

$$F_x(t) = \log \sum_a \pi_{\text{ref}}(a \mid x) \exp\{\eta(r(x, a) + t\Delta(x, a))\}, \quad 0 \leq t \leq 1,$$

and let $\pi_t(\cdot \mid x)$ be the Gibbs law generated by $r + t\Delta$. Finiteness of $\mathcal{A}$ permits differentiation term by term, giving

$$F'_x(t) = \eta \mathbb{E}_{\pi_t} \Delta, \quad F''_x(t) = \eta^2 \text{Var}_{\pi_t}(\Delta).$$

Substitution of the Gibbs formula into the conditional welfare gives

$$J_{\eta, r, x}(\pi_0) - J_{\eta, r, x}(\pi_1) = \eta^{-1} \{F_x(0) - F_x(1) + F'_x(1)\}.$$

Two integrations therefore yield

$$\begin{aligned} J_{\eta, r, x}(\pi_0) - J_{\eta, r, x}(\pi_1) &= \eta^{-1} \int_0^1 t F''_x(t) dt \\ &= \eta \int_0^1 t \text{Var}_{\pi_t}(\Delta) dt. \end{aligned}$$

Indeed,

$$F'_x(1) - \{F_x(1) - F_x(0)\} = \int_0^1 \{F'_x(1) - F'_x(s)\} ds = \int_0^1 t F''_x(t) dt.$$

Since $r, g \in [0, 1]^S$, one has $-1 \leq \Delta \leq 1$. Moreover,

$$\frac{\pi_t(a \mid x)}{\pi_0(a \mid x)} = \frac{e^{\eta t \Delta(x, a)}}{\mathbb{E}_{\pi_0} e^{\eta t \Delta(x, A)}} \leq \frac{e^{\eta t}}{e^{-\eta t}} = e^{2\eta t}.$$

Therefore

$$\text{Var}_{\pi_t}(\Delta) \leq \mathbb{E}_{\pi_t} \Delta^2 \leq e^{2\eta t} \mathbb{E}_{\pi_0} \Delta^2.$$

Substitution into the path identity gives, contextwise,

$$J_{\eta, r, x}(\pi_r) - J_{\eta, r, x}(\pi_g) \leq \eta \left(\int_0^1 t e^{2\eta t} dt \right) \mathbb{E}_{\pi_r} \Delta^2.$$

Integration by parts gives

$$\eta \int_0^1 t e^{2\eta t} dt = \frac{e^{2\eta}(2\eta - 1) + 1}{4\eta} = \kappa_\eta.$$

Averaging over ρ proves the quadratic branch.

For the square-root branch, first note that direct substitution of the Gibbs formula gives, for every policy π supported on π_{ref} ,

$$J_{\eta, g}(\pi_g) - J_{\eta, g}(\pi) = \eta^{-1} \mathbb{E}_\rho \text{KL}(\pi(\cdot \mid X) \parallel \pi_g(\cdot \mid X)) \geq 0.$$

Insert the welfare generated by g :

$$\begin{aligned} J_{\eta,r}(\pi_r) - J_{\eta,r}(\pi_g) &= J_{\eta,g}(\pi_r) - J_{\eta,g}(\pi_g) - \mathbb{E}_{\rho,\pi_r}\Delta + \mathbb{E}_{\rho,\pi_g}\Delta \\ &\leq \mathbb{E}_{\rho,\pi_r}|\Delta| + \mathbb{E}_{\rho,\pi_g}|\Delta|. \end{aligned}$$

The preceding likelihood-ratio calculation at $t = 1$ gives $\pi_g/\pi_r \leq e^{2\eta}$, so

$$\mathbb{E}_{\rho,\pi_g}|\Delta| \leq e^{2\eta}\mathbb{E}_{\rho,\pi_r}|\Delta|.$$

Cauchy–Schwarz then yields

$$J_{\eta,r}(\pi_r) - J_{\eta,r}(\pi_g) \leq (1 + e^{2\eta})\sqrt{\mathbb{E}_{\rho,\pi_r}\Delta^2}.$$

Taking the minimum of the two simultaneously valid bounds proves the lemma. $\square$

Theorem 6 (thm:no-shared-feature-slow-branch). *Fix $d \geq 4$, $\eta > 0$, $1 < D < C < e^\eta$, and any finite public experiment $\mathfrak{E}$ for which $\mathcal{M}_{\mathfrak{E}}(d, C, D, \eta) \neq \emptyset$. Define*

$$b_\eta = \frac{1}{2} \min\{2, \eta/2\}, \quad Q_{d,D,\eta} = \max\{600\kappa_\eta D(d+2), b_\eta, 1\}.$$

Then, for every $0 < \epsilon < 1$,

$$N_{\mathfrak{E}}^*(\epsilon; d, C, D, \eta) \leq \left\lceil \frac{8Q_{d,D,\eta}}{\epsilon} \log \left(\frac{8eQ_{d,D,\eta}}{\epsilon} \right) \right\rceil.$$

Consequently,

$$\lim_{\epsilon \downarrow 0} \frac{\epsilon^2}{d} N_{\mathfrak{E}}^*(\epsilon; d, C, D, \eta) = 0.$$

Thus no specified finite public experiment, genuinely shared-feature or otherwise, admits constants $c(C, D, \eta) > 0$ and $\epsilon_0(C, D, \eta) > 0$ for which $N_{\mathfrak{E}}^(\epsilon; d, C, D, \eta) \geq c(C, D, \eta)d/\epsilon^2$ for every $0 < \epsilon \leq \epsilon_0(C, D, \eta)$. The proposed shared-feature slow branch is therefore false.*

Proof. The hypotheses of the theorem are a special case of lem:fixed-experiment-fast-branch-inversion. The quantities b_η and $Q_{d,D,\eta}$ in that lemma are exactly the quantities displayed here. Therefore the lemma gives, for every $0 < \epsilon < 1$,

$$N_{\mathfrak{E}}^*(\epsilon; d, C, D, \eta) \leq \left\lceil \frac{8Q_{d,D,\eta}}{\epsilon} \log \left(\frac{8eQ_{d,D,\eta}}{\epsilon} \right) \right\rceil.$$

It also gives

$$\lim_{\epsilon \downarrow 0} \frac{\epsilon^2}{d} N_{\mathfrak{E}}^*(\epsilon; d, C, D, \eta) = 0.$$

If constants $c(C, D, \eta) > 0$ and $\epsilon_0(C, D, \eta) > 0$ satisfied the proposed lower bound throughout $0 < \epsilon \leq \epsilon_0(C, D, \eta)$, the last ratio would be at least $c(C, D, \eta)$ throughout that interval, contradicting its convergence to zero. This proves the target. In fact, the cited lemma proves the same conclusion without the strict off-diagonal restriction: it covers every nonempty finite exact shell, including $C = D > 1$ and the boundary shell. $\square$

Justification. The theorem rules out every positive d/ϵ^2 lower branch valid for all sufficiently small ϵ on a specified finite experiment. *Gap.* It does not exclude slower preasymptotic behavior on a restricted interval; its earlier temperature caveat is superseded by the all-temperature stabilization theorem. *Consumer.* It closes the proposed slow branch for both diagonal and genuinely shared-feature experiments.

Lemma 11 (lem:fixed-experiment-fast-branch-inversion). *Fix $d \geq 4$, $\eta > 0$, and any finite public experiment $\mathfrak{E}$ for which $\mathcal{M}_{\mathfrak{E}}(d, C, D, \eta) \neq \emptyset$. Define*

$$b_\eta = \frac{1}{2} \min\{2, \eta/2\}, \quad Q_{d,D,\eta} = \max\{600\kappa_\eta D(d+2), b_\eta, 1\}.$$

Then, for every $0 < \epsilon < 1$,

$$N_{\mathfrak{E}}^*(\epsilon; d, C, D, \eta) \leq \left\lceil \frac{8Q_{d,D,\eta}}{\epsilon} \log \left(\frac{8eQ_{d,D,\eta}}{\epsilon} \right) \right\rceil.$$

Consequently,

$$\lim_{\epsilon \downarrow 0} \frac{\epsilon^2}{d} N_{\mathfrak{E}}^*(\epsilon; d, C, D, \eta) = 0.$$

Thus no specified finite public experiment with a nonempty exact shell admits constants $c(C, D, \eta) > 0$ and $\epsilon_0(C, D, \eta) > 0$ for which $N_{\mathfrak{E}}^(\epsilon; d, C, D, \eta) \geq c(C, D, \eta)d/\epsilon^2$ for every $0 < \epsilon \leq \epsilon_0(C, D, \eta)$. This is an asymptotic conclusion and does not exclude slower preasymptotic behavior on a restricted accuracy interval.*

Proof. Let

$$a_n = d \log(en) + \log(2n^2).$$

The fast alternative in thm:erm-plugin-upper supplies the measurable learner $\hat{\pi}_{\text{ERM}}$, whose worst-case expected regularized regret over the fixed nonempty shell is at most

$$600\kappa_\eta D \frac{a_n}{n} + b_\eta n^{-2}.$$

Indeed, because $\delta_n = n^{-2}$, the complement term in that theorem is

$$\frac{\delta_n}{2} \min\{2, \eta/2\} = b_\eta n^{-2}.$$

By the infimum over learners in def:minimax-risk, the same expression bounds

$$\mathfrak{R}_{n,\mathfrak{E}}(\mathcal{M}_{\mathfrak{E}}(d, C, D, \eta)).$$

For every integer $n \geq 1$,

$$a_n = d + (d+2) \log n + \log 2 \leq (d+2) \log(2en).$$

Write $Q = Q_{d,D,\eta}$, fix $0 < \epsilon < 1$, and set

$$L = \log \left(\frac{8eQ}{\epsilon} \right), \quad n_\epsilon = \left\lceil \frac{8QL}{\epsilon} \right\rceil.$$

Since $Q \geq 1$, $0 < \epsilon < 1$, and $L \geq \log(8e) > 1$,

$$\frac{8QL}{\epsilon} \leq n_\epsilon \leq \frac{9QL}{\epsilon}.$$

Moreover, $L \geq \log(8e)$ implies $\log(9L/4) \leq L$. Indeed, $x \mapsto x - \log(9x/4)$ is increasing for $x > 1$ and is positive at $x = \log(8e)$. Hence

$$\log(2en_\epsilon) \leq \log \left(\frac{18eQL}{\epsilon} \right) = L + \log(9L/4) \leq 2L.$$

Using $600\kappa_\eta D(d+2) \leq Q$, $b_\eta \leq Q$, and the lower bound on n_ϵ , we get

$$600\kappa_\eta D \frac{a_{n_\epsilon}}{n_\epsilon} \leq 600\kappa_\eta D(d+2) \frac{2L}{8QL/\epsilon} \leq \frac{\epsilon}{4}$$

and

$$b_\eta n_\epsilon^{-2} \leq \frac{b_\eta \epsilon^2}{64Q^2L^2} \leq \frac{\epsilon}{64}.$$

The last inequality uses $b_\eta \leq Q$, $Q \geq 1$, $L > 1$, and $0 < \epsilon < 1$. Therefore

$$\mathfrak{R}_{n_\epsilon, \mathfrak{E}}(\mathcal{M}_\mathfrak{E}(d, C, D, \eta)) \leq \frac{17}{64} \epsilon < \epsilon.$$

By def:sample-complexity,

$$N_\mathfrak{E}^*(\epsilon; d, C, D, \eta) \leq n_\epsilon = \left\lceil \frac{8Q_{d,D,\eta}}{\epsilon} \log \left(\frac{8eQ_{d,D,\eta}}{\epsilon} \right) \right\rceil.$$

The upper estimate $n_\epsilon \leq 9QL/\epsilon$ now yields

$$0 \leq \frac{\epsilon^2}{d} N_\mathfrak{E}^*(\epsilon; d, C, D, \eta) \leq \frac{9Q}{d} \epsilon \log \left(\frac{8eQ}{\epsilon} \right) \rightarrow 0 \quad (\epsilon \downarrow 0),$$

because d , D , and η , and hence Q , are fixed. If the asserted quadratic lower bound held for all sufficiently small ϵ , the ratio on the left would be bounded below by $c(C, D, \eta) > 0$, which is impossible. No strict inequality between C and D was used. $\square$

Justification. This lemma explicitly inverts the plug-in fast branch on every nonempty finite exact shell. *Gap.* It is used for the asymptotic lower-bound impossibility, while the sharper universal all-temperature achievability statement is now supplied separately by the stabilization theorem. *Consumer.* It proves that no specified finite experiment can have a positive quadratic lower rate throughout all sufficiently small accuracies.

Proposition 5 (prop:zero-risk-nontrivial-shell). *For every $d \geq 4$, $\eta > 0$, and $1 < D \leq C < e^\eta$, there is a finite public experiment with bounded features and a nonempty exact shell $\mathcal{M}(d, C, D, \eta)$ such that every member of the shell has the same publicly computable Gibbs optimum. For this experiment, $\mathfrak{R}_n(\mathcal{M}(d, C, D, \eta)) = 0$ for every $n \in \mathbb{N}$, and $N^*(\epsilon; d, C, D, \eta) = 1$ for every $\epsilon > 0$.*

Proof. Fix $d \geq 4$, $\eta > 0$, and $1 < D \leq C < e^\eta$. Let $e_0, \dots, e_{d-1}$ be an orthonormal basis of $\mathbb{R}^d$, use the two actions $\{1, 0\}$, and set

$$q = \frac{e^\eta/C - 1}{e^\eta - 1}.$$

The inequalities $1 < C < e^\eta$ imply $q \in (0, 1)$. Introduce a context x_C with

$$\phi(x_C, 1) = e_0, \quad \phi(x_C, 0) = 0, \quad \pi_{\text{ref}}(1 \mid x_C) = q.$$

If $D < C$, also introduce a context x_A at which both action features equal e_0 . For each $j = 1, \dots, d-1$, introduce a context x_j at which both action features equal e_j . At every action-invariant context give both actions positive reference probability. These finite supports, this feature map, the reference policy, and η define a public experiment $\mathfrak{E}_0$. Every feature has Euclidean norm at most one.

We first exhibit a member of its exact shell. If $D < C$, put

$$a_{\text{anc}} = q \frac{C - D}{D - 1} > 0,$$

give x_C raw context mass one, x_A raw mass a_{anc} , and every x_j raw mass one, and normalize. If $D = C$, omit x_A and normalize the other positive raw masses. Take $\theta = e_0$ and let $Y = \phi(X, A)^\top \theta$ deterministically. At x_C , the Gibbs normalizer is

$$qe^\eta + 1 - q = e^\eta/C.$$

Hence the target-to-reference ratios of actions 1 and 0 are respectively C and $C/e^\eta < 1$. Every action-invariant context has ratio one, so $C_P = C$. The logging and target covariance matrices are diagonal in the chosen basis. If $D = C$, their generalized ratio in direction e_0 is $C = D$. If $D < C$, it is

$$\frac{qC + a_{\text{anc}}}{q + a_{\text{anc}}} = D,$$

because $a_{\text{anc}}(D-1) = q(C-D)$. Every other generalized eigenvalue is one. Every basis direction has positive logging mass, so $\Lambda_{\text{ref}} \succ 0$, and therefore $D_P = D$. The constructed law belongs to $\mathcal{M}_{\mathfrak{E}_0}(d, C, D, \eta)$, proving nonemptiness.

Now let P be an arbitrary member of this exact shell, and write $t = e_0^\top \theta$. Nonsingularity forces positive context mass on every x_j . The context x_C also has positive mass: otherwise every positive-mass context would be action-invariant, which would imply $C_P = 1$, contrary to $C > 1$. Bounded rewards at x_C give $0 \leq t \leq 1$. At that context the action-1 likelihood ratio is

$$K(t) = \frac{e^{\eta t}}{qe^{\eta t} + 1 - q},$$

whereas the action-0 ratio is $(qe^{\eta t} + 1 - q)^{-1} \leq 1$. All ratios at all other contexts equal one. Moreover,

$$K'(t) = \frac{\eta(1-q)e^{\eta t}}{(qe^{\eta t} + 1 - q)^2} > 0, \quad K(1) = C.$$

Thus the exact equality $C_P = C$ forces $t = 1$. Consequently every shell member has the same Gibbs optimum: at x_C it is the tilt generated by the publicly fixed reward pair $(1, 0)$, and at every other context it equals π_{ref} . Denote this common, publicly computable policy by $\pi^\dagger$.

The constant learner returning $\pi^\dagger$ belongs to $\mathcal{L}_n(\mathfrak{E}_0)$. By `lem:gibbs-regret-identity`, for every shell member its regret is

$$J_{\eta, P}(\pi_P^\star) - J_{\eta, P}(\pi^\dagger) = \eta^{-1} \mathbb{E}_\rho \text{KL}(\pi^\dagger(\cdot | X) \| \pi_P^\star(\cdot | X)) = 0.$$

Regret is nonnegative, so `def:minimax-risk` gives

$$\mathfrak{R}_{n, \mathfrak{E}_0}(\mathcal{M}_{\mathfrak{E}_0}(d, C, D, \eta)) = 0$$

for every $n \in \mathbb{N}$. Since the sample-size domain begins at one, `def:sample-complexity` yields $N_{\mathfrak{E}_0}^\star(\epsilon; d, C, D, \eta) = 1$ for every $\epsilon > 0$. $\square$

Justification. An explicit shared-feature public design forces every shell member to have the same publicly computable Gibbs optimum, so a constant learner has zero regret. *Gap.* The construction is deliberately uninformative and supplies no lower bound for a specified informative design. *Consumer.* It proves that any lower bound uniform over public experiments is at most the unavoidable sample-size floor of one.

Theorem 7 (`thm:fixed-design-feasibility-and-frontier-nonuniqueness`). *For a fixed public experiment $\mathfrak{E}$, nonemptiness of $\mathcal{M}_{\mathfrak{E}}(d, C, D, \eta)$ is equivalent to the explicit finite exponential nonlinear-semidefinite feasibility system in `lem:fixed-experiment-shell-certificate`. Nevertheless, the indices (d, C, D, η) do not determine the fixed-experiment minimax frontier. For every $d \geq 4$, $\eta > 0$, and $1 < D \leq C < e^\eta$, there is a finite public experiment $\mathfrak{E}_0$ with a nonempty exact (C, D) -shell on which*

$$\mathfrak{R}_{n, \mathfrak{E}_0}(\mathcal{M}_{\mathfrak{E}_0}(d, C, D, \eta)) = 0$$

for every n , and $N_{\mathfrak{E}_0}^\star(\epsilon; d, C, D, \eta) = 1$ for every $\epsilon > 0$. There is also a finite public experiment $\mathfrak{E}_1$ at the same indices and constants $c(C, D, \eta) > 0$ and $\epsilon_0(C, D, \eta) > 0$ for which

$$N_{\mathfrak{E}_1}^\star(\epsilon; d, C, D, \eta) \geq c(C, D, \eta) \frac{dD\eta}{\epsilon} \quad (0 < \epsilon \leq \epsilon_0(C, D, \eta)).$$

These conclusions include pairs satisfying $2D - 1 \geq e^\eta$. Consequently, any sample-complexity lower bound that is uniform over all fixed public experiments with a nonempty shell at the given indices is at most 1; no nontrivial such bound can exceed the unavoidable inequality $N_{\mathfrak{E}}^\star \geq 1$, and no such bound can diverge as $\epsilon \downarrow 0$. In particular, no uniform matching $\Omega(dD\epsilon^{-2})$ branch can hold as a function only of (d, C, D, η) .

Proof. For a fixed public experiment $\mathfrak{E}$, the equivalence between nonemptiness of $\mathcal{M}_{\mathfrak{E}}(d, C, D, \eta)$ and the stated finite exponential nonlinear-semidefinite system is exactly `lem:fixed-experiment-shell-certificate`, with the unindexed notation there interpreted as the fixed-experiment abbreviation from `def:common-experiment`.

Fix $d \geq 4$, $\eta > 0$, and $1 < D \leq C < e^\eta$. By `prop:zero-risk-nontrivial-shell`, there is a finite public experiment $\mathfrak{E}_0$ with a nonempty exact shell such that

$$\mathfrak{R}_{n, \mathfrak{E}_0}(\mathcal{M}_{\mathfrak{E}_0}(d, C, D, \eta)) = 0$$

for every $n \in \mathbb{N}$, and therefore

$$N_{\mathfrak{E}_0}^*(\epsilon; d, C, D, \eta) = 1$$

for every $\epsilon > 0$. By `prop:informative-fast-shell-all-indices`, there is another finite public experiment $\mathfrak{E}_1$, at the same values of (d, C, D, η) , and positive constants $c(C, D, \eta)$ and $\epsilon_0(C, D, \eta)$ such that

$$N_{\mathfrak{E}_1}^*(\epsilon; d, C, D, \eta) \geq c(C, D, \eta) \frac{dD\eta}{\epsilon}, \quad 0 < \epsilon \leq \epsilon_0(C, D, \eta).$$

The informative proposition assumes only $1 < D \leq C < e^\eta$, so it also covers pairs satisfying $2D - 1 \geq e^\eta$. The two experiments have identical numerical indices but different sample-complexity functions: the first is identically one, while the second diverges at the displayed pointwise local rate. Hence (d, C, D, η) does not determine the fixed-experiment minimax frontier.

Finally, let $B(\epsilon; d, C, D, \eta)$ be any lower bound asserted uniformly over all fixed public experiments having a nonempty exact shell at the displayed indices. Applying it to $\mathfrak{E}_0$ gives

$$B(\epsilon; d, C, D, \eta) \leq N_{\mathfrak{E}_0}^*(\epsilon; d, C, D, \eta) = 1.$$

Because $N_{\mathfrak{E}}^* \geq 1$ follows from its definition over $n \in \mathbb{N}$, one is the unavoidable floor and the largest possible uniform index-only lower bound. In particular, no such bound can diverge as $\epsilon \downarrow 0$, and a uniform matching $\Omega(dD\epsilon^{-2})$ branch is impossible. $\square$

Justification. The theorem combines the finite feasibility certificate with zero-risk and informative experiments sharing the same four numerical indices. *Gap.* It proves index insufficiency for lower difficulty; it does not supply uniform D - or η -dependence for the informative lower constant. *Consumer.* It prevents the universal clean upper bound from being misread as an index-only matching minimax frontier.

Lemma 12 (`lem:fixed-experiment-shell-certificate`). *Fix the public experiment $\mathfrak{E}$, and write $\mathcal{A}_x^+ = \{a \in \mathcal{A} : \pi_{\text{ref}}(a \mid x) > 0\}$. For $\vartheta \in \mathbb{R}^d$, define*

$$Z_{\vartheta}(x) = \sum_{a \in \mathcal{A}} \pi_{\text{ref}}(a \mid x) e^{\eta \phi(x, a)^{\top} \vartheta}, \quad w_{\vartheta}(x, a) = \frac{e^{\eta \phi(x, a)^{\top} \vartheta}}{Z_{\vartheta}(x)},$$

$$B_x = \sum_{a \in \mathcal{A}} \pi_{\text{ref}}(a \mid x) \phi(x, a) \phi(x, a)^{\top}, \quad G_x(\vartheta) = \sum_{a \in \mathcal{A}} \pi_{\text{ref}}(a \mid x) w_{\vartheta}(x, a) \phi(x, a) \phi(x, a)^{\top}.$$

Then $\mathcal{M}(d, C, D, \eta) \neq \emptyset$ if and only if the public feature map has Euclidean norm at most one at every cell and at least one of the following finitely many feasibility problems has a solution. Enumerate a nonempty set $I \subseteq \mathcal{X}$ and a cell (x_0, a_0) with $x_0 \in I$ and $a_0 \in \mathcal{A}_{x_0}^+$. The variables are $\vartheta \in \mathbb{R}^d$, $(\rho_x)_{x \in I}$, and $u \in \mathbb{R}^d$, and the constraints are

$$0 \leq \phi(x, a)^{\top} \vartheta \leq 1 \quad ((x, a) \in S), \quad \rho_x > 0 \quad (x \in I), \quad \sum_{x \in I} \rho_x = 1,$$

$$B(\rho) := \sum_{x \in I} \rho_x B_x \succ 0,$$

$$e^{\eta \phi(x, a)^{\top} \vartheta} \leq C Z_{\vartheta}(x) \quad (x \in I \text{ and } a \in \mathcal{A}_x^+), \quad e^{\eta \phi(x_0, a_0)^{\top} \vartheta} = C Z_{\vartheta}(x_0),$$

$$G(\vartheta, \rho) := \sum_{x \in I} \rho_x G_x(\vartheta), \quad DB(\rho) - G(\vartheta, \rho) \succeq 0,$$

$$\{DB(\rho) - G(\vartheta, \rho)\}u = 0, \quad u^\top u = 1.$$

There are at most $(2^{|\mathcal{X}|} - 1)|S|$ branches, and a branch has $2d + |I|$ real variables and only displayed exponential, linear, quadratic, and semidefinite constraints.

Proof. Suppose first that $P \in \mathcal{M}(d, C, D, \eta)$. Membership implies that the public feature map has Euclidean norm at most one at every cell. Let $I = \{x \in \mathcal{X} : \rho(x) > 0\}$, put $\rho_x = \rho(x)$, and take $\vartheta = \theta$. Bounded linear realizability gives $0 \leq \phi(x, a)^\top \vartheta \leq 1$ on S . Directly from the definitions,

$$\Lambda_{\text{ref}} = \sum_{x \in I} \rho_x B_x = B(\rho), \quad \Lambda_P^* = \sum_{x \in I} \rho_x G_x(\vartheta) = G(\vartheta, \rho).$$

Thus nonsingularity gives $B(\rho) \succ 0$. Because the supports are finite, $C_P = C$ says precisely that every active-cell ratio $w_\vartheta(x, a)$ is at most C and equality holds at at least one active cell. Choosing such a cell as (x_0, a_0) gives the displayed exponential inequalities and equality.

Let

$$H = B(\rho)^{-1/2} G(\vartheta, \rho) B(\rho)^{-1/2}.$$

The equality $D_P = D$ is $\lambda_{\max}(H) = D$. Since H is symmetric, this is equivalent to $H \preceq DI_d$ and $DI_d - H$ being singular. Congruence by $B(\rho)^{1/2}$ turns the first condition into $G(\vartheta, \rho) \preceq DB(\rho)$. Singularity is equivalent to the existence of a unit vector u in the kernel of $DB(\rho) - G(\vartheta, \rho)$. Hence the branch indexed by I, x_0, a_0 is feasible.

Conversely, suppose one displayed branch is feasible. Define a law by $P(X = x) = \rho_x$ for $x \in I$, by zero context mass outside I , by $P(A = a \mid X = x) = \pi_{\text{ref}}(a \mid x)$, and by the deterministic reward

$$Y = \phi(X, A)^\top \vartheta.$$

The public feature-norm condition and the first group of constraints make this a $[0, 1]$ -valued realizable linear reward law. The covariance identities above remain valid, and $B(\rho) \succ 0$ gives nonsingular logging geometry. The active-cell inequalities and the equality at (x_0, a_0) give $C_P = C$. The semidefinite and unit-kernel constraints imply $H \preceq DI_d$ and give an eigenvalue equal to D , so $D_P = D$. Therefore the constructed law belongs to $\mathcal{M}(d, C, D, \eta)$.

There are $2^{|\mathcal{X}|} - 1$ nonempty context supports and at most $|S|$ choices of the equality-attaining cell for each support. The variables consist of d coordinates of ϑ , $|I|$ context masses, and d coordinates of u , proving the final count. $\square$

Justification. The shell equalities $C_P = C$ and $D_P = D$ become a finite union of explicit exponential and semidefinite feasibility systems once $\mathfrak{E}$ is fixed. *Gap.* The certificate characterizes feasibility but does not by itself determine statistical difficulty or provide a decision algorithm. *Consumer.* It is the explicit fixed-design feasibility component of the headline theorem.

Theorem 8 (thm:clean-temperature-feature-upper). *Fix $c_0 = 64$. For every positive integer n and $g, f \in \mathcal{F}$, let π_g be the Gibbs policy generated by g , and define*

$$H_{g,f}(x) = \sum_{a \in \mathcal{A}} \pi_g(a \mid x) \{g(x, a) - f(x, a)\}^2, \quad \widehat{U}_n(g) = \sup_{f \in \mathcal{Q}_n} \frac{1}{n} \sum_{i=1}^n H_{g,f}(X_i).$$

Let $\tilde{g}_n$ be the sequentially lexicographically selected member of $\arg \min_{g \in \mathcal{Q}_n} \widehat{U}_n(g)$, and let $\hat{\pi}_{\text{stab}} = \pi_{\tilde{g}_n}$. This is a measurable member of $\mathcal{L}_n(\mathfrak{E})$, uses only $(\mathcal{D}_n, \mathfrak{E})$, and uses neither ρ , C , nor D as input. Put

$$a_n = d \log(en) + \log(2n^2),$$

$$b_n = \frac{8}{3n} [\log(4n^2) + 2d \log\{1 + 4n(\eta + 2)\}] + \frac{3}{n}, \quad V_n = D \left(3000 \frac{a_n}{n} + 3b_n \right).$$

For every nonempty exact shell and every positive integer n ,

$$\sup_{P \in \mathcal{M}_{\mathfrak{E}}(d, C, D, \eta)} \mathbb{E}_{P^{\otimes n}} [J_{\eta, P}(\pi_P^*) - J_{\eta, P}(\hat{\pi}_{\text{stab}})] \leq \min \left\{ \eta V_n, \sqrt{2V_n} \right\} + \delta_n \min \{2, \eta/2\}.$$

Moreover, for every $\epsilon \in (0, 1)$, the following explicit inversion holds. If $\eta \leq 2\epsilon$, then

$$N_{\mathfrak{E}}^*(\epsilon; d, C, D, \eta) = 1.$$

If $\eta > 2\epsilon$, define

$$q_{\epsilon} = dD \min\{\eta/\epsilon, \epsilon^{-2}\}, \quad \ell_{\epsilon} = 1 + \log(eq_{\epsilon}) + \log\{1 + (1 + \eta)q_{\epsilon}\},$$

$$n_0 = \lceil 10^8 q_{\epsilon} \ell_{\epsilon} \rceil.$$

Then

$$\sup_{P \in \mathcal{M}_{\mathfrak{E}}(d, C, D, \eta)} \mathbb{E}_{P^{\otimes n_0}} [J_{\eta, P}(\pi_P^*) - J_{\eta, P}(\hat{\pi}_{\text{stab}})] \leq \epsilon, \quad N_{\mathfrak{E}}^*(\epsilon; d, C, D, \eta) \leq n_0.$$

Consequently, with universal constants and whenever the displayed sample-size order is at least two,

$$N_{\mathfrak{E}}^*(\epsilon; d, C, D, \eta) \leq \tilde{O}(dD \min\{\eta/\epsilon, \epsilon^{-2}\}).$$

Thus the clean feature-coverage upper bound holds at every temperature; pointwise coverage C is not needed by the learner or by this upper bound.

Proof. Fix a positive integer n . We first verify that the stated learner is measurable. Regard $\mathcal{F}$ as a compact subset of the finite-dimensional Euclidean space $\mathbb{R}^S$, with coordinates ordered as in `def:prediction-polytope-learner`, and let z denote a realized sample. The map $(z, f) \mapsto L_n(z, f)$ is Borel measurable in z and continuous in f . Applying `lem:measurable-maximum-theorem` to the constant correspondence $z \mapsto \mathcal{F}$ and the objective $-L_n$ shows that the empirical-risk-minimizer correspondence is measurable, nonempty, and compact valued. Recursively minimizing its first coordinate, then its second coordinate, and so on applies the same lemma at each of finitely many steps. The final correspondence is a singleton, so $z \mapsto \hat{f}_z$ is Borel measurable.

Put

$$\Gamma(z) = \mathcal{Q}_n(z) = \left\{ f \in \mathcal{F} : \frac{1}{n} \sum_{i=1}^n \{f(X_i, A_i) - \hat{f}_z(X_i, A_i)\}^2 \leq 64 \frac{a_n}{n} \right\}.$$

This correspondence is nonempty and compact valued. It is weakly measurable: for every open $O \subseteq \mathcal{F}$, compact metrizability gives compact sets $K_m \subset O$ with $O = \bigcup_m K_m$, and $\{z : \Gamma(z) \cap O \neq \emptyset\}$ is the union over m of the measurable sets

$$\left\{ z : \min_{f \in K_m} \frac{1}{n} \sum_{i=1}^n \{f(X_i, A_i) - \hat{f}_z(X_i, A_i)\}^2 \leq 64 \frac{a_n}{n} \right\}.$$

Measurability of each displayed minimum follows from `lem:measurable-maximum-theorem`.

For fixed z , the map $(g, f) \mapsto P_n H_{g, f}$ is continuous on $\mathcal{F}^2$, since the action support is finite and every Gibbs normalizer is strictly positive. The measurable maximum theorem, with base point (z, g) , correspondence $\Gamma(z)$, and objective $P_n H_{g, f}$, shows that

$$(z, g) \mapsto \hat{U}_n(z, g) = \max_{f \in \Gamma(z)} P_n H_{g, f}$$

is Borel measurable. For fixed z , uniform continuity on the compact set $\mathcal{F}^2$ makes this supremum continuous in g . Applying the same theorem to $-\hat{U}_n$ over $g \in \Gamma(z)$, followed by recursive coordinate minimization, proves that the stipulated lexicographic selector $z \mapsto \tilde{g}_n(z)$ is Borel measurable. The finite Gibbs formula is continuous in its reward-vector input. Hence $\hat{\pi}_{\text{stab}} = \pi_{\tilde{g}_n}$ is measurable and belongs to $\mathcal{L}_n(\mathfrak{E})$. Its construction uses only $(\mathcal{D}_n, \mathfrak{E})$, and therefore uses neither ρ , C , nor D .

Fix $P \in \mathcal{M}_{\mathfrak{E}}(d, C, D, \eta)$, write $r = r_\theta$, and set

$$R_n = 600 \frac{a_n}{n}.$$

By lem:bounded-linear-localization, there is an event $\mathcal{E}_n^{\text{loc}}$ of probability at least $1 - \delta_n/2$ on which

$$r \in \mathcal{Q}_n, \quad \sup_{f \in \mathcal{Q}_n} \mathbb{E}_{\rho, \pi_{\text{ref}}}(f - r)^2 \leq R_n.$$

Apply lem:uniform-gibbs-radius-comparison with $\zeta = \delta_n/2$. Because $\delta_n = n^{-2}$, there is an event $\mathcal{E}_n^{\text{cmp}}$, of probability at least $1 - \delta_n/2$, on which, simultaneously for every $g, f \in \mathcal{F}$,

$$P_\rho H_{g,f} \leq 2P_n H_{g,f} + b_n, \quad P_n H_{g,f} \leq 2P_\rho H_{g,f} + b_n.$$

The comparison is simultaneous over the deterministic class $\mathcal{F}^2$, so it remains valid for the data-dependent set $\mathcal{Q}_n$ and selector $\tilde{g}_n$.

Work on $\mathcal{E}_n = \mathcal{E}_n^{\text{loc}} \cap \mathcal{E}_n^{\text{cmp}}$, which has probability at least $1 - \delta_n$. For every $f \in \mathcal{Q}_n$, the contrast $f - r$ belongs to $\text{span}(\phi)$. By lem:feature-coverage-domination and $D_P = D$,

$$P_\rho H_{r,f} = \mathbb{E}_{\rho, \pi_P^*}(r - f)^2 \leq D \mathbb{E}_{\rho, \pi_{\text{ref}}}(r - f)^2 \leq DR_n.$$

Since $r \in \mathcal{Q}_n$, the empirical comparison gives

$$\hat{U}_n(r) \leq 2 \sup_{f \in \mathcal{Q}_n} P_\rho H_{r,f} + b_n \leq 2DR_n + b_n.$$

The defining minimality of $\tilde{g}_n$ yields $\hat{U}_n(\tilde{g}_n) \leq \hat{U}_n(r)$. Again using $r \in \mathcal{Q}_n$ and the reverse comparison,

$$\begin{aligned} \mathbb{E}_{\rho, \pi_{\tilde{g}_n}}(\tilde{g}_n - r)^2 &= P_\rho H_{\tilde{g}_n, r} \\ &\leq 2P_n H_{\tilde{g}_n, r} + b_n \\ &\leq 2\hat{U}_n(\tilde{g}_n) + b_n \\ &\leq 4DR_n + 3b_n. \end{aligned}$$

At the true Gibbs endpoint, feature domination and localization also give

$$\mathbb{E}_{\rho, \pi_P^*}(\tilde{g}_n - r)^2 \leq D \mathbb{E}_{\rho, \pi_{\text{ref}}}(\tilde{g}_n - r)^2 \leq DR_n.$$

Define

$$A_n = \mathbb{E}_{\rho, \pi_P^*}(\tilde{g}_n - r)^2, \quad B_n = \mathbb{E}_{\rho, \pi_{\tilde{g}_n}}(\tilde{g}_n - r)^2.$$

Every nonempty shell has $D \geq 1$ by thm:feasible-index-region. Therefore

$$A_n + B_n \leq 5DR_n + 3b_n \leq D(5R_n + 3b_n) = D\left(3000 \frac{a_n}{n} + 3b_n\right) = V_n.$$

Apply `lem:two-endpoint-gibbs-conversion` with $g = \tilde{g}_n$. On $\mathcal{E}_n$,

$$\begin{aligned} J_{\eta,P}(\pi_P^\star) - J_{\eta,P}(\hat{\pi}_{\text{stab}}) &\leq \min\{\eta(A_n + B_n), \sqrt{A_n} + \sqrt{B_n}\} \\ &\leq \min\{\eta V_n, \sqrt{2V_n}\}, \end{aligned}$$

where $(\sqrt{A_n} + \sqrt{B_n})^2 \leq 2(A_n + B_n)$.

We also need a deterministic bound off $\mathcal{E}_n$. For arbitrary $r, g : S \rightarrow [0, 1]$, put $\Delta = g - r$, let π_t be the Gibbs policy generated by $r + t\Delta$, and define

$$F(t) = \eta^{-1} \mathbb{E}_\rho \log \sum_a \pi_{\text{ref}}(a \mid X) e^{\eta(r+t\Delta)(X,a)}.$$

Finite support permits differentiation, giving

$$F''(t) = \eta \mathbb{E}_\rho \text{Var}_{\pi_t}(\Delta).$$

The Gibbs variational identity and integration by parts yield

$$J_{\eta,r}(\pi_r) - J_{\eta,r}(\pi_g) = F'(1) - \{F(1) - F(0)\} = \eta \int_0^1 t \mathbb{E}_\rho \text{Var}_{\pi_t}(\Delta) dt \leq \eta/2,$$

because $\Delta \in [-1, 1]$ implies $\text{Var}_{\pi_t}(\Delta) \leq 1$. The slow part of `lem:two-endpoint-gibbs-conversion`, together with $|\Delta| \leq 1$, also bounds the regret by two. Averaging the good- and bad-event bounds and using $P^{\otimes n}(\mathcal{E}_n^c) \leq \delta_n$ proves, uniformly over the exact shell,

$$\mathbb{E}_{P^{\otimes n}} [J_{\eta,P}(\pi_P^\star) - J_{\eta,P}(\hat{\pi}_{\text{stab}})] \leq \min\{\eta V_n, \sqrt{2V_n}\} + \delta_n \min\{2, \eta/2\}.$$

Taking the supremum over P proves the finite-sample assertion.

Fix $\epsilon \in (0, 1)$. We now invert the bound. For $d \geq 4$ and $n \geq 1$, elementary logarithmic inequalities give

$$a_n \leq 2d \log(en), \quad \log(4n^2) \leq d \log(en),$$

$$\log\{1 + 4n(\eta + 2)\} \leq \log 8 + \log\{1 + n(1 + \eta)\}, \quad 9 \leq 3d \log(en).$$

Substitution into b_n and V_n gives

$$V_n \leq 6100 \frac{dD}{n} [\log(en) + \log\{1 + n(1 + \eta)\}]. \quad (1)$$

If $\eta \leq 2\epsilon$, the deterministic regret bound $\eta/2 \leq \epsilon$ holds already at $n = 1$. Hence $N_{\mathfrak{E}}^*(\epsilon; d, C, D, \eta) = 1$.

Suppose $\eta > 2\epsilon$, and define

$$q_\epsilon = dD \min\{\eta/\epsilon, \epsilon^{-2}\}, \quad \ell_\epsilon = 1 + \log(eq_\epsilon) + \log\{1 + (1 + \eta)q_\epsilon\}.$$

Since $D \geq 1$, $d \geq 4$, $\epsilon < 1$, and $\eta > 2\epsilon$, one has $q_\epsilon > 4$ and $\ell_\epsilon > 1$. Set $K_0 = 10^8$ and

$$n_0 = \lceil K_0 q_\epsilon \ell_\epsilon \rceil.$$

Then

$$K_0 q_\epsilon \ell_\epsilon \leq n_0 \leq 2K_0 q_\epsilon \ell_\epsilon.$$

Writing $s = (1 + \eta)q_\epsilon$, using $s \geq 1$, $\log q_\epsilon \leq \ell_\epsilon$, and $\log \ell_\epsilon \leq \ell_\epsilon$, we obtain

$$\log(en_0) \leq \{\log(2eK_0) + 2\}\ell_\epsilon,$$

$$\log\{1 + n_0(1 + \eta)\} \leq \{\log(3K_0) + 2\}\ell_\epsilon.$$

Consequently

$$\log(en_0) + \log\{1 + n_0(1 + \eta)\} \leq c_{K_0} \ell_\epsilon, \quad c_{K_0} = 4 + \log(6eK_0^2) < 44.$$

Applying (1) at n_0 gives

$$V_{n_0} \leq \alpha \frac{dD}{q_\epsilon}, \quad \alpha = \frac{6100c_{K_0}}{K_0} < \frac{1}{8}.$$

If the minimum defining q_ϵ is η/ϵ , then $\eta V_{n_0} \leq \alpha\epsilon \leq \epsilon/8$. If it is ϵ^{-2} , then $\sqrt{2V_{n_0}} \leq \sqrt{2\alpha}\epsilon \leq \epsilon/2$. Thus the good-event term is at most $\epsilon/2$ in either case.

The complement term is also smaller than $\epsilon/2$. In the first case, $n_0 \geq q_\epsilon \geq 4\eta/\epsilon$, so

$$n_0^{-2} \min\{2, \eta/2\} \leq \frac{\epsilon^2}{32\eta} < \frac{\epsilon}{64}.$$

In the second case, $n_0 \geq q_\epsilon \geq 4\epsilon^{-2}$, so

$$n_0^{-2} \min\{2, \eta/2\} \leq \frac{\epsilon^4}{8} \leq \frac{\epsilon}{8}.$$

Therefore the worst-case expected regret of $\hat{\pi}_{\text{stab}}$ at n_0 is at most ϵ . By def:minimax-risk and def:sample-complexity,

$$N_{\mathfrak{E}}^*(\epsilon; d, C, D, \eta) \leq n_0 = \lceil 10^8 q_\epsilon [1 + \log(eq_\epsilon) + \log\{1 + (1 + \eta)q_\epsilon\}] \rceil.$$

The displayed logarithms are exactly the factors suppressed by $\tilde{O}$, proving

$$N_{\mathfrak{E}}^*(\epsilon; d, C, D, \eta) \leq \tilde{O}(dD \min\{\eta/\epsilon, \epsilon^{-2}\}).$$

Neither the learner nor any bound in the argument uses C , which completes the proof. $\square$

Justification. An explicit measurable idealized global-optimization learner stabilizes the Gibbs-weighted confidence-set radius and attains the clean dD upper order at every temperature. *Gap.* The selector is defined by exact global optimization and measurable selection; no finite-precision algorithm or approximation-regret theorem is supplied. *Consumer.* It proves that pointwise coverage C is absent from the universal upper frontier.

Lemma 13 (lem:two-endpoint-gibbs-conversion). *Let $r, g : S \rightarrow [0, 1]$, put $\Delta = g - r$, and let π_r and π_g be their Gibbs policies relative to π_{ref} . Define*

$$A = \mathbb{E}_{\rho, \pi_r} \Delta^2, \quad B = \mathbb{E}_{\rho, \pi_g} \Delta^2.$$

Then

$$J_{\eta, r}(\pi_r) - J_{\eta, r}(\pi_g) \leq \min\{\eta(A + B), \sqrt{A} + \sqrt{B}\}.$$

Proof. Fix $x \in \mathcal{X}$, and restrict all action distributions to the nonempty support $\mathcal{A}_x^+ = \{a : \pi_{\text{ref}}(a | x) > 0\}$. Write $p = \pi_r(\cdot | x)$, $q = \pi_g(\cdot | x)$, and $\Delta(a) = g(x, a) - r(x, a)$. Both p and q are strictly positive on $\mathcal{A}_x^+$. Their Gibbs formulas imply

$$\frac{q(a)}{p(a)} = \frac{e^{\eta\Delta(a)}}{\mathbb{E}_p e^{\eta\Delta}}, \quad \log \frac{q(a)}{p(a)} = \eta\{\Delta(a) - c\}, \quad c = \eta^{-1} \log \mathbb{E}_p e^{\eta\Delta}.$$

Let $m_p = \mathbb{E}_p \Delta$ and $m_q = \mathbb{E}_q \Delta$. For $\psi(t) = \log \mathbb{E}_p e^{t\Delta}$, finite support makes ψ differentiable and convex, and exponential tilting gives $\psi'(0) = m_p$ and $\psi'(\eta) = m_q$. Hence the secant-slope inequalities for a differentiable convex function give

$$m_p = \psi'(0) \leq \frac{\psi(\eta) - \psi(0)}{\eta} = c \leq \psi'(\eta) = m_q.$$

Substitution of the Gibbs formula for p into conditional regularized welfare gives

$$J_{\eta, r, x}(p) - J_{\eta, r, x}(q) = \eta^{-1} \text{KL}(q \| p) = m_q - c.$$

Therefore

$$J_{\eta, r, x}(p) - J_{\eta, r, x}(q) \leq m_q - m_p \leq |m_q| + |m_p| \leq \sqrt{\mathbb{E}_q \Delta^2} + \sqrt{\mathbb{E}_p \Delta^2}.$$

This is the conditional slow inequality.

For the fast inequality, first observe that for all $u, v > 0$,

$$(u - v) \log(u/v) \leq \frac{u + v}{2} \{\log(u/v)\}^2.$$

Indeed, on writing $u/v = e^t$, the inequality reduces to $2 \tanh(|t|/2) \leq |t|$; the difference between the right and left sides vanishes at zero and has derivative $\tanh^2(s/2) \geq 0$ at $s = |t|$. Apply this inequality action by action with $u = q(a)$ and $v = p(a)$, and use the displayed likelihood-ratio formula. It follows that the conditional Jeffreys divergence satisfies

$$\text{KL}(q\|p) + \text{KL}(p\|q) \leq \frac{\eta^2}{2} \left\{ \mathbb{E}_p(\Delta - c)^2 + \mathbb{E}_q(\Delta - c)^2 \right\}.$$

Define

$$S(t) = \mathbb{E}_p(\Delta - t)^2 + \mathbb{E}_q(\Delta - t)^2.$$

The function S is convex. Since $c \in [m_p, m_q]$, convexity implies $S(c) \leq \max\{S(m_p), S(m_q)\}$. Direct expansion at either endpoint gives the same value,

$$S(m_p) = S(m_q) = \mathbb{E}_p \Delta^2 + \mathbb{E}_q \Delta^2 - 2m_p m_q.$$

The elementary inequality $-2m_p m_q \leq m_p^2 + m_q^2$, followed by Jensen's inequality under p and q , therefore yields

$$S(c) \leq 2 \left\{ \mathbb{E}_p \Delta^2 + \mathbb{E}_q \Delta^2 \right\}.$$

Since $\text{KL}(p\|q) \geq 0$, the preceding two displays and the conditional Gibbs identity show

$$J_{\eta,r,x}(p) - J_{\eta,r,x}(q) \leq \eta \left\{ \mathbb{E}_p \Delta^2 + \mathbb{E}_q \Delta^2 \right\}.$$

Finally average both conditional inequalities over $X \sim \rho$. The fast inequality becomes

$$J_{\eta,r}(\pi_r) - J_{\eta,r}(\pi_g) \leq \eta(A + B).$$

For the slow inequality, Cauchy-Schwarz gives

$$\mathbb{E}_\rho \sqrt{\mathbb{E}_{\pi_r}(\cdot|X) \Delta^2} \leq \sqrt{A}, \quad \mathbb{E}_\rho \sqrt{\mathbb{E}_{\pi_g}(\cdot|X) \Delta^2} \leq \sqrt{B}.$$

Thus $J_{\eta,r}(\pi_r) - J_{\eta,r}(\pi_g) \leq \sqrt{A} + \sqrt{B}$ as well. Taking the smaller of the two bounds proves the proposed sharpened statement. $\square$

Lemma 14 (lem:uniform-gibbs-radius-comparison). *For $g, f \in \mathcal{F}$, let*

$$H_{g,f}(x) = \sum_{a \in \mathcal{A}} \pi_g(a | x) \{g(x, a) - f(x, a)\}^2,$$

where π_g is the Gibbs policy generated by g . Let $X_1, \dots, X_n$ be independent with law ρ , write $P_\rho h = \mathbb{E}_\rho h(X)$ and $P_n h = n^{-1} \sum_{i=1}^n h(X_i)$, and fix $0 < \zeta < 1$. With probability at least $1 - \zeta$, simultaneously for every $g, f \in \mathcal{F}$,

$$P_\rho H_{g,f} \leq 2P_n H_{g,f} + b_n(\zeta), \quad P_n H_{g,f} \leq 2P_\rho H_{g,f} + b_n(\zeta),$$

where

$$b_n(\zeta) = \frac{8}{3n} \left[\log \frac{2}{\zeta} + 2d \log\{1 + 4n(\eta + 2)\} \right] + \frac{3}{n}.$$

Proof. Let $V = \text{span}(\mathcal{F})$, viewed as a linear subspace of $\mathbb{R}^S$ with the supremum norm, and put $r = \dim(V) \leq d$. Since every member of $\mathcal{F}$ takes values in $[0, 1]$, $\mathcal{F}$ lies in the unit ball of V . For $0 < v < 1$, a maximal v -separated subset of $\mathcal{F}$ is a v -net. The open supremum-norm balls of radius $v/2$ centered at its elements are disjoint and are contained in the ball of radius $1 + v/2$. Comparing r -dimensional Lebesgue volumes on V gives

$$N_{\mathcal{F}}(v) \leq \left(1 + \frac{2}{v}\right)^r \leq \left(1 + \frac{2}{v}\right)^d.$$

We next control the dependence of $H_{g,f}$ on its two arguments. Suppose $\|g - g'\|_{\infty} \leq v$, and define $g_t = (1-t)g + tg'$. Differentiation of the finite Gibbs formula yields, for every $x \in \mathcal{X}$ and $a \in \mathcal{A}$,

$$\frac{d}{dt} \pi_{g_t}(a \mid x) = \eta \pi_{g_t}(a \mid x) \left[(g' - g)(x, a) - \sum_{b \in \mathcal{A}} \pi_{g_t}(b \mid x) (g' - g)(x, b) \right].$$

Therefore

$$\sum_{a \in \mathcal{A}} \left| \frac{d}{dt} \pi_{g_t}(a \mid x) \right| \leq 2\eta v,$$

and integration over $t \in [0, 1]$ gives

$$\|\pi_g(\cdot \mid x) - \pi_{g'}(\cdot \mid x)\|_1 \leq 2\eta v.$$

If also $\|f - f'\|_{\infty} \leq v$, then $g - f$ and $g' - f'$ take values in $[-1, 1]$, whence

$$|(g - f)^2 - (g' - f')^2| \leq 2|(g - f) - (g' - f')| \leq 4v$$

pointwise. Splitting the difference between the Gibbs weights and the squared contrasts consequently shows

$$\|H_{g,f} - H_{g',f'}\|_{\infty} \leq (2\eta + 4)v.$$

Choose $v = \{n(2\eta + 4)\}^{-1}$. The Cartesian square of a v -net of $\mathcal{F}$ induces a $1/n$ -net for $\{H_{g,f} : g, f \in \mathcal{F}\}$ whose cardinality M satisfies

$$M \leq \{1 + 4n(\eta + 2)\}^{2d}.$$

It remains to prove and apply a fixed-function deviation inequality. Let $W_1, \dots, W_n$ be independent random variables in $[0, 1]$ with common mean μ . For $|\lambda| < 3$ and $|w| \leq 1$, expansion of the exponential and the inequality $k! \geq 2 \cdot 3^{k-2}$ for $k \geq 2$ give

$$e^{\lambda w} \leq 1 + \lambda w + \frac{\lambda^2 w^2}{2(1 - |\lambda|/3)}.$$

Apply this inequality to $w = W_i - \mu$. Since $\mathbb{E}(W_i - \mu) = 0$ and $\mathbb{E}(W_i - \mu)^2 \leq \mu$, independence and $1 + s \leq e^s$ imply the corresponding Bernstein moment-generating-function bounds for both signs. Exponential Markov's inequality, with the optimizing parameter for variance factor $n\mu$ and scale $1/3$, then gives, for every $t > 0$,

$$\Pr \left\{ |P_n W - \mu| > \sqrt{\frac{2\mu t}{n}} + \frac{t}{3n} \right\} \leq 2e^{-t}.$$

The elementary inequality $\sqrt{2\mu t/n} \leq \mu/2 + t/n$ shows that, outside this single two-sided event,

$$\mu \leq 2P_n W + \frac{8t}{3n}, \quad P_n W \leq 2\mu + \frac{8t}{3n}.$$

Apply this event to each of the M net functions and set

$$t = \log \frac{2M}{\zeta}.$$

For each fixed net function the event containing both comparisons fails with probability at most $2e^{-t}$. Hence the union bound gives all comparisons on the net simultaneously with probability at least $1 - 2Me^{-t} = 1 - \zeta$. For an

arbitrary $H_{g,f}$, choose a net function H_0 with $\|H_{g,f} - H_0\|_\infty \leq 1/n$. Replacing both its population and empirical means by those of $H_{g,f}$ costs at most $3/n$ in either comparison. Finally,

$$t \leq \log \frac{2}{\zeta} + 2d \log\{1 + 4n(\eta + 2)\}.$$

Thus, on the same event and simultaneously for every $g, f \in \mathcal{F}$,

$$\begin{aligned} P_\rho H_{g,f} &\leq 2P_n H_{g,f} + \frac{8}{3n} \left[\log \frac{2}{\zeta} + 2d \log\{1 + 4n(\eta + 2)\} \right] + \frac{3}{n}, \\ P_n H_{g,f} &\leq 2P_\rho H_{g,f} + \frac{8}{3n} \left[\log \frac{2}{\zeta} + 2d \log\{1 + 4n(\eta + 2)\} \right] + \frac{3}{n}. \end{aligned}$$

This is the proposed sharpened statement. $\square$

Theorem 9 (thm:fixed-design-index-insufficiency). *Fix $d \geq 4$, $\eta > 0$, and a public experiment $\mathfrak{E} = (\mathcal{X}, \mathcal{A}, \phi, \pi_{\text{ref}}, \eta)$, where the context law ρ remains law-specific. For every $1 < D \leq C < e^\eta$, nonemptiness of $\mathcal{M}_{\mathfrak{E}}(d, C, D, \eta)$ is equivalent to the explicit finite exponential nonlinear-semidefinite system in lem:fixed-experiment-shell-certificate. Whenever that shell is nonempty, the explicit measurable idealized global-optimization learner of thm:clean-temperature-feature-upper uses only $(\mathcal{D}_n, \mathfrak{E})$, not ρ , C , or D . For every $0 < \epsilon < 1$, the following inversion holds. If $\eta \leq 2\epsilon$, then $N_{\mathfrak{E}}^*(\epsilon; d, C, D, \eta) = 1$. If $\eta > 2\epsilon$, put*

$$q_\epsilon = dD \min\{\eta/\epsilon, \epsilon^{-2}\}, \quad \ell_\epsilon = 1 + \log(eq_\epsilon) + \log\{1 + (1 + \eta)q_\epsilon\}.$$

Then

$$N_{\mathfrak{E}}^*(\epsilon; d, C, D, \eta) \leq \lceil 10^8 q_\epsilon \ell_\epsilon \rceil = \tilde{O}(dD \min\{\eta/\epsilon, \epsilon^{-2}\}).$$

On a nonempty boundary shell, $N_{\mathfrak{E}}^*(\epsilon; d, 1, 1, \eta) = 1$ for every $\epsilon > 0$. The tuple (d, C, D, η) does not determine the fixed-design minimax frontier: for every $1 < D \leq C < e^\eta$, there are two finite public experiments $\mathfrak{E}_0$ and $\mathfrak{E}_1$ having nonempty exact shells at those same indices such that

$$N_{\mathfrak{E}_0}^*(\epsilon; d, C, D, \eta) = 1 \quad (\epsilon > 0),$$

whereas, for constants $c(C, D, \eta) > 0$ and $\epsilon_0(C, D, \eta) > 0$,

$$N_{\mathfrak{E}_1}^*(\epsilon; d, C, D, \eta) \geq c(C, D, \eta) \frac{dD\eta}{\epsilon} \quad \text{for } 0 < \epsilon \leq \epsilon_0(C, D, \eta).$$

The informative construction applies without the condition $2D - 1 < e^\eta$, but its constants are pointwise in (C, D, η) ; the invariant lower-rate content is $\Omega_{C,D,\eta}(d/\epsilon)$, not uniform dependence on D or η . Consequently, uniformly over public experiments, no lower bound based only on (d, C, D, η) can exceed one or diverge as $\epsilon \downarrow 0$. For every specified finite experiment with a nonempty exact shell and fixed (d, C, D, η) ,

$$\lim_{\epsilon \downarrow 0} \frac{\epsilon^2}{d} N_{\mathfrak{E}}^*(\epsilon; d, C, D, \eta) = 0,$$

so no positive d/ϵ^2 , or fixed- D dD/ϵ^2 , lower branch can hold for all sufficiently small ϵ .

Proof. Fix $d \geq 4$, $\eta > 0$, and the finite public experiment $\mathfrak{E}$. By def:common-experiment, the supports, feature map, reference policy, and inverse temperature are public, while ρ remains law-specific. By lem:fixed-experiment-shell-certificate, for every $1 < D \leq C < e^\eta$, the shell $\mathcal{M}_{\mathfrak{E}}(d, C, D, \eta)$ is nonempty if and only if at least one branch of the displayed finite exponential nonlinear-semidefinite system is feasible.

Suppose such a nontrivial shell is nonempty. The learner of thm:clean-temperature-feature-upper is a measurable element of $\mathcal{L}_n(\mathfrak{E})$ and uses only $(\mathcal{D}_n, \mathfrak{E})$, not ρ , C , or D . Fix $0 < \epsilon < 1$. If $\eta \leq 2\epsilon$, that theorem gives

$$N_{\mathfrak{E}}^*(\epsilon; d, C, D, \eta) = 1.$$

If $\eta > 2\epsilon$, put

$$q_\epsilon = dD \min\{\eta/\epsilon, \epsilon^{-2}\}, \quad \ell_\epsilon = 1 + \log(eq_\epsilon) + \log\{1 + (1 + \eta)q_\epsilon\}.$$

Its explicit inversion gives risk at most ϵ at

$$n_0 = \lceil 10^8 q_\epsilon \ell_\epsilon \rceil,$$

and therefore, by def:minimax-risk and def:sample-complexity,

$$N_{\mathfrak{E}}^*(\epsilon; d, C, D, \eta) \leq n_0 = \tilde{O}(dD \min\{\eta/\epsilon, \epsilon^{-2}\}).$$

This completes the clean inversion for every $0 < \epsilon < 1$. Independently of that restriction, on a nonempty boundary shell thm:zero-risk-boundary gives

$$N_{\mathfrak{E}}^*(\epsilon; d, 1, 1, \eta) = 1$$

for every $\epsilon > 0$.

Now fix $1 < D \leq C < e^\eta$. By thm:fixed-design-feasibility-and-frontier-nonuniqueness, there is a finite public experiment $\mathfrak{E}_0$ with a nonempty exact shell and

$$N_{\mathfrak{E}_0}^*(\epsilon; d, C, D, \eta) = 1 \quad (\epsilon > 0).$$

At the same indices, prop:informative-fast-shell-all-indices supplies a finite public experiment $\mathfrak{E}_1$ and positive constants $c(C, D, \eta)$ and $\epsilon_0(C, D, \eta)$ such that

$$N_{\mathfrak{E}_1}^*(\epsilon; d, C, D, \eta) \geq c(C, D, \eta) \frac{dD\eta}{\epsilon}, \quad 0 < \epsilon \leq \epsilon_0(C, D, \eta).$$

The informative proposition assumes only $1 < D \leq C < e^\eta$, so it includes pairs satisfying $2D - 1 \geq e^\eta$. Since its constant may depend on (C, D, η) , its parameter-invariant content is the pointwise local rate $\Omega_{C,D,\eta}(d/\epsilon)$, rather than uniform lower dependence on D or η . The two experiments prove that (d, C, D, η) does not determine fixed-experiment sample complexity.

If $L(\epsilon; d, C, D, \eta)$ is any lower bound asserted uniformly over all public experiments with a nonempty exact shell at the displayed indices, applying it to $\mathfrak{E}_0$ forces

$$L(\epsilon; d, C, D, \eta) \leq N_{\mathfrak{E}_0}^*(\epsilon; d, C, D, \eta) = 1.$$

Thus no such index-only bound can exceed one or diverge as $\epsilon \downarrow 0$.

Finally, fix a specified finite experiment with a nonempty exact shell and hold (d, C, D, η) fixed. The boundary case is immediate because its sample complexity equals one. In the nontrivial case, for all sufficiently small $\epsilon > 0$, one has simultaneously $0 < \epsilon < 1$, $\eta > 2\epsilon$, and $\eta\epsilon < 1$. Hence the valid clean inversion applies and

$$q_\epsilon = dD\eta/\epsilon, \quad \ell_\epsilon = O_{d,D,\eta}(\log(1/\epsilon)).$$

It follows that

$$N_{\mathfrak{E}}^*(\epsilon; d, C, D, \eta) = O_{d,D,\eta}(\epsilon^{-1} \log(1/\epsilon)).$$

Consequently

$$0 \leq \frac{\epsilon^2}{d} N_{\mathfrak{E}}^*(\epsilon; d, C, D, \eta) = O_{d,D,\eta}(\epsilon \log(1/\epsilon)) \longrightarrow 0.$$

A positive d/ϵ^2 lower bound, or a positive dD/ϵ^2 lower bound with fixed D , therefore cannot hold for every sufficiently small ϵ . $\square$

Justification. This theorem synchronizes fixed-design feasibility and nonuniqueness with the exact all-temperature stabilized upper inversion. *Gap.* The clean upper is universal, whereas its informative lower comparison is only pointwise and local in d/ϵ . *Consumer.* It records why the public experiment remains essential even though C is statistically irrelevant for universal achievability.

Theorem 10 (thm:full-feasible-frontier-answer). *Fix $d \geq 4$, $\eta > 0$, and a public experiment $\mathfrak{E}$. Nonemptiness of $\mathcal{M}_{\mathfrak{E}}(d, C, D, \eta)$ is equivalent to feasibility of one of the finitely many explicit exponential nonlinear-semidefinite systems displayed in lem:fixed-experiment-shell-certificate. If the boundary shell $\mathcal{M}_{\mathfrak{E}}(d, 1, 1, \eta)$ is nonempty, then $N_{\mathfrak{E}}^*(\epsilon; d, 1, 1, \eta) = 1$ for every $\epsilon > 0$. Whenever a nontrivial fixed-design shell is nonempty, the explicit measurable idealized global-optimization learner of thm:clean-temperature-feature-upper uses only $(\mathcal{D}_n, \mathfrak{E})$, not ρ , C , or D . For every $0 < \epsilon < 1$, the following inversion holds. If $\eta \leq 2\epsilon$, then $N_{\mathfrak{E}}^*(\epsilon; d, C, D, \eta) = 1$. If $\eta > 2\epsilon$, define*

$$q_{\epsilon} = dD \min\{\eta/\epsilon, \epsilon^{-2}\}, \quad \ell_{\epsilon} = 1 + \log(eq_{\epsilon}) + \log\{1 + (1 + \eta)q_{\epsilon}\},$$

$$n_0 = \lceil 10^8 q_{\epsilon} \ell_{\epsilon} \rceil.$$

Then the learner has worst-case expected regularized regret at most ϵ at sample size n_0 , and

$$N_{\mathfrak{E}}^*(\epsilon; d, C, D, \eta) \leq n_0 = \tilde{O}(dD \min\{\eta/\epsilon, \epsilon^{-2}\}).$$

The tuple (d, C, D, η) does not, however, determine the fixed-design minimax frontier. For every $1 < D \leq C < e^{\eta}$, there are finite public experiments $\mathfrak{E}_0$ and $\mathfrak{E}_1$ with nonempty exact shells at the same indices such that

$$N_{\mathfrak{E}_0}^*(\epsilon; d, C, D, \eta) = 1 \quad (\epsilon > 0),$$

whereas, for constants $c(C, D, \eta) > 0$ and $\epsilon_0(C, D, \eta) > 0$,

$$N_{\mathfrak{E}_1}^*(\epsilon; d, C, D, \eta) \geq c(C, D, \eta) \frac{dD\eta}{\epsilon} \quad (0 < \epsilon \leq \epsilon_0(C, D, \eta)).$$

The informative construction includes pairs with $2D - 1 \geq e^{\eta}$, but its constant is pointwise in (C, D, η) ; its invariant content is $\Omega_{C, D, \eta}(d/\epsilon)$, not uniform lower dependence on D or η . Consequently every lower bound asserted uniformly over all public experiments and based only on (d, C, D, η) is at most one. Moreover, for each specified finite experiment with a nonempty exact shell and fixed (d, C, D, η) ,

$$\lim_{\epsilon \downarrow 0} \frac{\epsilon^2}{d} N_{\mathfrak{E}}^*(\epsilon; d, C, D, \eta) = 0,$$

so no positive quadratic lower branch can hold throughout all sufficiently small accuracies. Thus the clean upper bound is universal over every nonempty exact shell, while lower matching is only pointwise and local in the d/ϵ rate on the constructed informative experiment.

Proof. Fix $d \geq 4$, $\eta > 0$, and the finite public experiment $\mathfrak{E}$. By lem:fixed-experiment-shell-certificate, the shell $\mathcal{M}_{\mathfrak{E}}(d, C, D, \eta)$ is nonempty if and only if at least one of the finitely many displayed exponential nonlinear-semidefinite systems is feasible. This proves the feasibility assertion.

Suppose first that the boundary shell $\mathcal{M}_{\mathfrak{E}}(d, 1, 1, \eta)$ is nonempty. The boundary conclusion in thm:fixed-design-index-insufficiency gives

$$N_{\mathfrak{E}}^*(\epsilon; d, 1, 1, \eta) = 1$$

for every $\epsilon > 0$.

Now suppose that a nontrivial fixed-design shell is nonempty. By thm:clean-temperature-feature-upper, the stabilized global-optimization rule $\hat{\pi}_{\text{stab}}$ is a measurable member of $\mathcal{L}_n(\mathfrak{E})$. Its construction uses only $(\mathcal{D}_n, \mathfrak{E})$, and therefore uses neither the law-specific context distribution ρ nor the shell labels C and D .

Fix $0 < \epsilon < 1$. If $\eta \leq 2\epsilon$, the deterministic part of thm:clean-temperature-feature-upper yields

$$N_{\mathfrak{E}}^*(\epsilon; d, C, D, \eta) = 1.$$

If $\eta > 2\epsilon$, define

$$q_{\epsilon} = dD \min\{\eta/\epsilon, \epsilon^{-2}\}, \quad \ell_{\epsilon} = 1 + \log(eq_{\epsilon}) + \log\{1 + (1 + \eta)q_{\epsilon}\},$$

and

$$n_0 = \lceil 10^8 q_{\epsilon} \ell_{\epsilon} \rceil.$$

The explicit inversion in `thm:clean-temperature-feature-upper` gives

$$\sup_{P \in \mathcal{M}_{\mathfrak{E}}(d, C, D, \eta)} \mathbb{E}_{P \otimes n_0} [J_{\eta, P}(\pi_P^*) - J_{\eta, P}(\hat{\pi}_{\text{stab}})] \leq \epsilon.$$

Because $\hat{\pi}_{\text{stab}} \in \mathcal{L}_{n_0}(\mathfrak{E})$, taking the infimum over learners in the minimax risk and then applying the sample-complexity definition gives

$$N_{\mathfrak{E}}^*(\epsilon; d, C, D, \eta) \leq n_0.$$

The two logarithms in ℓ_ϵ are precisely the polylogarithmic factors suppressed by $\tilde{O}$, so

$$N_{\mathfrak{E}}^*(\epsilon; d, C, D, \eta) \leq \tilde{O}(dD \min\{\eta/\epsilon, \epsilon^{-2}\}).$$

This establishes the asserted universal upper bound on its declared domain $0 < \epsilon < 1$.

It remains to compare fixed experiments at common indices. Fix $1 < D \leq C < e^\eta$. By `thm:fixed-design-index-insufficiency`, there are finite public experiments $\mathfrak{E}_0$ and $\mathfrak{E}_1$, each having a nonempty exact shell at (d, C, D, η) , such that

$$N_{\mathfrak{E}_0}^*(\epsilon; d, C, D, \eta) = 1 \quad (\epsilon > 0),$$

whereas there are constants $c(C, D, \eta) > 0$ and $\epsilon_0(C, D, \eta) > 0$ for which

$$N_{\mathfrak{E}_1}^*(\epsilon; d, C, D, \eta) \geq c(C, D, \eta) \frac{dD\eta}{\epsilon} \quad (0 < \epsilon \leq \epsilon_0(C, D, \eta)).$$

The same theorem states that the informative construction applies throughout $1 < D \leq C < e^\eta$, including pairs with $2D - 1 \geq e^\eta$. Since the positive constant may depend on (C, D, η) , the invariant rate statement is the pointwise local bound $\Omega_{C, D, \eta}(d/\epsilon)$; it does not furnish lower dependence on D or η uniformly over those indices. In particular, the two experiments have identical index tuples but different sample-complexity frontiers, so (d, C, D, η) cannot determine the fixed-design minimax frontier.

More explicitly, let $L(\epsilon; d, C, D, \eta)$ be any lower bound claimed uniformly over all public experiments whose exact shell at the displayed indices is nonempty. Applying the claim to $\mathfrak{E}_0$ forces

$$L(\epsilon; d, C, D, \eta) \leq N_{\mathfrak{E}_0}^*(\epsilon; d, C, D, \eta) = 1.$$

Thus an experiment-uniform index-only lower bound cannot exceed one and cannot diverge as $\epsilon \downarrow 0$.

Finally, `thm:fixed-design-index-insufficiency` gives, for every specified finite experiment with a nonempty exact shell and fixed (d, C, D, η) ,

$$\lim_{\epsilon \downarrow 0} \frac{\epsilon^2}{d} N_{\mathfrak{E}}^*(\epsilon; d, C, D, \eta) = 0.$$

This limit concerns sufficiently small positive ϵ , which eventually belongs to $(0, 1)$, so it is consistent with the declared domain of the explicit inversion. If a positive quadratic lower branch held for all sufficiently small ϵ , then the displayed normalized quantity would be bounded below by a positive constant, contradicting the limit. Hence no such branch persists. The universal conclusion is therefore the clean feature-coverage upper bound, whereas the available matching lower-rate conclusion is only pointwise and local on the informative experiment. $\square$

Justification. This theorem integrates fixed-design feasibility, exact inversion of the universal clean upper bound, experiment-index insufficiency, and the pointwise informative lower construction. *Gap.* The upper result is closed for every nonempty exact shell. Lower matching is only pointwise and local in d/ϵ on one constructed informative experiment; no uniform D - or η -dependence and no quadratic lower branch are proved. *Consumer.* It is the canonical answer separating universal feature-coverage achievability from experiment-specific lower difficulty.

Lemma 15 (`lem:gibbs-pessimism-conversion`). *Let $r, g : S \rightarrow [0, 1]$, put $\Delta = g - r$, and suppose $\Delta \leq 0$ pointwise. Write*

$$J_{\eta, r}(\pi) = \mathbb{E}_{\rho, \pi} r(X, A) - \eta^{-1} \mathbb{E}_{\rho} \text{KL}(\pi(\cdot \mid X) \parallel \pi_{\text{ref}}(\cdot \mid X)).$$

For $\gamma \in [0, 1]$, let $\pi_\gamma(\cdot | x)$ be the Gibbs policy generated by $r + \gamma\Delta$, so $\pi_0 = \pi_r$ and $\pi_1 = \pi_g$. Then

$$J_{\eta,r}(\pi_r) - J_{\eta,r}(\pi_g) \leq \min \left\{ \mathbb{E}_{\rho, \pi_r}[-\Delta], \frac{\eta}{2} \mathbb{E}_{\rho, \pi_r} \Delta^2 \right\}.$$

Proof. Fix a context x and define

$$H_x(\gamma) = \log \sum_a \pi_{\text{ref}}(a | x) \exp\{\eta(r(x, a) + \gamma\Delta(x, a))\}.$$

Differentiation is legitimate because the action set is finite, and gives

$$H'_x(\gamma) = \eta \mathbb{E}_{\pi_\gamma(\cdot | x)} \Delta, \quad H''_x(\gamma) = \eta^2 \text{Var}_{\pi_\gamma(\cdot | x)}(\Delta).$$

The conditional Gibbs-regret identity can be written as

$$J_{\eta,r,x}(\pi_0) - J_{\eta,r,x}(\pi_1) = \frac{1}{\eta} \{H_x(0) - H_x(1) + H'_x(1)\}.$$

Since $H'_x(1) \leq 0$,

$$J_{\eta,r,x}(\pi_0) - J_{\eta,r,x}(\pi_1) \leq \frac{H_x(0) - H_x(1)}{\eta} = \int_0^1 \mathbb{E}_{\pi_\gamma}[-\Delta] d\gamma.$$

Moreover,

$$\frac{d}{d\gamma} \mathbb{E}_{\pi_\gamma} \Delta = \eta \text{Var}_{\pi_\gamma}(\Delta) \geq 0.$$

Thus $\mathbb{E}_{\pi_\gamma}[-\Delta] \leq \mathbb{E}_{\pi_0}[-\Delta]$, proving the slow inequality after averaging over ρ .

For the fast inequality, Taylor's theorem applied at $\gamma = 1$ gives a $\bar{\gamma}_x \in [0, 1]$ such that

$$H_x(0) = H_x(1) - H'_x(1) + \frac{1}{2} H''_x(\bar{\gamma}_x).$$

Consequently

$$J_{\eta,r,x}(\pi_0) - J_{\eta,r,x}(\pi_1) = \frac{\eta}{2} \text{Var}_{\pi_{\bar{\gamma}_x}}(\Delta) \leq \frac{\eta}{2} \mathbb{E}_{\pi_{\bar{\gamma}_x}} \Delta^2.$$

It remains to move the intermediate policy to the target endpoint. Put $G_x(\gamma) = \mathbb{E}_{\pi_\gamma} \Delta^2$. Direct differentiation yields

$$G'_x(\gamma) = \eta \left\{ \mathbb{E}_{\pi_\gamma} \Delta^3 - \mathbb{E}_{\pi_\gamma} \Delta^2 \mathbb{E}_{\pi_\gamma} \Delta \right\}.$$

If U, U' are independent with the law of $\Delta(x, A)$ under π_γ , then

$$2 \text{Cov}(U^2, U) = \mathbb{E}[(U^2 - U'^2)(U - U')] = \mathbb{E}[(U - U')^2(U + U')] \leq 0,$$

because $U, U' \leq 0$. Hence $G'_x(\gamma) \leq 0$, so $G_x(\bar{\gamma}_x) \leq G_x(0)$. Averaging over ρ proves

$$J_{\eta,r}(\pi_r) - J_{\eta,r}(\pi_g) \leq \frac{\eta}{2} \mathbb{E}_{\rho, \pi_r} \Delta^2.$$

Combining the two bounds proves the lemma. □

7 Interpretation

Feature coverage D is sufficient for the universal clean reverse-KL upper bound, so additional pointwise coverage C is statistically irrelevant for that achievability guarantee. Prior work already shows more generally that coverage summaries can overstate or incompletely describe offline difficulty. The specific conclusion here is that even the simultaneous exact indices (d, C, D, η) fail to determine the fixed-public-experiment reverse-KL frontier: two experiments at those indices can have sample complexity one and a diverging pointwise local d/ϵ lower rate. Reference support in the displayed learner class is solely a finite-objective representation: under extended reverse KL, every off-support output at a positive-mass context has infinite regret, and public measurable support repair proves equality with the full-simplex minimax problem.

Technical internal limitation (diagnostic only)

The stabilized learner is explicit and measurable but is an idealized global-optimization procedure; computational efficiency is not established. The informative lower constant is allowed to depend on (C, D, η) , so the construction proves neither uniform lower dependence on D or η nor a quadratic dD/ϵ^2 branch. For every specified finite experiment, the quadratic normalization tends to zero as $\epsilon \downarrow 0$. The reference-supported decision representation is not an additional statistical limitation: it has exactly the same minimax value as the full-simplex extended-reverse-KL formulation.

8 Honest scope

The universal mathematical result is the all-temperature feature-coverage upper bound over every nonempty fixed-design exact reverse-KL shell. Relative to Zhao et al., the lawful sharpening is only the bounded shared-linear spectral specialization: under directional richness, D_P is the largest eigenvalue while Zhao et al.’s single-policy quantity is the trace, but the bounded prediction polytope alone does not guarantee that identity, and the present theorem does not dominate their general-function, sufficiently-small-accuracy, high-probability result. Relative to Li–Ma–Srebro, the novelty claim concerns reverse-KL Gibbs welfare and second-moment spectral coverage, not the origin of offline linear pessimism. The fixed-design theorem specializes, rather than supersedes, the broader coverage-insufficiency and information-complexity perspectives of Nguyen–Tang–Arora, Chen et al., and Vannella et al. Lower matching remains limited to a pointwise local $\Omega_{C,D,\eta}(d/\epsilon)$ result on one informative experiment, while another experiment at the same indices has zero minimax risk; there is no uniform D - or η -dependent converse and no persistent quadratic lower branch. Finally, the stabilizing learner is defined by exact measurable global optimization over compact prediction polytopes; no finite-precision algorithm or approximation-regret theorem is supplied. Throughout, the standard extended reverse-KL convention governs the full policy simplex. Restricting the written minimax infimum and diagnostic criterion to reference-supported policies only selects the finite-objective representation; the measurable public repair shows that this convention changes neither the minimax value nor any headline rate.

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